

SPARSE OBSERVATION ALGEBRAS AND CERTIFIED MEMORY ACROSS EXPONENT COLLISIONS

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ABSTRACT. We determine persistent finite-state memory for positive sparse monomial experiments, uniformly through additive exponent collisions. The sharp regret profile is a maximum of fractional powers of Vandermonde products, truncated by the attainable past dimension and maximized over checkpoints. For certified finite moment data, one causal filter realizes this profile using integer transition tables and dyadic outputs. Its numerical scale is selected from a two-sided covering certificate computed on approximate feasible histories, without determining exponent equalities or receiving the profile as input. If the profile is $\Xi_N(M, a)$, the selected precision is $\frac{1}{2} \log_2(1/\Xi_N(M, a)) + O(1)$ binary places. A separated future chain bounds this uniformly by $\log_2(M + 1)/((r - 1)\lfloor N/2 \rfloor) + O(1)$. The proof combines collision-uniform attainment with finite numerical compilation and an a posteriori stopping rule. Persistent state, numerical advice, and read-only program size are accounted for separately. The intrinsic classification holds for every full-support prior; effectiveness requires the stated numerical input interface. For a common moment name of accuracy δ , a further stable realization achieves regret $O(\Xi_N + \delta^2)$ in every compatible experiment in a specified dominated prior family. A positive rank-two experiment shows that the uncertainty term is sharp.

1. INTRODUCTION

The number of exactly distinguishable posterior directions can jump when an additive relation between observation exponents disappears. The persistent memory needed at a fixed prediction accuracy need not jump: new directions may have arbitrarily small amplitude. We study this question for positive sparse monomial experiments, and determine how their sharp memory order is realized by finite numerical programs. The numerical scale can itself be selected from finite data, without first identifying an additive collision stratum.

Let $A = \{0 = a_0 < a_1 < \dots < a_{r-1} = D\}$, let $W_A = \text{span}\{t^a : a \in A\}$, and let mA be the set of sums of exactly m members of A , with $0A = \{0\}$. Write $h_A(m) = |mA|$. For the exact information statements the parameter interval is $I = [l, u]$, $0 \leq l < u < \infty$, and the prior μ has full support. For collision-uniform statements we take $I = [0, 1]$. A positive finite detector spans W_A , and a command in $[\eta, 1 - \eta]^J$ retains or rejects each detector symbol. Every reported symbol, including rejection, is counted as a trial. Its likelihood is bounded below by a fixed positive number κ . The precise experiment and common future-only equivalence are defined in Section 2.

The one-step exponents range over a fixed compact subset K of the strictly ordered positive chamber. One fixed full-column-rank coefficient matrix specifies the detector on K , with uniform positivity. Neither semialgebraicity of K nor a density for μ is assumed. The horizon $N \geq 2$ and prior are fixed. Choose a fixed $H \geq \max\{1, N \max_K a_{r-1}\}$. For each future length m , retain all nonconstant formal homogeneous labels $\alpha \in \mathbb{N}_0^r$, $|\alpha| = m$, including labels that have the same

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exponent. Their normalized nodes are $x_\alpha = H^{-1}\alpha \cdot a$ and their number is $q_m = \binom{m+r-1}{r-1} - 1$. Set

$$\mathcal{V}_{m,\ell}(a) = \max_{|F|=\ell} \prod_{\{\alpha,\beta\} \subset F} |x_\alpha - x_\beta|, \quad \mathcal{V}_{m,1} = 1,$$

where F ranges over subsets of formal labels. Zero volumes remain zero. Put $p_{n,m} = \min\{n(r-1), q_m\}$ and

$$\Psi_{n,m}(M, a) = \max_{1 \leq \ell \leq p_{n,m}} (\mathcal{V}_{m,\ell}(a)/M)^{2/\ell}, \quad \Xi_N(M, a) = \max_{1 \leq n < N} \Psi_{n,N-n}(M, a).$$

The checkpoint profile is zero at an endpoint. At each checkpoint the query is selected independently after compression from a fixed finite spanning menu of m -trial product probes. The excess squared loss is the average squared error of its conditional event probabilities. The notation $\bar{R}_{M;n,m}^a$ denotes optimal unconditional checkpoint regret under independent uniform acquisition commands; $R_{M;n,m}^{a,\max}$ denotes optimal worst-history regret. Their causal counterparts $\mathfrak{R}_{M,N}^{a,\text{av}}$ and $\mathfrak{R}_{M,N}^{a,\max}$ minimize the maximum checkpoint criterion over one stage-compatible filter. Only one checkpoint and its remaining query trials are executed in a run.

Theorem 1.1 (Sharp causal memory and certified numerical realization). *Under the preceding fixed-experiment hypotheses, with $r \geq 2$, the checkpoint and causal regret laws have orders $\Psi_{n,m}$ and Ξ_N , respectively, uniformly in $a \in K$ and integers $M \geq 1$. Suppose the command cube is rationally specified and certified finite approximations to the cell coefficients and prior moments through formal degree N are available at requested tolerances, together with coefficient and positivity bounds. An algorithm constructs integer transition tables and dyadic output tables with at most M persistent labels at each stage and maximum worst-history regret at most $C\Xi_N(M, a)$. For every such filter with a fixed memoryless command-coding rule, the maximum unconditional checkpoint regret under the common exploration law is at least $c\Xi_N(M, a)$.*

The algorithm requires no collision stratum, additive-gap lower bound, or value of Ξ_N as input. Its selected numerical precision satisfies

$$b(M, a) = \frac{1}{2} \log_2(1/\Xi_N(M, a)) + O(1),$$

and a sufficient read-only program size is $O(M\Xi_N(M, a)^{-J/2} \log(M+1))$ bits. In particular, setting $d_0 = (r-1)\lfloor N/2 \rfloor$, one may uniformly use

$$b(M) \leq d_0^{-1} \log_2(M+1) + O(1), \quad P(M) = O(M^{1+J/d_0} \log(M+1)).$$

These numerical bounds are sufficient; no precision or program-length converse is asserted. The running filter retains only its index, not a posterior vector, historical tape, or numerical oracle. For computable numerical inputs the synthesis is uniform and terminating. All constants may depend on the fixed prior and horizon, but not on a or M .

For $r = 1$, all report factors are constant in the hidden parameter; the posterior-dependent state is a singleton and one label suffices. The exact information classification below includes this case.

The geometric assertions are proved in Theorems 5.3 and 5.4. The finite-table construction is given in Section 6; its stronger resource conclusions and adaptive stopping rule are proved in Theorem 7.9 and Corollary 7.2. The earlier fine-mesh bounds remain valid consequences and are recorded with their proofs. For an arbitrary, possibly noncomputable full-support prior, the intrinsic classification is unchanged. The effective statement uses supplied numerical advice, rather than an algorithm for obtaining unsupplied noncomputable moments. With an external clock the changing state has M values; storing the clock costs a factor depending only on N . The program and temporary workspace are accounted for separately.

1.1. Structure of the proof. The first step is a pairing between the attainable product tangent and the complete future test flag. At an interior binomial tuple, the product tangent contains $n(r-1)+1$ distinct monomials. Strict mixed-moment positivity holds for every full-support prior. Complete Hermite blocks retain this positivity at exact collisions; normalization removes exactly one rank. The local image carries an unconditional volume minorization which includes the probability of acquiring the failure prefix. A separate bounded-format argument covers the entire reachable image with the attainable dimension truncated at $p_{n,m}$.

The determinant profile has a classical finite spectral meaning: Proposition 4.2 identifies it, up to fixed constants, with the exterior singular-value profile of a Vandermonde matrix. That statement alone does not imply statistical attainment. The arbitrary-prefix Hermite statement in Lemma 4.1 is likewise classical [12, Proposition 7]. The normalized product, acquired-history probability, and compatible transitions provide the additional statistical structure.

The numerical construction never uses singular coordinates. It forms finite lists of feasible histories, evaluates their raw states to a certified accuracy, and applies classical farthest-first selection [22]. Every transition is evaluated by appending a report and command to an actual representative history. Approximate vectors outside the reachable set are not passed to a posterior update. Positive evidence certifies evaluation from a finite moment table even when coincident formal labels receive different approximate values.

The final step makes the numerical scale observable. If r is the computed covering radius of an approximate history list with mesh and evaluation tolerance h , and e is the largest true M -centre covering radius, then

$$\max\{0, r/2 - h\} \leq e \leq r + (A+2)h.$$

The rule $r \geq 4(A+2)h$ therefore stops at $h \asymp e$ without knowing e . Attainable geometry identifies $e^2 \asymp \Xi_N$. This gives a finite-data interpretation of the full profile, not only of one of its power-law floors. A separated future chain gives the uniform precision bound. Transition and decoder residuals provide separate finite checks on the program actually returned.

For the intersecting two-parameter family, the resulting program-size phases track the full three-term regret profile. High-order contact may place a nonzero exponent gap far below the chosen mesh, without changing the guarantee. The algorithm does not identify the contact order or collision tree. The stable common-advice theorem is then proved in Section 8. Complete observation algebra, affine, five- and seven-trial, decision, and mechanical proofs are given in the appendices. Their separate assumptions are retained.

1.2. Imperfect numerical descriptions. The geometric law has a further consequence when a finite moment name is compatible with more than one experiment. On a dominated family of full-support priors, the constants remain uniform. Pairing identical histories proves Hausdorff stability of the attainable state sets in the raw coordinates; optimal covering radii are therefore Lipschitz in the finite moment data. A common-name stopping rule yields one M -label program, not a family selected using an unsupplied true model, with regret $O(\Xi_N(M, a) + \delta^2)$ in every compatible experiment. These assertions and their precise uniformity hypotheses are stated in Lemma 8.5, Theorem 8.7, and Corollary 8.8.

The extra term has a matching elementary lower witness. A fixed positive rank-two detector admits two full-support priors with the same δ -accurate finite moment name and with a physical prediction gap of order δ after an identical positive-probability acquisition event. Proposition 8.9 gives regret of order $M^{-2} + \delta^2$ for this experiment. The two-point lower-bound method is elementary; the assertion being added is its compatibility with the same physical, budgeted prediction problem and the collision-uniform upper construction. It is not a claim that every finite-precision implementation has this uncertainty model.

1.3. Relation to finite approximation and information states. Finite approximation of a controlled model and finite realization of a prediction state are related, but their conclusions and numerical inputs must be distinguished. We make the comparison at the level of particular statements, rather than regarding finite-state approximation as a new principle.

Saldi, Yüksel and Linder [24, Theorem 5.2] in the explicitly linked arXiv version 3 bound the discounted performance loss of policies obtained from finite approximating Markov decision processes by $O(n^{-1/d})$. The result uses compact Euclidean state space, a Lipschitz cost, a transition kernel Lipschitz in the Wasserstein distance, and $\beta K_2 < 1$. Their finite model uses state quantization and averaged transition kernels; their Section 6 also studies order optimality. Our theorem does not strengthen that discounted-control conclusion. Here the supplied data are certified evaluations of feasible histories; the output is an integer transition program with at most a prescribed M labels at each stage. The scale selected in Theorem 7.4 is the attainable covering radius itself, not a prescribed ambient-dimensional power. The collision law additionally identifies that scale and supplies a matching prediction converse for the same experiment.

Kara and Yüksel [5, Assumption 4 and Theorem 12] obtain an exponentially decreasing discounted policy-loss bound for finite-window feedback after a warm-up period. Their hypotheses include transition regularity, a domination condition, a Dobrushin-type filter stability coefficient $\alpha < 1$, and a restriction on the discount. The finite-window description retains recent observations and actions. Our finite-horizon construction does not require filter contraction, but makes no corresponding infinite-horizon assertion. It enumerates histories only offline and retains an index, not a recent observation window. Its effective input is a finite coefficient–moment table in the monomial case. Window length, persistent-label budget, and numerical program length are different resources; no comparison of their optima follows simply by identifying these names.

Subramanian, Sinha, Seraj and Mahajan [8, Definition 7 and Theorem 9] give an all-history value approximation and policy-loss bound from reward and transition errors of an approximate information state. Their recursive error allowance combines reward error, an integral-probability-metric transition error, and the next-stage allowance; the policy bound has twice that allowance. They also treat action compression in Theorem 17. Thus neither an all-history error recurrence nor action compression is new here. The additional conclusion proved here is a finite synthesis with a prescribed label budget: approximate feasible histories give a computable two-sided covering certificate, which selects its own numerical mesh. In the monomial experiment the certificate is tied to the full collision-sensitive attainable profile by Lemma 7.8, with both the finite program and its numerical advice separately accounted for.

These distinctions concern the displayed theorems, not an assertion that one of the cited theories is contained in another. The classical farthest-first estimate supplies the finite certificate; the attained posterior geometry supplies its interpretation as the sharp prediction scale. Their combination yields the resource phases of Corollary 7.10 without an input identifying which phase is present. The geometric and numerical assertions require separate proofs, and the specializations in the appendices retain their own hypotheses.

Proposition 8.3 isolates a narrower logical comparison: a truthful residual bound for a supplied table is not a construction theorem at the optimal covering scale. Its positive counterpart is Theorem 8.2. This does not assert that a new finite-approximation principle is needed. The non-routine geometric input is the collision-uniform attainment of the entire exterior profile by positive histories, for every fixed full-support prior, together with a global cover of the same order. One consequence beyond table-size exponents is Corollary 8.8: a single imperfect moment name produces one program achieving $O(\Xi_N + \delta^2)$ simultaneously in all compatible experiments in a stated prior envelope. Proposition 8.9 identifies a positive experiment where the additional

uncertainty term is necessary. These are finite-horizon prediction statements; no improvement to the cited infinite-horizon control results is asserted.

1.4. The underlying information classification. The statements below retain the exact-state and arbitrary-prior conclusions. The new numerical theorem strengthens their realization, not their assumptions on the mathematical prior.

Theorem 1.2 (Intrinsic attainable resolution and causal memory). *Under these hypotheses there are constants $0 < c \leq C < \infty$, depending only on the fixed experiment, K , N and μ , with the following properties for every known calibration $a \in K$ and every integer $M \geq 1$.*

At each checkpoint $n + m = N$, the optimal M -message squared-prediction regret is between $c\Psi_{n,m}(M, a)$ and $C\Psi_{n,m}(M, a)$, both unconditionally under independent uniform acquisition commands and in the minimax sense over admitted histories. The query is sampled independently after the index is formed, from the ordered products of one fixed attainable failure-factor basis; all its m trials are counted.

There exists one deterministic stage-compatible transducer, allowed to depend on a and M , storing at most M persistent labels after every report, whose regret at every checkpoint and on every history is at most $C\Xi_N(M, a)$. Conversely every such transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under the same common exploration law. Independent coding randomization does not remove the lower bound. The clock and fixed real-valued program are read-only as in Definition 2.4.

The constants are uniform across all additive collision strata of K . No calibration-blind codebook or uniformity over all full-support priors is part of the assertion.

Theorem 1.3 (Sparse finite-horizon information law). *Suppose the positive finite detector spans W_A and every lookup entry ranges over $[\eta, 1 - \eta]$, where $0 < \eta < 1/2$. For $n, m \geq 0$ put*

$$(1) \quad d_A(n, m) = \min\{n(r - 1), h_A(m) - 1\}.$$

Two attainable posteriors after n reports have the same law under every common future-only continuation of length m if and only if their moments at the exponents in mA agree. The prediction map on all-failure command histories has generic rank $d_A(n, m)$, and it has a smooth attainable patch of this dimension with a continuous section to histories. The smallest Euclidean dimension of a sufficient encoding continuous on each known-report history stratum is exactly $d_A(n, m)$.

For a total budget N , these dimensions can be attained simultaneously by a causal exact-real state. Its maximal size is

$$(2) \quad D_A(N) = \max_{0 \leq n \leq N} d_A(n, N - n).$$

Only prior moments with exponents in NA are needed for its updates.

2. CALIBRATED EXPERIMENTS AND THEIR OPERATIONAL QUOTIENT

All spaces carry their Borel sigma-algebras. Deterministic continuation, transition and decision rules are Borel measurable; randomized rules are Borel probability kernels. Every external random seed is independent of the hidden parameter and the acquired history unless explicitly charged as part of that history.

Let $k_1, \dots, k_K \in W_A$ be strictly positive on I , satisfy $\sum_j k_j = 1$, and span W_A . On each trial a fresh raw symbol has these probabilities conditional on the same fixed parameter t . The raw trials are conditionally independent. A command $g \in \mathcal{G} = [\eta, 1 - \eta]^K$ accepts raw symbol

j with probability g_j , using an independent prepared uniform variable. A rejected symbol is reported as $\dagger$. Thus the report alphabet is $\mathcal{X} = \{1, \dots, K, \dagger\}$ and its likelihood factors are

$$(3) \quad f_{g,j} = g_j k_j, \quad f_{g,\dagger} = F_g = \sum_j (1 - g_j) k_j.$$

A single command is applied at every possible parameter value. Every trial is counted, including every rejected report. There is a constant $\kappa > 0$ such that $\kappa \leq f_{g,x}(t) \leq 1$ for all g, x, t . Indeed one may take $\kappa = \eta \min_{j,t} k_j(t)$.

Lemma 2.1 (Attainable interior and realizability). *The failure family is precisely the image of the rejection cube under $u \mapsto \sum_j u_j k_j$. It contains a neighborhood of the constant $1/2$ in W_A . Conversely, every finite exponent set A as above has a positive normalized r -cell realization with this property.*

Proof. The displayed map is a surjective linear map onto W_A . Choose a linear right inverse $L : W_A \rightarrow \mathbb{R}^K$. For every v satisfying $\|Lv\|_\infty < 1/2 - \eta$, rejection vector $u = \mathbf{1}/2 + Lv$ realizes $1/2 + v$ inside the cube. This proves both the interior statement and that the same physical command implements it. For existence, choose $\delta > 0$ with $\delta \sum_{a \in A \setminus \{0\}} u^a < 1/r$ and set

$$k_a(t) = 1/r + \delta t^a \quad (a \neq 0), \quad k_0(t) = 1/r - \delta \sum_{a \neq 0} t^a.$$

They are positive and sum to one. Their span contains one and then each $t^a = (k_a - 1/r)/\delta$, so has dimension r . These are calibrated categorical experiments. No mechanical origin is asserted for this general realization; the separate Lorentz realization is derived in Section G. $\square$

A history $h = (g_1, x_1, \dots, g_n, x_n)$ has posterior

$$(4) \quad P_h(t) \mu(dt), \quad P_h = \frac{\prod_{i=1}^n f_{g_i, x_i}}{\mu(\prod_{i=1}^n f_{g_i, x_i})}.$$

All commands are known inputs when received. They cease to be available once discarded by a memory-limited filter. A continuous history encoding means continuity in $(g_1, \dots, g_n)$ separately for each fixed report word. It need not be injective on histories and need not identify all histories with the same posterior.

Definition 2.2 (Common future-only continuation). At a checkpoint a continuation has the same fixed task data in the two experiments. Its commands are functions only of the reports and commands *since that checkpoint*, and any fresh independent random seed. Its first command is the same in both experiments. It may not read the original prefix, a prefix-dependent seed, or any other uncharged historical tape. Two posteriors are m -equivalent when all such length- m continuations have the same future report law.

A separate payoff automaton referring to past observations is task memory, not something removed by posterior equivalence. Losses that inspect t directly are also not included without adding their test functions. This quantifier matters: two different constant failure commands leave the same prior, but a rule that reads and repeats the original command does not meet Definition 2.2.

Proposition 2.3 (Exact future test space). *The linear span of length- m future likelihood tests is*

$$(5) \quad W_A^m = \text{span}\{t^s : s \in mA\}.$$

Two posteriors are m -equivalent if and only if their moments at the exponents in $mA \setminus \{0\}$ agree. A finite list of actually executable, nonadaptive all-failure probes spans the same space.

Proof. Choose finitely many attainable failures spanning W_A , using Lemma 2.1. All their m -fold products are executable all-failure event probabilities. Multilinearity makes their span the span of all products of m members of W_A , which is exactly (5). Independence is conditional on t ; one does not replace the shared parameter by fresh parameters between trials.

For any fixed future word a deterministic continuation chooses a known sequence of commands along that word. Its likelihood is a product of m members of W_A , even when the chosen sequence differs on other branches. Equality of the stated moments therefore implies equality of all word probabilities. There are finitely many report words. Integrating an independent continuation seed proves the randomized case. Conversely, equivalence includes all the spanning all-failure probes; their predictions determine all the moments in (5). The constant moment is one. Only the selected probe is executed and all its m trials are consumed, regardless of earlier nonfailure reports. $\square$

A fixed positive nonconstant denominator or an arbitrary nonmonomial factor space is not silently substituted in (5). Such models have their own product and test pairings. The theorem below addresses every sparse monomial space W_A , not every finite-dimensional space of continuous functions.

2.1. Persistent state and prediction loss.

Definition 2.4 (Finite-state filter). An M -state filter has transitions

$$i_{n+1} = U_n(i_n, g_{n+1}, x_{n+1}), \quad i_n \in \{1, \dots, M\}.$$

It reads only its previous index, the current command and current report. Past commands, past reports and retained command-generation or update seeds are unavailable outside the index. Fresh private coins may be used and then discarded. At a checkpoint the decoder reads only i_n and a newly revealed independent query. The known clock permits stage-dependent transition and output tables. Real command inputs and read-only real calibration constants are allowed; arithmetic workspace, execution time and finite-precision implementation of these fixed functions are not the memory resource of this definition. In particular, this is a finite-state transducer, not an algorithm that repeatedly rereads an exact prefix.

For each $1 \leq n < N$ fix a finite spanning probe list $H_{n,1}, \dots, H_{n,s_n^q}$ as in Proposition 2.3, using $m = N - n$ trials. To avoid confusion, s_n^q is the number of queries, not the exact state s_n . After the checkpoint state is formed, choose an independent uniform query J . Announce $a \in [0, 1]$, execute that probe using its m fresh trials, and let Y be its all-failure indicator. All N trials are counted. The conditional squared-loss excess over the full-history predictor is

$$(6) \quad \mathcal{R}(a, p) = \frac{1}{s_n^q} \sum_j (a_j - p_j)^2, \quad p_j = \mu(P_h H_{n,j}).$$

This follows by expanding $\mathbb{E}[(a - Y)^2 \mid h, J]$; decoder randomization is averaged on the right. Only one selected checkpoint and one probe are executed in a run. Simultaneous bounds over checkpoints do not count the unexecuted probes as physical trials.

3. ATTAINABLE TRANSVERSALITY AND THE EXACT MEMORY PROFILE

We first prove the positivity statement that prevents a prior-dependent loss of rank. No density assumption on the prior is needed.

Lemma 3.1 (Strict mixed-moment positivity). *Let $\alpha_1 < \dots < \alpha_s$ and $\beta_1 < \dots < \beta_s$ be nonnegative real numbers. For every full-support probability on I ,*

$$(7) \quad \det[\mu(t^{\alpha_i + \beta_j})]_{i,j=1}^s > 0.$$

Consequently a rectangular moment matrix indexed by two increasing finite sets of distinct nonnegative exponents has rank the smaller cardinality.

Proof. For increasing positive points $t_1 < \dots < t_s$, the generalized Vandermonde determinant $\det[t_j^{\alpha_i}]$ is positive. Here is a self-contained sign argument. Put $t_j = e^{x_j}$. A nonzero combination of s exponentials with distinct ordered exponents has at most $s - 1$ distinct zeros: factor out the first exponential, differentiate, and use Rolle's theorem inductively. Thus the determinant cannot vanish on the connected region $x_1 < \dots < x_s$. When the x_j coalesce to x_0 from ordered nearby points, its first nonzero Taylor term has the sign of the Wronskian

$$e^{x_0 \sum_i \alpha_i} \prod_{i < j} (\alpha_j - \alpha_i)$$

times the ordinary positive Vandermonde in the x_j , divided by positive factorials. Its sign is therefore positive everywhere in that region. The same holds for the β determinant.

Expanding determinants and applying Fubini gives the finite determinant integration identity

$$\det[\mu(t^{\alpha_i + \beta_j})] = \frac{1}{s!} \int_{I^s} \det[t_j^{\alpha_i}] \det[t_j^{\beta_i}] \prod_{j=1}^s \mu(dt_j).$$

The integrand is nonnegative, by permutation symmetry and continuity at zero or repeated points. Choose s disjoint ordered subintervals in $(l, u) \cap (0, \infty)$. Each has positive prior mass by full support; on their product the integrand is strictly positive. The integral is strictly positive. Applying this to any square minor proves the last assertion. This is the classical total-positivity mechanism [7], with its needed form proved explicitly here. $\square$

Lemma 3.2 (A monomial tangent inside the attainable cube). *For every $n \geq 1$, choose distinct, sufficiently small positive numbers $c_1, \dots, c_n$. The failure factors $F_i = (1 + c_i t^D)/2$ are simultaneously attainable at interior commands. At this tuple the differential image of the unnormalized product map $W_A^n \rightarrow \text{span}\{t^s : s \in nA\}$ is*

$$(8) \quad T_{A,n} = \text{span}\{t^b : b \in B_{A,n}\},$$

where

$$(9) \quad B_{A,n} = \{jD : 0 \leq j \leq n\} \cup \bigcup_{a \in A \setminus \{0, D\}} \{a + jD : 0 \leq j < n\}.$$

In particular $\dim T_{A,n} = n(r - 1) + 1$, and the product itself belongs to this tangent space.

Proof. Attainability follows by applying the fixed right inverse in Lemma 2.1 to $c_i t^D/2$ and taking all c_i small enough. The factors and their implementing commands lie in an open subset of the allowed cube. Constants 2^{-n} do not change a differential image, so omit them in the remaining calculation. Put $z = t^D$, $P(z) = \prod_i (1 + c_i z)$ and $P_i(z) = P(z)/(1 + c_i z)$. These n polynomials form a basis of the polynomials in z of degree at most $n - 1$. Indeed evaluation at $z = -1/c_j$ makes all P_i except P_j zero, and $P_j(-1/c_j) \neq 0$.

The product differential is $\sum_i \dot{F}_i \prod_{j \neq i} F_j$. Variations in the constant direction give all polynomials in z through degree $n - 1$; variations in t^D give their multiples by z . Together they give all z^j , $0 \leq j \leq n$. Variations in an interior exponent a give precisely $t^a z^j$, $0 \leq j < n$. The listed exponent sets are disjoint: two numbers $a + jD$ and $a' + j'D$ with $0 < a, a' < D$ agree only if $a = a'$ and $j = j'$, and none is a multiple of D . Their total cardinality is $(n + 1) + n(r - 2) = n(r - 1) + 1$. Every summand of the differential has this form, proving equality rather than only inclusion in (8). The product is obtained by varying one factor along itself, so is in the tangent. $\square$

Theorem 3.3 (Generic normalized pairing rank). *For fixed $n, m \geq 1$, the map from n all-failure commands to the moment vector $(\mu(t^s P_h))_{s \in mA \setminus \{0\}}$ has generic rank $d_A(n, m)$ in (1). This holds for every full-support prior, including priors without a density. At an interior attainable tuple it has a smooth $d_A(n, m)$ -dimensional local image and a continuous local section to histories. A projection onto $d_A(n, m)$ of the moment or spanning-probe coordinates maps such a section onto an open Euclidean ball.*

Proof. Let $P = \prod_i F_i$, $Z = \mu P > 0$, and include the constant test in the linear map

$$L : T_{A,n} \longrightarrow \mathbb{R}^{h_A(m)}, \quad LQ = (\mu(t^s Q))_{s \in mA}.$$

At the tuple from Lemma 3.2, Lemma 3.1 gives

$$\text{rank } L = \min\{n(r-1) + 1, h_A(m)\}.$$

Let e_0 extract the coordinate of exponent zero, and put $p = L(P)/Z$. The derivative of the normalized prediction map on this tangent is

$$(10) \quad Q \longmapsto Z^{-1}(LQ - p e_0(LQ)).$$

Now $e_0 p = 1$ and $p \in L(T_{A,n})$. The linear map $v \mapsto v - p e_0 v$, restricted to $L(T_{A,n})$, has kernel exactly $\text{span}\{p\}$ and image in the zero-constant-coordinate hyperplane. It loses *exactly* one rank. Deleting that identically zero coordinate therefore gives rank $\min\{n(r-1), h_A(m) - 1\}$.

At every tuple the normalized product map is invariant under independent rescaling of its n nonzero factors, so its rank is at most $n(r-1)$. The number of nonconstant tests gives the other upper bound. Thus the attained rank is globally maximal. The command-to-factor linear map is surjective at every interior command; the same maximal rank is attained in command coordinates. Its Jacobian entries are rational functions of the commands with strictly positive polynomial denominator Z^2 . A maximal minor nonzero at the displayed tuple has a nonzero polynomial numerator. Its zero set is a proper real algebraic set with empty interior and zero Lebesgue measure. This proves genericity throughout the command cube, not only along the binomial family.

Choose $d = d_A(n, m)$ independent output coordinates and d command coordinates at that tuple, and freeze the complementary command coordinates. The inverse function theorem gives a local inverse for the selected output projection; shrink to a ball with compact closure. The full image of this section is a smooth local graph. A spanning list of probes is an injective linear image of the moment coordinates, so the same conclusion holds after choosing independent probe coordinates. All section histories are feasible all-failure histories. Their probability conditional on each parameter is at least η^n ; no failure evidence has been dropped. $\square$

Notice that when $d < h_A(m) - 1$ the attainable image can be curved. The proof does not replace it by its affine hull. The lower-bound ball is in a selected output projection, and its lift is a local graph, not necessarily a flat ball in the ambient prediction space.

Theorem 3.4 (Minimal continuous encoding and causal attainment). *The minimal dimension stated in Theorem 1.3 is $d_A(n, m)$. For a fixed N , there are compact exact state sets $\mathcal{S}_n \subset \mathbb{R}^{d_A(n, N-n)}$ and continuous-in-command causal updates attaining this dimension at every n . Their state represents the posterior-dependent future law, not an uncharged past-dependent task.*

Proof. Compose a sufficient continuous encoding with the section in Theorem 3.3. Different points in the projected ball have different predictions, so this composition is injective. A continuous injection of an open d -ball into $\mathbb{R}^k$, $k < d$, is impossible: embedding $\mathbb{R}^k$ as a coordinate plane of $\mathbb{R}^d$ would contradict invariance of domain. The decoder need not be continuous for this argument.

For the global upper bound, normalize each observed factor by its own prior integral, $\tilde{f}_i = f_i/\mu f_i$. Each normalized factor is in the affine hyperplane $\mu f = 1$ of W_A , with $r - 1$ coordinates. The chronologically ordered n factors determine the posterior product and use $n(r - 1)$ real coordinates. This representation may distinguish permutations; sufficiency does not forbid that. Alternatively, retain the $h_A(m) - 1$ nonconstant posterior moments in mA . Proposition 2.3 proves this alternative sufficient on every history. Use whichever representation is smaller. Both are continuous on each fixed-report stratum. Finite unions of compact continuous images of the command cube give compact state sets. The cases $n = 0$ or $m = 0$ have a single class and zero coordinates.

For causal implementation retain normalized factors while $n(r - 1) \leq h_A(N - n) - 1$. The left side increases and the right side nonincreases, so there is at most one changeover. When the inequality first fails, multiply the retained factors and compute the required moments. Afterward, if $m = N - n$ and the new report factor is $f(t) = \sum_{a \in A} f_a t^a$, update by

$$(11) \quad M'_s = \frac{\sum_{a \in A} f_a M_{s+a}}{\sum_{a \in A} f_a M_a}, \quad s \in (m - 1)A \setminus \{0\}, \quad M_0 = 1.$$

All indices $s + a$ belong to mA . The denominator is the report probability and is at least κ . At the changeover the integrands are products of n factor monomials and a remaining-test monomial, so their exponents are in $nA + (N - n)A = NA$. Normalizing each factor requires only moments in $A \subset NA$. Thus a finite read-only table $(\mu(t^s))_{s \in NA}$ suffices for all updates. Formula (11) then gives the exact shrinking state to time N . $\square$

For arbitrary nonmonomial spaces W , the starting identity remains (10), with the true product tangent and W^m . What is special here is the explicit attainable tangent (8) and the strictly positive mixed pairing. No dimension formula depending only on $\dim W$ is inferred outside this class.

4. FINITE INTERPOLATION AND THE ATTAINED PAIRING

The finite spectral profile used below has a classical interpretation. We isolate it from the assertion that the corresponding directions are excited by actual histories. The first two statements also occur in the technical note accompanying the referee report of 6 September 2026; we include complete proofs and make no priority claim for them.

Lemma 4.1 (Arbitrary-prefix Hermite data). *Let $y_1, \dots, y_p$ be an ordered multiset of real numbers, and let q_y be the multiplicity of y in the full multiset. The functionals*

$$f \mapsto [y_1, \dots, y_j]f, \quad 1 \leq j \leq p,$$

form a basis of the space generated by $f \mapsto f^{(k)}(y)$, $0 \leq k < q_y$. Repetitions need not be adjacent. For positive nodes, application to $f(z) = t^{Hz}$ gives exactly the complete blocks $t^{Hy}(\log t)^k$, $0 \leq k < q_y$.

Proof. A prefix divided difference depends only on the Hermite data at that prefix, so belongs to the stated p -dimensional space. On polynomials $1, z, \dots, z^{p-1}$ its values vanish in degrees below $j - 1$ and equal one in degree $j - 1$. The resulting matrix is triangular with unit diagonal, so these p functionals are independent. Substitution of t^{Hz} transforms differentiation at a node into multiplication by the nonzero factor H^k and the indicated logarithmic function. This is the Newton-Hermite prefix principle; see [12, Proposition 7]. It does not assert attainable posterior rank. $\square$

Proposition 4.2 (Exterior spectral form of determinant volumes). *Fix q and nodes $x_1, \dots, x_q \in [0, 1]$, retaining repetitions. Let $W = (x_i^{j-1})_{i,j=1}^q$ and let $\sigma_1 \geq \dots \geq \sigma_q \geq 0$ be its singular values. Define $\mathcal{V}_\ell$ as the maximal absolute Vandermonde product over ℓ labels, with $\mathcal{V}_1 = 1$. Then*

$$c_q \mathcal{V}_\ell \leq \prod_{j=1}^{\ell} \sigma_j(W) \leq C_q \mathcal{V}_\ell, \quad 1 \leq \ell \leq q.$$

The constants are independent of all gaps, and the assertion includes coincident nodes. Equivalently the middle expression is the operator norm of $\bigwedge^\ell W$.

Proof. Use the finite Leja ordering of Lemma 5.1, and let d_j be its nonincreasing pivots. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$, and let T have as its j th column the coefficients of P_{j-1} in the power basis. It is upper triangular with diagonal one. Elementary symmetric polynomial bounds on nodes in $[0, 1]$ bound T uniformly in q . Writing $T = I + U$ with $U^q = 0$ gives $T^{-1} = \sum_{k=0}^{q-1} (-U)^k$, hence a uniform bound on its inverse.

Evaluation gives $WT = LD$, $D = \text{diag}(d_j)$. If s is the number of distinct sites, the active leading s -block of L is lower triangular with diagonal ± 1 , bounded entries, and bounded inverse. Replace each inactive column by the corresponding coordinate vector. The resulting matrix $\tilde{L}$ and its inverse are uniformly bounded: it is block lower triangular, with the active block and an identity block on its diagonal. Because inactive entries of D are zero, $W = \tilde{L}DT^{-1}$ still holds. Multiplication by these uniformly bounded invertible matrices compares the ordered singular values of W with the ordered d_j , including zero singular values. Finally Lemma 5.1 gives

$$\prod_{j=1}^{\ell} d_j \leq \mathcal{V}_\ell \leq \ell! \prod_{j=1}^{\ell} d_j.$$

This proves the proposition without dividing by a vanishing pivot. The Leja lemma is a finite interpolation calculation proved below before its use in the statistical theorem, so this comparison introduces no circular dependence on attainment or memory. $\square$

4.1. Positivity and global covering. The next two lemmas are the analytic inputs to the intrinsic theorem. Their complete proofs are placed here, before the application, while the special-case arguments using them are collected in the appendices.

Lemma 4.3 (Strict confluent mixed pairing). *Let $\alpha_1 < \dots < \alpha_s$ be nonnegative, and let λ run through distinct nonnegative numbers with multiplicities q_λ summing to s . Include the columns $t^\lambda (\log t)^k / k!$, $0 \leq k < q_\lambda$, in this order within each increasing λ cluster. Assume $q_0 = 1$ if zero occurs. The matrix of their integrals against $t^{\alpha_i} \mu(dt)$ has strictly positive determinant. Consequently the corresponding rectangular pairing has full column rank whenever there are at least as many distinct row exponents as columns.*

Proof. Put $t = e^x$. The functions become $e^{\lambda x} x^k / k!$. A nonzero linear combination with total multiplicity s has at most $s - 1$ distinct zeros. Indeed divide by the first exponential and differentiate q_{λ_1} times. Rolle's theorem reduces a hypothetical s zeros to at least $s - q_{\lambda_1}$ zeros. The remaining terms are exponential polynomials with distinct nonzero exponents and the same remaining polynomial degrees: the operator $(D + \lambda - \lambda_1)^{q_{\lambda_1}}$ is invertible on polynomials of each fixed degree bound. Induction gives a contradiction. If no remaining term occurs, the original function is an exponential times a polynomial of degree less than s , and the assertion is immediate.

Thus the evaluation determinant at distinct ordered points never vanishes. Its sign is positive. One can determine the sign by letting ordered simple exponents approach the prescribed repeated

exponents and dividing by their positive within-cluster Vandermonde factors. Equivalently, the Wronskian of the displayed exponential-polynomial basis has sign

$$\prod_{\lambda < \lambda'} (\lambda' - \lambda)^{q_{\lambda} q_{\lambda'}} > 0;$$

the exponential factor and the factorial normalizations are positive. Nonvanishing fixes this sign throughout the connected ordered-point domain.

Use the determinant integration identity from Lemma 3.1, with one determinant replaced by this confluent evaluation determinant. Both determinants have the same positive sign on ordered distinct positive points. The integrand is nonnegative elsewhere by symmetry and continuity. Separated ordered subintervals have positive prior mass, so the integral is strictly positive. The logarithmic functions are bounded at zero under the stated hypotheses, which also justifies Fubini. Select any s rows for the rectangular assertion. $\square$

A bounded semialgebraic format means a bound on the number of variables, polynomial relations and their degrees in a defining formula; arbitrary real coefficients are allowed. Quantified formulas of fixed format can also be used, since real quantifier elimination gives a bound depending only on this format. Neither a coefficient norm nor a lower bound on a nonzero coefficient is part of the definition.

Lemma 4.4 (Bounded-format sets in a thin rectangle). *Let S be a nonempty compact semialgebraic subset of $\prod_{i=1}^q [-a_i, a_i]$, where $a_1 \geq \dots \geq a_q \geq 0$. Suppose its format is bounded and its dimension is at most p , with $1 \leq p \leq q$. There is a constant C depending only on the format and q such that, for every $\varepsilon > 0$,*

$$(12) \quad \mathcal{N}(S, \varepsilon) \leq C \sum_{j=0}^p \varepsilon^{-j} \prod_{i=1}^j a_i.$$

The empty product is one. Consequently S has a covering by at most M representatives belonging to S , of radius at most

$$(13) \quad C' \max_{1 \leq j \leq p} \left(M^{-1} \prod_{i=1}^j a_i \right)^{1/j}.$$

If all a_i vanish, one representative suffices with radius zero.

Proof. We use two classical real-geometric facts. First, intersections of a bounded-format semialgebraic set with arbitrary affine subspaces have a uniform bound on their number of connected components. The bound is independent of the coefficients; one formulation is the regularity statement in [18, Lemma 2.18], together with real quantifier elimination. Second, the real Vitushkin–Ivanov entropy inequality is

$$(14) \quad \mathcal{N}(S, \varepsilon) \leq C_q \sum_{j=0}^q \varepsilon^{-j} V_j(S), \quad V_j(S) = c_{q,j} \int b_0(S \cap L) \, dL.$$

For $j > 0$ the integral is over affine subspaces of codimension j , with normalized rotational measure and Lebesgue translation measure; $V_0(S) = b_0(S)$. This is the classical real inequality in [16, Theorem 3.5], also recalled as equations (4)–(5) in [17]. No nonarchimedean result is used here.

For $j > p$, almost every such affine section is empty: its translation parameter lies outside a projection of dimension at most $p < j$. Thus $V_j(S) = 0$. For $1 \leq j \leq p$, write an affine section

as $L = u + E^\perp$, $u \in E$, for a j -plane E . If it meets S , then u belongs to the projection onto E of the containing rectangle. The component bound, denoted B , therefore gives

$$V_j(S) \leq c_{q,j} B \int_{G(q,j)} \text{vol}_j \left(\text{proj}_E \prod_i [-a_i, a_i] \right) dE.$$

This projected rectangle is a zonotope. Its volume is the sum, over j -element subsets of the coordinate directions, of their parallelepiped volumes. Each corresponding orthogonal-projection minor has absolute value at most one. Hence its volume is at most $2^j \binom{q}{j} a_1 \cdots a_j$. Also $V_0(S) \leq B$. Substitution into (14) proves (12), including zero-width axes. This argument explains why the products stop at the dimension of S , not at the number of sides of its ambient rectangle.

To respect the integer budget, let e be the maximum in (13) without its constant. If $a_1 > 0$, then $e > 0$ and $a_1 \cdots a_j \leq M e^j$ for every $j \leq p$. Choose $L \geq 2$ with $C \sum_{j=1}^p L^{-j} \leq 1/2$. For $M \geq 2C$, (12) at radius Le is at most $C + M/2 \leq M$. Discard empty balls and replace each center by a point of S in its ball; the radius at most doubles. For $M < 2C$, a single point of S covers it at radius $2\sqrt{q}a_1$, while $e \geq a_1/M \geq a_1/(2C)$. Increasing the constant proves the assertion for these budgets as well. A nearest-representative map, with a fixed tie convention, is Borel. If $a_1 = 0$, the set is a singleton. $\square$

5. INTRINSIC RESOLUTION AT ARBITRARY EXPONENT COLLISIONS

The affine hypothesis can be removed altogether. The appropriate invariant is a family of maximal Vandermonde products on the future exponent set. It is defined at a collision itself, does not require a chosen path into that collision, and determines both attainable checkpoint resolution and causal memory. We first establish a uniformly conditioned Newton flag, then prove its attainable transversality. The covering and causal arguments are given separately, so that a local flag is never substituted for a global description of the reachable set.

5.1. Calibration chamber and determinant volumes. Fix $r \geq 2$ and a compact subset K of the open chamber

$$\{(a_1, \dots, a_{r-1}) : 0 < a_1 < \cdots < a_{r-1}\}, \quad a_0 = 0.$$

The parameter of the experiment is $t \in [0, 1]$, with one fixed full-support prior μ . The known calibration is $a \in K$. One fixed matrix $C = (c_{ji})$ of column rank r defines the cells

$$k_j^a(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i}, \quad \sum_j k_j^a(t) = 1, \quad k_j^a(t) \geq k_* > 0.$$

The command cube is $[\eta, 1 - \eta]^J$, $0 < \eta < 1/2$, as before. These assumptions are imposed on all of K ; no regularity or semialgebraicity of K is required. Compactness inside the chamber separates the one-step exponents, but *does not* separate their additive sums. The finitely many report likelihoods have a common positive lower bound κ .

Fix $N \geq 2$, and choose once and for all $H \geq \max\{1, N \max_{a \in K} a_{r-1}\}$. For $m \geq 1$ write

$$\Lambda_m = \{\alpha \in \mathbb{N}_0^r : |\alpha| = m\}, \quad \Lambda_m^+ = \Lambda_m \setminus \{m e_0\}, \quad q_m = |\Lambda_m^+|, \quad x_\alpha(a) = H^{-1} \sum_i \alpha_i a_i.$$

Here $\mathbb{N}_0$ includes zero. Formal labels are retained even when their values coincide. All x_α , $\alpha \in \Lambda_m^+$, lie in $[a_*/H, 1]$, where $a_* = \min_{a \in K} a_1 > 0$. For $1 \leq \ell \leq q_m$ define

$$(15) \quad \mathcal{V}_{m,\ell}(a) = \max_{\substack{J \subset \Lambda_m^+ \\ |J| = \ell}} \prod_{\substack{\alpha, \beta \in J \\ \alpha < \beta}} |x_\alpha(a) - x_\beta(a)|, \quad \mathcal{V}_{m,1} = 1.$$

The ordering inside the product is immaterial. The maximum is zero if fewer than ℓ distinct positive future exponents remain. Put

$$(16) \quad \begin{aligned} p_{n,m} &= \min\{n(r-1), q_m\}, \\ \Psi_{n,m}(M, a) &= \max_{1 \leq \ell \leq p_{n,m}} \left(\frac{\mathcal{V}_{m,\ell}(a)}{M} \right)^{2/\ell}, \\ \Xi_N(M, a) &= \max_{1 \leq n \leq N} \Psi_{n, N-n}(M, a). \end{aligned}$$

Set $\Psi_{0,m} = \Psi_{n,0} = 0$. Changing H to any other fixed admissible value changes only comparison constants. In particular the invariant does not depend on a convention for measuring exponents.

Use the failure-factor basis $1/2$ and $1/2 + \delta t^{a_i}$, $1 \leq i < r$, realized by the fixed right inverse of C for one sufficiently small $\delta > 0$. Its ordered m -fold products are the query menu, sampled uniformly after the memory index is formed. Expansion in the symbolic variables $X_0, \dots, X_{r-1}$ gives a fixed matrix G_m of column rank $|\Lambda_m|$: an invertible linear change of variables on homogeneous polynomials of degree m is invertible. Consequently the physical query norm and the norm of the formal raw moment vector are uniformly equivalent. Coincident exponent values impose relations on attainable vectors, not a loss of column rank of this *formal* coefficient matrix.

5.2. A uniformly conditioned flag. Order the positive nodes greedily. Choose the first label by a fixed rule; for example take its least node, breaking ties by a fixed label order. Having selected $x_1, \dots, x_{j-1}$, select a remaining label maximizing

$$\prod_{i < j} |x - x_i|, \quad d_1 = 1, \quad d_j = \prod_{i < j} |x_j - x_i|.$$

Break every tie by the same label rule. After all distinct values have been selected, continue through the remaining labels; their d_j are zero. This is finite Leja ordering. The ordering and its use in Newton interpolation are classical [20, 21]; the estimates needed here are proved below, including their zero-pivot form.

Lemma 5.1 (Newton scales and maximal determinant volumes). *For every calibration, $1 = d_1 \geq d_2 \geq \dots \geq d_{q_m} \geq 0$. There is a matrix L_a , whose entries have absolute value at most one, such that*

$$(17) \quad (t^{Hx_i})_{i=1}^{q_m} = L_a \operatorname{diag}(d_j) (g_j^a(t))_{j=1}^{q_m}, \quad g_j^a(t) = [x_1, \dots, x_j](z \mapsto t^{Hz}).$$

Divided differences at repeated nodes have their Hermite values. If s is the number of distinct positive nodes, the first s rows and columns of L_a form a lower triangular matrix with diagonal entries ± 1 ; its inverse has a bound depending only on q_m . Thus, for any vector v ,

$$(18) \quad c_m \|\operatorname{diag}(d_j)v\| \leq \|L_a \operatorname{diag}(d_j)v\| \leq C_m \|\operatorname{diag}(d_j)v\|.$$

Moreover, for every $1 \leq \ell \leq q_m$,

$$(19) \quad \prod_{j=1}^{\ell} d_j \leq \mathcal{V}_{m,\ell}(a) \leq \ell! \prod_{j=1}^{\ell} d_j.$$

All these assertions hold at exact coincidences.

Proof. Every distance is at most one. The maximization defining d_j and the removal of a candidate imply $d_{j+1} \leq d_j$. Put $P_{j-1}(z) = \prod_{i < j} (z - x_i)$. For $d_j > 0$ set $L_{ij} = P_{j-1}(x_i)/d_j$; for $d_j = 0$ set that column to zero. A previously selected node gives zero when $i < j$, and the greedy choice gives $|L_{ij}| \leq 1$ on all remaining nodes. The leading nonzero block is lower triangular with diagonal ± 1 . Back substitution bounds its inverse in a fixed dimension. If

$d_j = 0$, all remaining node values already occur among the earlier ones, so P_{j-1} vanishes on every node. Newton's interpolation identity, evaluated at the nodes, is therefore exactly (17), also when there are repeats. Only the first s scaled coordinates can be nonzero, proving (18).

For $\ell \leq s$, evaluation of the monic polynomials $P_0, \dots, P_{\ell-1}$ on any ℓ chosen nodes has determinant equal to their Vandermonde determinant. Its matrix is the selected rows of $L_{:,1:\ell}$ times $\text{diag}(d_1, \dots, d_\ell)$. The Leibniz formula bounds the first determinant by $\ell!$. For the first ℓ selected nodes its absolute value is one, giving both bounds in (19). For $\ell > s$, both sides vanish. No division by a zero pivot has been made. $\square$

The next assertion is the attainable part of the argument. Bounds for a Vandermonde matrix alone would not prove it.

Lemma 5.2 (Uniform attainable Newton flags). *For every $n, m \geq 1$ with $n + m \leq N$, let $z_a(h) = (\mathbb{E}[g_j^a(t) \mid h])_{j=1}^{q_m}$ and $p = p_{n,m}$. At one common interior all-failure command tuple, the first p coordinates of z_a have a surjective differential. Its least row singular value is bounded below uniformly over $a \in K$ and every ordering of the formal nodes, including limiting orderings at collisions. Under independent uniform acquisition commands, their all-failure subprobability law dominates β times uniform probability on a translated p -cube of side 2ρ , where $\beta, \rho > 0$ are uniform.*

Proof. We first justify continuity at all repeated-node configurations, without an assumption on how a collision is approached. The Hermite–Genocchi formula [12] expresses $g_j^a(t)$ as the integral of $(H \log t)^{j-1} t^{Hz}$ over the simplex of convex combinations of $x_1, \dots, x_j$; the simplex has volume $1/(j-1)!$. Since $Hx \geq a_* > 0$,

$$|g_j^a(t)| \leq \frac{H^{j-1}}{(j-1)!} t^{a_*} |\log t|^{j-1}, \quad 0 < t \leq 1.$$

The right side is bounded and tends to zero at $t = 0$. The same argument with higher logarithmic powers gives continuity, uniform in t , as the nodes vary in their compact range. Define $g_j^a(0) = 0$.

For any initial multiset of p nodes, the p divided-difference functionals are independent on polynomials of degree at most $p-1$: their values on $1, z, \dots, z^{p-1}$ form an upper triangular matrix with diagonal one. They span the Hermite evaluation functionals at the distinct nodes, with all derivative orders below the corresponding multiplicities. Applied to t^{Hz} , their span is therefore the complete exponential-polynomial space

$$\text{span}\{t^{H\lambda}(\log t)^k : \lambda \text{ in the multiset, } 0 \leq k < m_\lambda\}.$$

This argument permits repeated nodes at nonadjacent positions. Adjoin the constant function. The resulting $p+1$ tests form a complete confluent Chebyshev space, not a selection of isolated high derivatives.

Choose the common binomial factors $F_i = (1 + c_i t^{a_{r-1}})/2$, with distinct sufficiently small positive c_i . Their command coefficients are independent of a . By Lemma 3.2 their product tangent has $n(r-1) + 1$ distinct monomials. The mixed and confluent pairing of Lemmas 3.1 and 4.3 has rank $p+1$ against the selected tests, at every node configuration. In the normalized differential

$$Q \mapsto (\mu F)^{-1}(LQ - v e_0 LQ), \quad v = LF/\mu F,$$

the product belongs to the tangent and $e_0 v = 1$. Thus exactly the constant direction is lost and the differential onto the other p coordinates is surjective.

For each fixed permutation of formal labels this derivative is continuous on K , including coincidences, by the uniform bounds just proved and positivity of evidence. There are finitely many permutations. Compactness therefore supplies a common positive least row singular

value. First and second derivatives in command coordinates are uniformly bounded because the numerator is a polynomial and the evidence is at least κ^n .

To record the probability step explicitly, complete the derivative's row space by orthonormal kernel coordinates. A ball of uniform radius in command space gives a square map whose derivative differs from its invertible central derivative by at most half its least singular value. The contraction proof of the inverse theorem gives a fixed-size product box in output and kernel coordinates, with its inverse inside the command cube. The inverse Jacobian determinant is bounded below by the reciprocal of a uniform power of the forward derivative bound. The command density is constant, and the joint density of those commands and the all-failure word is at least that density times η^n . Integrating over the kernel box proves the stated subprobability minorization. No prior density is involved and no failure evidence is discarded. $\square$

5.3. The global attainable law.

Theorem 5.3 (Collision-uniform checkpoint classification). *Under the assumptions of this section there are $0 < c \leq C < \infty$ such that, for every $a \in K$ and integer $M \geq 1$,*

$$(20) \quad c\Psi_{n,m}(M, a) \leq \bar{R}_{M;n,m}^a \leq R_{M;n,m}^{a,\max} \leq C\Psi_{n,m}(M, a).$$

The unconditional lower law uses independent uniform acquisition commands. The upper code is deterministic, has at most M reachable representatives, and works on every admitted history. The lower law permits the independent coding randomization of Definition 2.4. Constants may depend on K, C, η, μ, n, m , but not on a or M .

Proof. Let $D_a = \text{diag}(d_j)$. For one report word each coordinate of $z_a(h)$ has the form $N_j(u)/Z(u)$, with numerator and evidence polynomials of degree at most n in the command variables and $Z(u) \geq \kappa^n$. Thus

$$S_a = \{D_a z_a(h) : |h| = n\}$$

is compact and has coefficient-independent bounded semialgebraic format: it is a finite union of projections of the formulas $Z(u)y_j = d_j N_j(u)$ on the command cube. Prior integrals and node values are real coefficients in these formulas. They need not depend semialgebraically on a .

Individually normalizing the n report factors puts them in affine hyperplanes of dimension $r - 1$. The moment map is rational in those coordinates. Hence $\dim S_a \leq p$. Lemma 5.2 also gives $|z_{a,j}| \leq B$, so S_a lies in the rectangle with half-widths $Bd_1 \geq \dots \geq Bd_{q_m}$. By Lemma 4.4, it has at most M reachable representatives with covering radius bounded by

$$C \max_{\ell \leq p} \left(M^{-1} \prod_{j \leq \ell} d_j \right)^{1/\ell}.$$

The physical metric equivalence follows from (18) and the full-column-rank product-probe matrix G_m . A fixed tie rule makes nearest-representative encoding Borel. Squaring and applying (19) gives the upper bound in (20), with the integer budget already enforced by Lemma 4.4.

For the lower bound, first average any randomized decoder conditional on its label; squared loss cannot increase. Orthogonally project its query centers onto the affine physical query span, apply a fixed left inverse of G_m , and then the inverse of the leading nonzero block of L_a . These maps have uniformly bounded norms. Projection onto any first $\ell \leq \min(p, s)$ coordinates reduces the squared-error problem to the marginal of the cube in Lemma 5.2, scaled by $d_1, \dots, d_\ell$. The probability that M radius- b balls cover this uniform rectangle is at most

$$C_\ell M b^\ell / (d_1 \cdots d_\ell).$$

Choose $b = c_\ell(d_1 \cdots d_\ell/M)^{1/\ell}$ with the constant small enough that at least half the minorized mass remains. Its mean squared error is at least a fixed multiple of b^2 . An encoder's randomized assignment cannot improve upon the distance to the nearest of its centers. Independent public coding randomness can be fixed and then averaged, since it does not generate the acquisition law. Maximize over ℓ . For $\ell > s$ the determinant volume is zero, so no additional term is needed. Equation (19) now gives the lower bound. The unconditional optimum is bounded above by the minimax optimum, completing the proof. $\square$

Theorem 5.4 (Causal classification across all collision strata). *For the same fixed experiment and horizon, there exist $c, C > 0$ such that for every known $a \in K$ and every integer $M \geq 1$ there exists a single, stage-compatible deterministic M -state transducer with maximum-history, maximum-checkpoint regret at most $C\Xi_N(M, a)$. Every M -state transducer has maximum unconditional checkpoint regret at least $c\Xi_N(M, a)$ under one common independent uniform exploration law. Equivalently,*

$$(21) \quad c\Xi_N(M, a) \leq \mathfrak{R}_{M,N}^{a,\text{av}} \leq \mathfrak{R}_{M,N}^{a,\text{max}} \leq C\Xi_N(M, a).$$

The transducer may depend on a and M . Constants are uniform in both; codebooks are not asserted to be calibration-blind. It stores only its representative index. The clock, known calibration and fixed real-valued program are read-only, as in Definition 2.4.

Proof. At stage n let $m = N - n$ and store a representative in the reachable raw moment image indexed by Λ_m , including the constant coordinate $v_{me_0} = 1$. For a new factor $f(t) = \sum_i f_i(g, x)t^{a_i}$, the exact update is

$$(22) \quad v'_\alpha = \frac{\sum_i f_i(g, x)v_{\alpha+e_i}}{\sum_i f_i(g, x)v_{e_i+(m-1)e_0}}, \quad \alpha \in \Lambda_{m-1}.$$

Every index on the right belongs to Λ_m . Coincident labels have equal moment values and cause no ambiguity. On the segment joining any two reachable moment vectors, the denominator is a report probability under a posterior mixture and is at least κ . Coefficients are uniformly bounded and raw moments belong to $[0, 1]$. Differentiating the quotient gives a Lipschitz constant independent of all exponent gaps.

At every stage Theorem 5.3 provides a raw reachable codebook of cardinality at most M and covering radius at most $C_n\Psi_{n,N-n}(M, a)^{1/2}$. Update a representative by (22), then quantize in the next reachable codebook. Updating a reachable representative produces a reachable next state, so this quantizer is always evaluated on its domain. If e_n is the raw error, then

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_{n+1}\Psi_{n+1,N-n-1}(M, a)^{1/2}.$$

The finite recurrence bounds every e_n by $C_N\Xi_N(M, a)^{1/2}$. The query matrix converts this to the asserted squared-regret bound. The exact prefix appears only in this error analysis, not in the transducer. No divided-difference coordinate, reciprocal pivot or stored command tape is used in a transition.

Conversely, at each checkpoint the streaming index is an M -message encoder of the prefix. All command laws used in Theorem 5.3 are prefixes of one common exploration law. Each transducer therefore satisfies every checkpoint lower bound; taking their maximum and then the infimum proves the lower bound in (21). Finitely many checkpoint constants can be replaced by their positive minimum. One checkpoint and its remaining query trials are executed in a run, as before. $\square$

Corollary 5.5 (Intrinsic bit law). *Let $B_*(\varepsilon, a)$ be the least integer persistent bit budget for maximum-history streaming regret at most ε in this experiment. Uniformly for $0 < \varepsilon \leq 1$ and*

$a \in K$,

$$(23) \quad B_*(\varepsilon, a) = \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n, N-n}}} \left[\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} + \log_2 \mathcal{V}_{N-n, \ell}(a) \right]_+ + O(1).$$

A zero determinant volume contributes zero to this maximum, by interpreting its logarithm as $-\infty$ before taking the positive part. The same law holds for the unconditional maximum-checkpoint criterion. At each fixed calibration the small-error slope is $D_A(N)/2$.

Proof. With $M = 2^B$, solving each inequality $(\mathcal{V}_{m, \ell}/M)^{2/\ell} \leq \varepsilon$ gives the displayed lower bound on B . Constants in (21) change it by at most a bounded amount, since ℓ is bounded. Integer rounding costs at most one further bit. For fixed a , the largest nonzero index at checkpoint (n, m) is $\min\{n(r-1), |mA| - 1\} = d_A(n, m)$, proving the slope assertion. $\square$

5.4. Collision orders and a finite tree formula. The determinant form has a consequence not requiring interpolation coordinates: it classifies arbitrary orders of tangency to a collision. It also gives a finite optimization rule for all resolution phases.

Theorem 5.6 (Collision-tree determination of memory phases). *Let $a(\theta) \in K$, $0 \leq \theta \leq \theta_0$, be a known path. For a fixed m , identify formal sums that are identical along the path. Suppose that for every remaining pair of labels*

$$|x_\alpha(\theta) - x_\beta(\theta)| \asymp \theta^{w_{\alpha\beta}}, \quad w_{\alpha\beta} \in [0, \infty), \quad 0 < \theta \leq \theta_0.$$

After decreasing θ_0 this hypothesis holds for every real analytic path whose nonidentical sums are analytic at zero. Define

$$(24) \quad \beta_{m, \ell} = \min_{\substack{J \text{ a set of remaining labels} \\ |J| = \ell}} \sum_{\{\alpha, \beta\} \subset J} w_{\alpha\beta}, \quad \beta_{m, 1} = 0.$$

Then $\mathcal{V}_{m, \ell}(a(\theta)) \asymp \theta^{\beta_{m, \ell}}$. For unavailable cardinalities the volume is zero. Consequently the checkpoint law is

$$(25) \quad \overline{R}_{M; n, m}^{a(\theta)} \asymp \max_{\ell \leq p_{n, m}} \theta^{2\beta_{m, \ell}/\ell} M^{-2/\ell},$$

with zero-volume terms omitted; the streaming law is its maximum over checkpoints. The exponents $\beta_{m, \ell}$ depend only on the pairwise collision orders, and can be computed by a finite tree allocation rule.

Proof. Each fixed subset product has order equal to the sum of its pair orders. There are finitely many subsets, so their maximum has the smallest such order. This proves the determinant assertion and then (25) by Theorems 5.3 and 5.4. Constants may depend on the path's nonzero leading coefficients; the intrinsic determinant theorem, in contrast, is already uniform on K .

For completeness we give the allocation rule and its justification. The triangle inequality for node differences implies

$$w_{\alpha\gamma} \geq \min\{w_{\alpha\beta}, w_{\beta\gamma}\}.$$

Indeed a smaller order on the left would contradict the upper bound by the sum of the two differences on the right as $\theta \downarrow 0$. Thus thresholds on w define nested equivalence classes. Make a rooted tree from these classes, with root height zero; an internal cluster C has height $h_C = \min_{\alpha \neq \beta \in C} w_{\alpha\beta}$, and its children are the equivalence classes for orders strictly greater than h_C . Split the root at orders greater than zero. Singleton children are leaves. Redundant equal-height vertices can be omitted. For an internal nonroot vertex put $\Delta_C = h_C - h_{\text{parent}(C)}$.

The sum of these increments on the common ancestor chain of two leaves is exactly their pair order. Hence, for a selected leaf set J ,

$$(26) \quad \sum_{\{\alpha, \beta\} \subset J} w_{\alpha\beta} = \sum_{C \text{ internal, } C \neq \text{root}} \Delta_C \binom{|J \cap C|}{2}.$$

Set $F_C(0) = F_C(1) = 0$ for a leaf and $F_C(k) = +\infty$ for $k > 1$. For an internal vertex, including the root with $\Delta_{\text{root}} = 0$, compute

$$(27) \quad F_C(k) = \Delta_C \binom{k}{2} + \min_{\sum_D \text{ child of } C \text{ } k_D = k} \sum_{D \text{ child of } C} F_D(k_D).$$

Equation (26) proves this recurrence by induction. It gives $F_{\text{root}}(\ell) = \beta_{m,\ell}$. Successive finite min-convolutions compute the table; no optimization over measures, posterior encoders or codebooks is involved. For analytic paths the pair orders are the finite orders of their first nonzero Taylor coefficients, after identical sums have been identified. This proves the assertion. $\square$

For the affine family of Section B, different limiting clusters have pair order zero, and distinct members of a cluster have pair order one. Thus a choice of k nodes from that cluster costs $\binom{k}{2}$; its successive costs are $0, 1, \dots, s-1$. Minimizing their sum picks the smallest costs from all clusters. Therefore $\beta_{m,\ell} = \nu_{m,1} + \dots + \nu_{m,\ell}$. This recovers Theorem D.1, while its separate proof remains available. Higher orders of contact between distinct branches require the pair orders above and are not encoded by the affine multiplicity list.

5.5. Intersecting collision loci and arbitrary tangency order. Here is a two-parameter consequence with different ranks on intersecting loci and arbitrarily deep resolution phases along tangent approaches. It illustrates the geometry of the determinant theorem rather than requiring a separate asymptotic analysis for each path.

Corollary 5.7 (A two-parameter collision arrangement). *Let $A_{u,v} = \{0, 1, 2+u, 3+v\}$, with $|u|, |v| \leq 1/16$, and take the four cells*

$$k_0(t) = \frac{1}{4} + \frac{1}{16}(t + t^{2+u} + t^{3+v}), \quad k_i(t) = \frac{1}{4} - \frac{1}{16}t^{a_i}, \quad 1 \leq i \leq 3.$$

Use any fixed admitted command cube and a fixed full-support prior. For total horizon five set $\rho = \max\{|u|, |v|\}$ and $\tau = \min\{|u|, |v-u|, |v-2u|\}$. Then, uniformly on the square,

$$(28) \quad \mathfrak{R}_{M,5}^{u,v} \asymp \max \left\{ M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau)^{2/9} M^{-2/9} \right\}$$

for either streaming criterion. The exact peak dimension is nine off the three collision lines, eight on any such line away from their intersection, and six at $(0,0)$.

Along $u = \theta$, $v = \theta + \theta^k$ with an integer $k \geq 2$, the three regimes have respective orders

$$M^{-1/3}, \quad \theta^{1/2} M^{-1/4}, \quad \theta^{(4+2k)/9} M^{-2/9}.$$

Their crossover budgets have orders θ^{-6} and $\theta^{-(8k-2)}$, with regrets of orders θ^2 and θ^{2k} , respectively. These are comparison orders, not exact integer thresholds.

Proof. The cells sum to one, lie between $3/16$ and $7/16$, and their coefficient matrix has rank four: its last three rows separate the three nonconstant coefficients and the sum of its rows gives the constant. They therefore satisfy the compact-chamber hypotheses.

At checkpoint $n = 3, m = 2$ the nine positive formal exponents are

$$1, \quad 2, \quad 2+u, \quad 3+u, \quad 3+v, \quad 4+2u, \quad 4+v, \quad 5+u+v, \quad 6+2v.$$

They form six uniformly separated groups, with three within-group gaps $|u|, |v-u|, |v-2u|$. Denote these gaps in decreasing order by $g_1 \geq g_2 \geq g_3$. For $\ell \leq 6$ the determinant volumes

are comparable to one. For $\ell = 6 + j$, $1 \leq j \leq 3$, they are comparable to $g_1 \cdots g_j$: choose one exponent from every group and a second from the j groups with largest gaps for the lower bound. Any selection of $6 + j$ exponents must double at least j groups; all other factors are bounded, proving the upper bound. Fixed powers of H only change constants. Since $p_{3,2} = 9$, its profile is

$$\max\{M^{-1/3}, g_1^{2/7} M^{-2/7}, (g_1 g_2)^{1/4} M^{-1/4}, (g_1 g_2 g_3)^{2/9} M^{-2/9}\}.$$

The signed differences satisfy $(v - u) - (v - 2u) = u$, so the largest absolute gap is at most twice the second largest. Moreover $g_1 \asymp \rho$, hence $g_2 \asymp \rho$ and $g_3 = \tau$. The seven-dimensional term is the geometric interpolation, with weights $3/7$ and $4/7$, of the six- and eight-dimensional terms, up to uniform constants. This proves the expression in (28) for the checkpoint, also on the collision lines.

At the other checkpoints, the attainable dimensions are bounded by three, six and three, respectively. Since every normalized gap is at most one, their squared covering errors are at most $CM^{-1/3}$. The causal classification therefore gives the same entire-horizon profile and the checkpoint-three lower bound supplies its converse. Counting distinct two-fold exponents gives the three exact peak dimensions asserted.

Finally, on the stated path $\rho \asymp \theta$ and $\tau \asymp \theta^k$. Substitution gives the three displayed terms. Equality in order of the first two gives $M = \theta^{-6}$, at regret θ^2 ; equality in order of the last two gives $M = \theta^{-(8k-2)}$, at regret θ^{2k} . As $k > 1$, these budgets are strictly separated as $\theta \downarrow 0$ and all three regimes occur. The exponent k records contact with a collision locus, not an additional assumed observable jet. $\square$

6. COMPILATION INTO FINITE NUMERICAL TABLES

A covering theorem supplies a finite set, but does not in general supply an algorithm for choosing that set. Moreover, a table with finitely many entries may still contain noncomputable real numbers. We separate these issues. The construction below uses finite grids of feasible histories, ordinary rational comparisons, and approximations to a finite moment table. Neither rank detection nor a decision whether two exponents coincide is among its operations.

6.1. A compilation theorem for compact state systems. Let $T \geq 1$ be fixed, let $U = [\eta, 1 - \eta]^J$, with rational $0 < \eta < 1/2$, and let $\mathcal{X}$ be finite. For $0 \leq n \leq T$ let $S_n \subset [-B, B]^{d_n}$ be compact, with S_0 a singleton. There are transition maps

$$T_{n,x} : S_n \times U \longrightarrow S_{n+1}, \quad S_{n+1} = \bigcup_{x \in \mathcal{X}} T_{n,x}(S_n, U).$$

Norms in this section are maximum norms. Suppose

$$(29) \quad \|T_{n,x}(s, u) - T_{n,x}(s', u')\|_\infty \leq L_n \|s - s'\|_\infty + G_n \|u - u'\|_\infty.$$

Each stage has finitely many queries, with a prediction map $Q_n : S_n \rightarrow [0, 1]^{b_n}$ of Lipschitz constant K_n . The bounds and all dimensions are fixed, independent of the state budget. Write $H_n(w, u_1, \dots, u_n)$ for the exact state of the report word $w \in \mathcal{X}^n$. Iteration of (29) shows that its command dependence is A_n -Lipschitz, where

$$A_0 = 0, \quad A_{n+1} = L_n A_n + G_n.$$

For $M \geq 1$, define the unrestricted-centre covering radius

$$e_n(M) = \inf_{z_1, \dots, z_M \in \mathbb{R}^{d_n}} \sup_{s \in S_n} \min_i \|s - z_i\|_\infty.$$

No spectral representation or dimension formula for $e_n(M)$ is assumed.

Definition 6.1 (Finite evaluation data). Given a rational grid history and a positive rational tolerance τ , an evaluation procedure returns a dyadic vector within τ of its exact state. It also returns dyadic query predictions within τ of the exact predictions at any chosen representative history. This is an offline procedure. In its effective version it is a specified algorithm with access to the numerical input oracles of the experiment. In a noneffective version the finitely many approximations are supplied data. Neither version gives an oracle to the running transducer.

Let $U_h \subset U$ be a finite rational h -net in the maximum norm, with $|U_h| \leq C_U(1 + h^{-1})^J$. At runtime an input includes an index of any $\bar{u} \in U_h$ satisfying $\|u - \bar{u}\|_\infty \leq h$. The guarantee below holds for every such choice. In particular it does not require deciding equality of an arbitrary real input to a rounding boundary. An input approximation, followed by rounding that rational approximation and clipping to U , supplies this interface with a fixed constant adjustment of h . For comparison with a memory lower bound, input coding is a fixed memoryless Borel function of the current command, or uses fresh independent rounding coins. It is not an additional history-dependent channel. The upper estimate itself holds for every name meeting the error bound.

Lemma 6.2 (Greedy covering from approximate samples). *Let S be a compact subset of a normed space. Suppose a finite indexed list of points $s_j \in S$ is an a -net of S , and $\|\tilde{s}_j - s_j\| \leq \tau$. Farthest-first selection of at most M points from the rational list $(\tilde{s}_j)$, with fixed tie breaking, selects indices I for which*

$$\sup_{s \in S} \min_{i \in I} \|s - s_i\| \leq 2e_S(M) + a + 4\tau.$$

If there are at most M indices, select all of them. Distinct indices may have identical approximate vectors.

Proof. Let r be the final covering radius of the approximate list. When the list has more than M indices, the selected M points and a farthest point have mutual distances at least r . Every cover by M balls of radius b must put two of these $M + 1$ points in the same ball, and therefore $r \leq 2b$. This also handles $r = 0$. A cover of S with radius $e_S(M) + \epsilon$ covers the approximate list with radius $e_S(M) + \epsilon + \tau$, allowing its centres to lie outside S . Thus $r \leq 2e_S(M) + 2\tau$ after $\epsilon \downarrow 0$. For a given $s \in S$, choose s_j within a , and then a selected approximate point within r of $\tilde{s}_j$. The triangle inequality gives $a + 2\tau + r$. The case of a list of size at most M is immediate. This is the classical farthest-first argument [22]; the approximation allowance is recorded explicitly. $\square$

Theorem 6.3 (Effective compilation without loss of covering order). *Under (29) and Definition 6.1, there is a finite procedure producing stage-dependent integer transition tables and dyadic output tables, with at most M state indices at each stage, such that, on every input-report history,*

$$(30) \quad \|\hat{Q}_n - Q_n(s_n)\|_\infty \leq C_T \left(\max_{1 \leq j \leq n} e_j(M) + h + \tau \right).$$

The running procedure reads only its previous index, the current finite input code and report, and, at a checkpoint, the new query label. It stores no numerical state vector, representative history, or oracle. The constant depends only on the displayed bounds and the fixed horizon. The synthesis procedure does not need $e_n(M)$ or a covering certificate as input.

Proof. Enumerate all pairs of a report word in $\mathcal{X}^n$ and a command word in U_h^n . They are feasible histories by the hypotheses on the state system. Their exact images form an $A_n h$ -net of S_n . Evaluate every image with error at most τ and apply Lemma 6.2. Associate each

selected index with the *exact* state $c_{n,i}$ of its selected history, for analysis only. The stored approximations $\tilde{c}_{n,i}$ have error at most τ . The exact representatives cover S_n with radius

$$(31) \quad R_n \leq 2e_n(M) + A_n h + 4\tau.$$

For each representative history at stage n , each $\bar{u} \in U_h$ and each x , append $(\bar{u}, x)$ to that history and evaluate the resulting exact state $y = T_{n,x}(c_{n,i}, \bar{u})$ with error at most τ . Choose the next index by minimizing the maximum-norm distance between this approximation and the stored approximations at stage $n + 1$. All comparisons are between rational numbers. If j is selected, the triangle inequality, applied to a nearest exact centre, gives

$$\|y - c_{n+1,j}\|_\infty \leq R_{n+1} + 4\tau.$$

This specifies one integer for every entry of a finite transition table. The offline representative history is not a runtime variable. In particular no update is applied to an approximate vector outside the reachable set: y is defined and evaluated through an actual history.

Let $E_n = \|s_n - c_{n,i_n}\|_\infty$, where s_n denotes the actual uncompressed state, again only in the analysis. For any admitted finite input code,

$$(32) \quad E_0 = 0, \quad E_{n+1} \leq L_n E_n + G_n h + 2e_{n+1}(M) + A_{n+1} h + 8\tau.$$

Induction over the fixed horizon bounds E_n by a constant times the right-hand quantity in (30). Store a dyadic approximation to $Q_n(c_{n,i})$ with error at most τ and clip it to $[0, 1]$. Clipping cannot increase its error to a probability. The query error is at most $K_n E_n + \tau$, proving the theorem. There are finitely many evaluations and comparisons, so the procedure terminates whenever the evaluation procedure does. Nonuniqueness of representatives or ties does not require an equality test on exact data. $\square$

Proposition 6.4 (Size of the finite program). *Put $V = |U_h|$ and $X = |\mathcal{X}|$. The transition table has at most $TMVX$ entries, each taking $\lceil \log_2(M + 1) \rceil$ bits. If outputs are written to b binary places, the output tables take at most $M \sum_n b_n(b + 1)$ bits, in addition to fixed metadata. Candidate enumeration uses at most $\sum_{n=0}^T (XV)^n$ histories. With cached candidate approximations, greedy selection uses $O(M \sum_{n=0}^T (XV)^n)$ fixed-dimensional distance operations, and transition selection uses $O(TM^2VX)$ such operations.*

Proof. The transition arguments are the stage, current representative, finite command code and report. Counting their combinations proves the first assertion. A query table has one b -place probability per representative and query. At stage n there are $(XV)^n$ grid histories. Maintain for each candidate its smallest distance to a chosen centre; one new centre updates this array with one pass, repeated at most M times. For each transition, comparison against at most M next centres gives the last count. Exact integer comparison, maximum and subtraction suffice. These counts exclude the running time of an arbitrary numerical oracle. $\square$

With a supplied clock the only persistent field has at most M values. An autonomous machine can retain the pair (n, i_n) , using at most $1 + (T + 1)M$ values. The current input code, table address and returned output are transient data. They require $O(\log M + \log V + b + \log(T + 1))$ bits for fixed dimensions. The finite read-only program is a separate, explicitly counted resource; it is not claimed to fit into $\log M$ bits.

The fixed fine-mesh realization below is retained in full. Its sufficient precision and program bounds are strengthened, without changing its construction or converse, in Section 7.

6.2. Finite data for positive monomial experiments. Return to the experiment of Section 5, with $I = [0, 1]$, and put $T = N - 1$. Use the full formal raw moment vector $v_n = (\mathbb{E}[t^{\alpha \cdot a} \mid h])_{|\alpha|=N-n}$, including its constant entry. Coincident labels remain present. Its dimension is bounded by a constant depending only on r, N .

Choose a fixed bound

$$B_f \geq 1 + \sup_{g,x} \sum_{i=0}^{r-1} |f_i(g, x)|, \quad f_{g,x}(t) = \sum_i f_i(g, x) t^{a_i}.$$

This uses the coefficient realization, not exponent separation. For two reachable raw vectors, the update numerator and denominator have differences at most $B_f \|v - v'\|_\infty$. Their denominators are at least κ , and a numerator divided by its own denominator belongs to $[0, 1]$. Subtracting the two quotients gives a state Lipschitz bound $2B_f/\kappa$. For fixed posterior, the difference of report factors at two commands is at most $J\|g - g'\|_\infty$ pointwise, so the command bound is $2J/\kappa$. Consequently (29) holds uniformly on the calibration set. Physical query predictions are fixed linear combinations of raw moments, with uniformly bounded coefficients. For effective evaluation we fix a sufficiently small positive rational δ in the canonical probe basis $1/2, 1/2 + \delta t^{a_i}$. Its formal query coefficients are then rational. Alternatively certified approximations to a chosen fixed query coefficient table are included among the numerical inputs. This verifies every metric hypothesis of Theorem 6.3.

Lemma 6.5 (Certified finite evaluation from prior moments). *For a fixed horizon, dyadic approximations to the coefficients of the cell matrix and to*

$$(33) \quad m_\gamma(a) = \int_0^1 t^{\gamma \cdot a} d\mu(t), \quad \gamma \in \mathbb{N}_0^r, \quad |\gamma| = N,$$

suffice to evaluate every rational-command history state and every fixed finite product-query prediction with a prescribed error $\tau > 0$. Required input accuracy is $c_N \min\{\tau, 1\}$ for a positive constant determined by supplied coefficient and evidence bounds. The number of input moment entries is independent of the history grid and of M . No equality decision on the exponents is required.

Proof. Expand an n -report product in formal homogeneous variables as $P(X) = \sum_{|\beta|=n} p_\beta X^\beta$. Its coefficient sum satisfies $\sum_\beta |p_\beta| \leq B_f^n$. A coordinate of the remaining raw state is

$$(34) \quad v_\alpha = \frac{\sum_{|\beta|=n} p_\beta m_{\alpha+\beta}}{\sum_{|\beta|=n} p_\beta m_{\beta+(N-n)e_0}}, \quad |\alpha| = N - n.$$

The denominator is the true prefix evidence and is at least κ^n . The formula remains exact when distinct formal labels have equal exponents. There is no cancellation-based inference of rank.

A perturbation of size δ in the input coefficients changes each factor coefficient vector by at most a fixed multiple of δ . The telescoping identity for products bounds the coefficient-sum error in P by $n(B_f + 1)^{n-1} C \delta$, for δ sufficiently small. Each prior moment lies in $[0, 1]$, and there are finitely many formal labels. Thus numerator and denominator errors are at most $D_N \delta$, where D_N is obtained from the finite bounds just displayed. Take $D_N \delta \leq \kappa^N/2$. The approximate denominator is then positive. If a true numerator is A , its denominator is Z , and $0 \leq A \leq Z$, direct subtraction gives

$$\left| \frac{\tilde{A}}{\tilde{Z}} - \frac{A}{Z} \right| \leq \frac{|\tilde{A} - A| + |\tilde{Z} - Z|}{\tilde{Z}} \leq 4D_N \kappa^{-N} \delta.$$

Choose δ to make this at most $\tau/2$ and round the result to a dyadic within $\tau/2$. Fixed product-query predictions are linear combinations of these raw entries, or the same product

ratios; increase the accuracy constant by their fixed coefficient bounds. This gives the claimed finite procedure. $\square$

Theorem 6.6 (Finite-precision realization of the intrinsic law). *Fix the experiment, the full-support prior, the horizon and compact calibration set of Section 5. Suppose the command cube is rationally specified, and positive rational lower bounds for the report likelihoods and positive exponents, together with upper coefficient bounds, are supplied. For each known calibration a and integer $M \geq 1$, finite approximations to the data in Lemma 6.5 produce a deterministic finite-table transducer with at most M persistent labels per stage and*

$$(35) \quad \sup_{n, \mathbf{h}} \mathcal{R}(\hat{Q}_n, Q_n(\mathbf{h})) \leq C\{\Xi_N(M, a) + h^2 + \tau^2\}.$$

Here h is the command mesh and τ the certified error of the offline state and output evaluations. Constants are uniform in a, M . For meshes and evaluation errors at most $c/(M+1)$, this transducer has regret at most $C'\Xi_N(M, a)$, while every such transducer with the memoryless input interface has maximum unconditional checkpoint regret at least $c'\Xi_N(M, a)$ under the same exploration law as in the intrinsic theorem.

All numerical input and output precisions can be chosen $\log_2(M+1) + O(1)$ binary places. The transition and output program has $O(M^{J+1} \log(M+1))$ bits for the fixed experiment. Its synthesis uses polynomially many fixed-dimensional rational operations in M , with exponent allowed to depend on J, N . If the coefficients and moments have computable approximation procedures, the synthesis is a single uniform terminating algorithm using those procedures offline. No computability assumption is needed for the preceding existential intrinsic law for arbitrary full-support priors.

Proof. Theorem 5.3, its reachable-centre construction, and the fixed physical/raw metric comparison imply

$$e_n(M) \leq C_n \Psi_{n, N-n}(M, a)^{1/2}.$$

The raw update calculation above and Lemma 6.5 verify all assumptions of Theorem 6.3. Squaring (30), using $(u+v+w)^2 \leq 3(u^2+v^2+w^2)$, and averaging over the finite query menu prove (35). For $N \geq 2$, the $\ell = 1$ term at every nontrivial checkpoint gives

$$\Xi_N(M, a) \geq M^{-2}.$$

Hence command and evaluation errors of order $(M+1)^{-1}$ are absorbed without determining any individual pivot or collision scale.

A dyadic grid with this mesh has $O(M^J)$ points. Coefficient and prior moment errors of size $c_N/(M+1)$ suffice by Lemma 6.5; the required binary precision is the stated logarithmic quantity. Proposition 6.4 gives the storage and arithmetic counts. Dyadic input coefficients may be converted to rational product ratios of $O_N(\log(M+1))$ bits, and all stored approximations have that precision. Integer comparisons are finite, including exact ties.

Finally the finite-table filter, even when its command input is a quantized approximation, is an admissible filter with at most M labels in the original experiment when input coding is memoryless. The exact-input filter can first apply that fixed coding rule. It has less, not more, information than one reading the exact command. The lower bound in Theorem 5.4 therefore applies unchanged. This is a lower bound for one filter over the common exploration law, not a combination of independently redesigned checkpoint filters. $\square$

Corollary 6.7 (Calibration accuracy independent of collision gaps). *Suppose $a_i \geq a_* > 0$ for $i > 0$, the calibration entries are computable reals, and the prior has a procedure that approximates $\int t^b d\mu$ for positive rational b to requested accuracy. The preceding compiler needs*

only $\log_2(M+1) + O(1)$ binary places of the calibration and of the resulting moments. The bound does not involve the smallest nonzero additive exponent gap and is valid at exact collisions.

Proof. For $b, b' \geq a_*/2$, the mean value theorem and $\sup_{0 < t \leq 1} t^{a_*/2} |\log t| = 2/(ea_*)$ give

$$\left| \int (t^b - t^{b'}) d\mu \right| \leq \frac{2}{ea_*} |b - b'|.$$

Every nonconstant exponent in (33) is at least a_* . Approximate its constituent calibration entries so that its error is $O((M+1)^{-1})$ and below $a_*/2$. Evaluate the corresponding positive rational exponent to the same accuracy. The constant entry is exactly one. There are finitely many labels, so the constants can be chosen uniformly. This argument neither compares two sums nor asks whether their difference vanishes. $\square$

Corollary 6.8 (Digital realization of high-contact phases). *For the two-parameter family in Corollary 5.7, the same three-term regret law is attained by finite numerical transition and output tables under the finite-data assumptions above. Along $u = \theta$, $v = \theta + \theta^k$, the required numerical precision is $\log_2(M+1) + O(1)$, with no extra term proportional to $\log(1/|v - u|)$. In particular the compiler remains valid before that smallest separation can be numerically resolved.*

Proof. Apply Theorem 6.6 and Corollary 6.7. The three-term expression and its contact orders are those of Corollary 5.7. The compiler never identifies the three collision lines or computes k ; its accuracy is set by M , and its performance follows from the complete attainable cover. The physical experiment and its full-history loss baseline are unchanged. $\square$

The construction is not a new interpolation theorem or a new farthest-first approximation algorithm. It connects a collision-uniform attainability theorem to a finite numerical state machine with a counted program. The distinction matters: a purely spectral identity supplies neither the attainable cover nor the positive-likelihood update bound. The horizon is fixed throughout. The program can be much larger than the persistent state, and the running time of a general moment oracle has not been bounded. For a noncomputable prior, finite advice exists at each precision but a uniform algorithm for producing that advice is not asserted. These distinctions leave the arbitrary-prior mathematical classification intact.

7. CERTIFIED PRECISION AND PROFILE-ADAPTIVE COMPILATION

The geometric profile has a second use besides specifying the error of an M -state filter. It determines a sufficient numerical scale for constructing that filter. We first obtain a scale uniform over the calibration chamber. We then recover the actual covering scale from finite approximate histories, without evaluating a determinant or identifying a collision stratum. All precision and program-length estimates in this section are upper resource bounds. The matching converse remains the converse for persistent labels.

The separated-chain observation and its uniform resource consequence were recorded in the preceding technical note [25].

7.1. A separated family of future tests. Set $k = \lfloor N/2 \rfloor$ and

$$d_0 = (r-1)k, \quad \delta_K = \min_{a \in K} \min_{1 \leq i < r} (a_i - a_{i-1}), \quad \bar{\delta} = \delta_K/H.$$

We assume $r \geq 2$, so $d_0 \geq 1$. Compactness in the strictly ordered chamber gives $\delta_K > 0$. This is a one-step separation, not a lower bound on nonzero additive gaps.

Proposition 7.1 (Separated-chain floor). *For every $a \in K$ and every integer $M \geq 1$,*

$$(36) \quad \Xi_N(M, a) \geq \bar{\delta}^{d_0-1} M^{-2/d_0}.$$

Proof. Put $m = N - k$. The future sumset mA contains

$$C_m = \{jD : 0 \leq j \leq m\} \cup \{jD + a_i : 0 \leq j < m, 1 \leq i \leq r - 2\}.$$

Each displayed exponent is a sum of at most m members of A ; padding by zeros produces a formal degree- m label. In the interval $[jD, (j+1)D]$ the chain is

$$jD < jD + a_1 < \cdots < jD + a_{r-2} < (j+1)D.$$

Thus C_m has $m(r-1) + 1$ distinct elements, with consecutive gaps at least δ_K . For $r = 2$ there are no interior terms, and the same count and separation hold. There are at least d_0 positive elements, since $m \geq k$. In particular $q_m \geq d_0$ and $p_{k,m} = d_0$. Choose d_0 positive elements and one formal label for each. Every pair of their normalized nodes is separated by at least $\bar{\delta}$. Their Vandermonde product is at least $\bar{\delta}^{d_0(d_0-1)/2}$. Keeping this one term in $\Psi_{k,m}$ proves (36). Other determinant terms, including all collision-sensitive ones, have not been removed. $\square$

Corollary 7.2 (Uniform sufficient numerical resources). *Under the finite-data hypotheses of Theorem 6.6, the complete regret order $\Xi_N(M, a)$ is realized with at most M persistent labels per stage using*

$$(37) \quad b(M) = \left\lceil \frac{\log_2(M+1)}{d_0} \right\rceil + O(1), \quad P(M) = O(M^{1+J/d_0} \log(M+1))$$

binary places and read-only program bits, respectively. With $T = N - 1$, a sufficient synthesis count is

$$O(M^{1+JT/d_0} + M^{2+J/d_0})$$

fixed-dimensional rational operations, excluding the running time of numerical input procedures. The constants have the same fixed-experiment and calibration-uniform scope as the intrinsic theorem.

Proof. Take h, τ bounded by a fixed multiple of $(M+1)^{-1/d_0}$. Proposition 7.1 gives

$$h^2 + \tau^2 \leq CM^{-2/d_0} \leq C\bar{\delta}^{1-d_0} \Xi_N(M, a).$$

Theorem 6.6 therefore gives the full upper law, not merely its separated-chain term. Its converse for memoryless input coding is unchanged. Lemma 6.5 transfers this tolerance to the finitely many coefficient and moment inputs with only a fixed additive change of precision. The grid has $O(M^{J/d_0})$ points. Counting transition entries, output entries, candidates, and nearest-centre comparisons as in Proposition 6.4 proves the remaining statements. The variable part of b is the least integer j for which $2^{jd_0} \geq M+1$, and can be found by integer comparisons. Neither δ_K nor an additive-gap inverse is required to choose it. $\square$

7.2. An observable certificate for the covering scale. Return to the general compact systems of Section 6. Write $e(M) = \max_{1 \leq n \leq T} e_n(M)$ and choose a supplied rational upper bound $A \geq \max_n A_n$. Use coordinate product grids with a fixed indexing convention, so their mesh exponent is finite metadata rather than a list of real commands. The constants L_n, G_n, K_n may likewise be replaced by supplied rational upper bounds. For a command h -net, let $\tilde{S}_n$ be the indexed list of approximate grid-history states, each within τ of the corresponding true state. Let I_n be the indices chosen by farthest-first selection with budget M , and put

$$(38) \quad r_n = \max_{z \in \tilde{S}_n} \min_{i \in I_n} \|z - \tilde{s}_{n,i}\|_\infty, \quad r = \max_{1 \leq n \leq T} r_n.$$

These numbers are rational and obtained from the finite data. The true states of selected histories are denoted by $c_{n,i}$, only in the analysis and in offline verification.

Lemma 7.3 (Two-sided finite certificate). *For each noninitial stage, and without any positivity assumption on $e_n(M)$,*

$$(39) \quad \max\{0, r_n/2 - \tau\} \leq e_n(M) \leq r_n + A_n h + 2\tau.$$

The selected true states cover S_n with radius at most the right-hand side. In particular, if $\tau = h$ and $c = A + 2$, then

$$(40) \quad \max\{0, r/2 - h\} \leq e(M) \leq r + ch.$$

Proof. If the approximate list has more than M entries, the selected M points together with a farthest point have mutual distances at least r_n . A cover of the true set by M balls of radius $e_n(M) + \epsilon$ covers this list by balls of radius $e_n(M) + \epsilon + \tau$. Two of the $M + 1$ points belong to one ball, hence $r_n \leq 2e_n(M) + 2\epsilon + 2\tau$. Let ϵ decrease to zero. If there are at most M entries, $r_n = 0$ and the lower inequality is immediate. Duplicate vectors cause no change in either argument. For the upper inequality, approximate an arbitrary history by a grid history. Its state changes by at most $A_n h$. Moving to the approximate sample, then to a selected approximate centre, and then to its true state costs at most $\tau + r_n + \tau$. This proves the covering claim and hence the upper inequality. Taking maxima proves (40). $\square$

Theorem 7.4 (A posteriori precision selection). *Assume the compact-system and finite-evaluation hypotheses of Theorem 6.3, and suppose $e(M) > 0$. For $b = 0, 1, \dots$, take $h_b = \tau_b = 2^{-b}$, construct the lists above, and stop at the first b satisfying*

$$(41) \quad r_b \geq 4ch_b, \quad c = A + 2.$$

Every comparison is rational. The procedure terminates without knowing $e(M)$. At termination it produces an M -label integer/dyadic program with all-history query error at most $C_{Te}(M)$. Moreover

$$(42) \quad h_b \asymp \min\{1, e(M)\}, \quad b = \log_2(2 + e(M)^{-1}) + O(1),$$

where the constants depend only on the fixed system bounds. The final finite certificate satisfies

$$(43) \quad \frac{3r_b}{8} \leq e(M) \leq \frac{5r_b}{4}.$$

A sufficient read-only program size is

$$(44) \quad O\left(M(1 + e(M)^{-1})^J \log(M + 1) + M \log(2 + e(M)^{-1})\right).$$

Proof. By (40), $r_b \geq e(M) - ch_b$. Consequently the test succeeds whenever $h_b \leq e(M)/(5c)$; such a b exists. At the first successful stage, $h_b \leq r_b/(4c) \leq r_b/8$, since $c \geq 2$. Substituting in the lower and upper bounds of (40) proves (43). It also gives $h_b \leq e(M)/(2c - 1)$, using $r_b \leq 2e(M) + 2h_b$.

If $b > 0$, failure at $b - 1$ and $h_{b-1} = 2h_b$ imply

$$e(M) \leq r_{b-1} + ch_{b-1} < 5ch_{b-1} = 10ch_b.$$

Thus $e(M)/(10c) < h_b \leq e(M)/(2c - 1)$ in this case. If $b = 0$, then $h_b = 1$ and the stopping condition and lower certificate give $e(M) \geq 2c - 1$. Since $e(M)$ is also bounded by a constant depending on B , the asserted comparison with $\min\{1, e(M)\}$ follows in all cases. Taking logarithms proves the precision statement.

Use the final selected histories in the transition construction of Theorem 6.3. Equation (30) and $h_b = \tau_b \leq Ce(M)$ give the all-history error bound. The number of state indices is not increased during this construction: refinement stages are offline calculations, not persistent states. The final command grid has $O((1 + e(M)^{-1})^J)$ points. The transition and output counts

in Proposition 6.4, with $b + O(1)$ places for outputs, give (44). No comparison of exact state coordinates, singular values, or exponents occurs in the stopping rule. $\square$

For $J \geq 1$, the total count over all dyadic refinements is bounded, up to fixed constants, by the last refinement count. Indeed the candidate and transition bounds are geometric sums in 2^{bJ} . A sufficient count is

$$O\left(M(1 + e(M)^{-1})^{JT} + M^2(1 + e(M)^{-1})^J\right).$$

This counts rational operations, not oracle running time or unit-cost operations on unrestricted real numbers. The number of refinements is $O(\log(2 + e(M)^{-1}))$. For fixed M the procedure does not decide whether an arbitrary compact set has zero covering radius.

Corollary 7.5 (An absolute-tolerance version). *For any rational $\zeta \in (0, 1]$, stop instead when either (41) holds or $h_b \leq \zeta$. No assumption that $e(M) > 0$ is needed. The resulting all-history query error is at most $C_T(e(M) + \zeta)$, and $h_b \asymp \min\{1, e(M) + \zeta\}$.*

Proof. Termination follows from $h_b \downarrow 0$. If the radius condition holds, the previous theorem's stopping-stage estimates apply; if it holds at $b > 0$, failure of both tests at $b - 1$ also gives $h_b > \zeta/2$. If the radius condition fails at termination, the certificate gives $e(M) < 5ch_b$ and the tolerance condition gives $h_b \leq \zeta$; for $b > 0$ one again has $h_b > \zeta/2$. The initial-stage case is bounded by the fixed state diameter. In all cases $h_b \asymp \min\{1, e(M) + \zeta\}$. Apply (30) with $\tau_b = h_b$. $\square$

7.3. Verification of an individual finite program. A small covering radius does not check that a stored decoder is correct. The following residual certificate checks transitions and outputs separately. It is useful also when a compilation uses sharper, local numerical bounds than the universal covering estimate.

Proposition 7.6 (Transition and decoder residual certificate). *Consider any integer transition table $j(n, i, u, x)$ and any stored probability vectors $q_{n,i}$. Suppose certified evaluations give $\tilde{c}_{n,i}$, $\tilde{y}_{n,i,u,x}$ and $\tilde{Q}_{n,i}$ within τ of, respectively, $c_{n,i}$, $T_{n,x}(c_{n,i}, \bar{u})$ and $Q_n(c_{n,i})$. Define the rational residual bounds*

$$\begin{aligned} d_n &= \max_{i,u,x} \|\tilde{y}_{n,i,u,x} - \tilde{c}_{n+1,j(n,i,u,x)}\|_\infty + 2\tau, \\ o_n &= \max_i \|q_{n,i} - \tilde{Q}_{n,i}\|_\infty + \tau. \end{aligned}$$

Put $D_0 = 0$ and $D_{n+1} = L_n D_n + G_n h + d_n$. For every actual history and every admitted current input name,

$$(45) \quad \|\hat{Q}_n - Q_n(s_n)\|_\infty \leq K_n D_n + o_n.$$

The formula applies to a perturbed or corrupted table as well as to a correctly compiled one; the measured residuals, and therefore the certificate, then change.

Proof. Two state evaluation errors give $\|T_{n,x}(c_{n,i}, \bar{u}) - c_{n+1,j}\|_\infty \leq d_n$. One output evaluation error gives $\|q_{n,i} - Q_n(c_{n,i})\|_\infty \leq o_n$. The transition Lipschitz bound and $\|u - \bar{u}\|_\infty \leq h$ therefore imply inductively $\|s_n - c_{n,i_n}\|_\infty \leq D_n$. The query Lipschitz bound and the output inequality prove (45). All maxima are over finite tables; the justification for their numerical error allowances is the supplied evaluation certificate, not an assumption that approximate vectors are reachable states. $\square$

7.4. Finite input errors without consistency tests. We record explicit constants for the finite moment interface. They are not chosen to optimize numerical conditioning. Their purpose is to certify the data actually used by the offline algorithm.

Lemma 7.7 (A finite coefficient–moment certificate). *Suppose each true report factor has coefficient ℓ^1 norm at most $B \geq 1$, and the coefficient matrix of J cells and r monomials is supplied with entrywise error at most ϵ . Every prior moment through formal degree N is supplied with error at most ϵ . Retain all formal labels, even when their exponents coincide. Put $C_f = Jr$ and assume $C_f\epsilon \leq 1$. For a prefix of length n , set*

$$D_n^{\text{in}} = nC_f(B+1)^{n-1} + (B+1)^n,$$

where the first term is zero when $n = 0$. The computed prefix evidence $\tilde{Z}$ and every computed raw-moment numerator $\tilde{A}$ have absolute errors at most ϵD_n^{in} . If $\epsilon D_n^{\text{in}} \leq \kappa^n/2$, then

$$(46) \quad \tilde{Z} \geq \kappa^n/2 > 0, \quad \left| \frac{\tilde{A}}{\tilde{Z}} - \frac{A}{Z} \right| \leq \frac{2\epsilon D_n^{\text{in}}}{\tilde{Z}} \leq 4\epsilon D_n^{\text{in}} \kappa^{-n}.$$

For a true probability query with exact formal coefficient ℓ^1 norm at most B_Q , replace the numerator error by $B_Q\epsilon D_n^{\text{in}}$ and the first numerator 2 on the right by $B_Q + 1$.

Proof. For an accepted cell or a rejection mixture, command entries have absolute value at most one. The error in a report factor has ℓ^1 norm at most $Jr\epsilon$. Thus every approximate factor has norm at most $B + 1$. Telescoping the product of n factors in the coefficient algebra, whose ℓ^1 norm is submultiplicative, bounds the product error by $nC_f\epsilon(B+1)^{n-1}$. Integrating the product error against true moments costs no further factor, since all true moments lie in $[0, 1]$. Integrating the approximate product against the moment errors costs at most $\epsilon(B+1)^n$. Shifting its formal indices for a raw moment changes neither coefficient norm. This proves both error estimates. Since $Z \geq \kappa^n$, the asserted positivity follows. Finally, with $q = A/Z \in [0, 1]$,

$$\frac{\tilde{A}}{\tilde{Z}} - q = \frac{(\tilde{A} - A) - q(\tilde{Z} - Z)}{\tilde{Z}}.$$

The two numerator errors prove (46). Multiplication by an exact query polynomial costs its coefficient norm, and the same identity proves the query version. Clipping an approximate probability to $[0, 1]$ does not increase its error. $\square$

The approximation table is not required to satisfy positivity identities or equality between coincident formal moments. Those properties hold for the true experiment; the error certificate is what is required of its numerical data. Formal products of degree less than N can be padded by the zero exponent to degree N . Conversely, keeping all degrees through N gives the same finite interface. For a prescribed evaluation tolerance τ , choosing $\epsilon \leq c_N\tau$ and allowing a fixed number of additional dyadic places proves the required certificate. Thus none of the refinement steps depends on deciding exact equality of exponent sums.

7.5. Recovery of the entire collision profile.

Lemma 7.8 (Covering and prediction scales). *For the raw attainable state sets of the monomial experiment,*

$$(47) \quad e_n(M)^2 \asymp \Psi_{n, N-n}(M, a), \quad e(M)^2 \asymp \Xi_N(M, a),$$

uniformly over $a \in K$ and $M \geq 1$.

Proof. The global reachable-centre construction in Theorem 5.3 gives the upper inequality. For the reverse inequality take any M raw centres with covering radius $e_n(M) + \epsilon$. The fixed formal query matrix is a bounded linear map from raw coordinates to physical prediction coordinates. Send the index of a nearest raw centre and apply this matrix at the decoder; clip each resulting probability to $[0, 1]$. The worst-history mean squared error is at most $C(e_n(M) + \epsilon)^2$. This is an admissible one-checkpoint code, irrespective of whether its raw centres are reachable. The worst-history lower bound in Theorem 5.3 gives $c\Psi_{n,N-n} \leq C(e_n(M) + \epsilon)^2$. Let ϵ decrease to zero. There are finitely many checkpoints, so taking maxima proves the second assertion with uniform constants. This argument uses neither the new stopping rule nor its causal upper bound, and hence does not assume the conclusion to be proved. $\square$

Theorem 7.9 (Profile-adaptive finite realization). *Under the finite-data hypotheses of Theorem 6.6, with numerical data available at successively requested tolerances, one algorithm produces an M -label finite numerical filter without receiving Ξ_N , a collision stratum, or any additive-gap lower bound. Its maximum worst-history regret is at most $C\Xi_N(M, a)$, and the original common-exploration lower bound is $c\Xi_N(M, a)$ for every such filter with a memoryless input interface. The algorithm selects a precision satisfying*

$$(48) \quad b(M, a) = \frac{1}{2} \log_2(1/\Xi_N(M, a)) + O(1),$$

and has sufficient read-only program size

$$(49) \quad P(M, a) = O\left(M \Xi_N(M, a)^{-J/2} \log(M + 1)\right).$$

Equation (48) describes the precision selected by this algorithm, up to a uniform additive constant. It is not a lower bound on the precision of every possible realization.

Proof. By Lemma 7.8 and Proposition 7.1, $e(M) > 0$. All constants and finite evaluation hypotheses for the raw system were verified in Section 6; Lemma 7.7 provides explicit error allowances at each refinement. Apply Theorem 7.4 and square its all-history query bound. Equation (47) gives the desired upper law. The state budget remains exactly the same upper budget M , so memoryless input coding is simulated by an exact-command filter and the intrinsic streaming converse applies unchanged.

Raw states lie in a unit cube, hence $e(M) \leq 1/2$. Also $M^{-2} \leq \Xi_N(M, a) \leq 1$. Taking logarithms of (47) and using (42) proves (48). Equations (44) and (36) absorb the output term into (49), and give $b(M, a) \leq d_0^{-1} \log_2(M + 1) + O(1)$. The supplied dyadic inputs and rounded outputs have $b + O(1)$ -place accuracy. The exhibited exact rational evaluator uses $O(b + 1)$ -bit intermediates, with a multiplicative constant depending on the fixed horizon and input format; it need not have only an additive overhead. Lemma 8.10 makes this distinction explicit. If moments are obtained from approximate calibration entries, the derivative estimate in Corollary 6.7 transfers the same tolerance. The synthesis is uniform when those input procedures are computable; otherwise it is a finite-advice construction with the same error bound. $\square$

Corollary 7.10 (Program-size phases at intersecting collision lines). *In the four-cell, five-trial family of Corollary 5.7, write $\rho = \max\{|u|, |v|\}$ and $\tau_c = \min\{|u|, |v - u|, |v - 2u|\}$. Put*

$$F = \max\{M^{-1/3}, \rho^{1/2} M^{-1/4}, (\rho^2 \tau_c)^{2/9} M^{-2/9}\}.$$

The adaptive algorithm realizes regret of order F using $b = \frac{1}{2} \log_2(1/F) + O(1)$ places. A sufficient program bound is

$$(50) \quad O\left(\min\{M^{5/3}, \rho^{-1} M^{3/2}, (\rho^2 \tau_c)^{-4/9} M^{13/9}\} \log(M + 1)\right),$$

where a reciprocal term with zero denominator is omitted. In particular $b \leq \frac{1}{6} \log_2(M+1) + O(1)$ throughout the parameter square.

Proof. Here $r = J = 4$, $N = 5$, and $d_0 = 6$. The intrinsic two-parameter theorem gives $\Xi_N \asymp F$. Substitute this comparison in (48) and (49). The identity $MF^{-2} = \min\{M^{5/3}, \rho^{-1}M^{3/2}, (\rho^2\tau_c)^{-4/9}M^{13/9}\}$ proves (50), with the stated zero convention. The separated-chain term gives the uniform precision bound. $\square$

Along $u = \theta$, $v = \theta + \theta^k$, $k \geq 2$, the three sufficient program scales are respectively $M^{5/3}$, $\theta^{-1}M^{3/2}$, and $\theta^{-(4k+8)/9}M^{13/9}$, times $\log(M+1)$. Their crossovers are the same $M \asymp \theta^{-6}$ and $M \asymp \theta^{-(8k-2)}$ as for the error profile. This is a consequence for numerical realizations, not a new classification of collision arrangements or a converse for program length. In particular, the algorithm need not determine k or resolve $|v - u|$ before selecting its numerical scale.

8. VERIFIED CONSTRUCTION AND STABILITY OF FINITE ADVICE

The intrinsic law concerns an experiment, whereas a numerical certificate concerns specified data or a specified program. We now relate these statements without identifying them. The resulting construction is stable under a common, imperfect numerical description of several experiments. In particular, an exact additive relation need not be resolved when its effect lies below the accuracy of the supplied moments.

8.1. A construction certificate independent of the program residual. Use the compact finite-horizon system of Section 6. The numerical value attached to each grid history is fixed during one compilation. Let $\tilde{s}_n(w)$ approximate its true state to accuracy τ in the maximum norm, including numerical rounding. Denote the selected histories by $w_{n,1}, \dots, w_{n,k_n}$, $k_n \leq M$, and put

$$\tilde{c}_{n,i} = \tilde{s}_n(w_{n,i}), \quad r_n = \max_{w \in \mathcal{H}_n^h} \min_{i \leq k_n} \|\tilde{s}_n(w) - \tilde{c}_{n,i}\|_\infty.$$

Here $\mathcal{H}_n^h$ is the complete finite list of grid histories. A history extending a selected history by a grid command and a report belongs to the next such list. We order the lists once and use the first index whenever a minimum or maximum is attained more than once.

Definition 8.1 (Finite construction contract). A table satisfies the construction contract if its representatives are the farthest-first selection from the fixed numerical lists, its state counts agree with those representatives, every transition satisfies

$$(51) \quad j(n, i, u, x) = \min_{k \leq k_{n+1}} \operatorname{argmin} \|\tilde{s}_{n+1}(w_{n,i}(u, x)) - \tilde{c}_{n+1,k}\|_\infty,$$

and every output is the prescribed clipped dyadic rounding of the numerical query evaluation at its representative history. All numerical lists and their radii r_n are fixed before a table is checked. The histories and the contract are offline objects, not persistent state.

Theorem 8.2 (Construction-to-covering implication). *Suppose the fixed state and query evaluations have errors at most τ , the command grid has radius h , and a table satisfies Definition 8.1. If E_n bounds the distance from a true state to the true representative selected by the running filter, then one may take*

$$(52) \quad E_0 = 0, \quad E_{n+1} = L_n E_n + G_n h + r_{n+1} + 2\tau.$$

The query error at stage n is at most $K_n E_n + \tau$. Moreover,

$$(53) \quad r_n \leq 2e_n(M) + 2\tau, \quad \max_{i,u,x} \|T_n(c_{n,i}, \bar{u}, x) - c_{n+1,j(n,i,u,x)}\|_\infty \leq r_{n+1} + 2\tau.$$

The right side of the second inequality is fixed by the numerical history lists, not increased in response to a supplied transition table. Consequently, at fixed horizon the table has worst-history query error $O(e(M) + h + \tau)$.

Proof. The true initial state is the representative of the empty history, so $E_0 = 0$ even when its numerical name is imperfect. For a stored representative and a grid command, the numerical target in (51) belongs to $\mathcal{H}_{n+1}^h$. Its distance to the selected numerical centre is therefore at most r_{n+1} . Passing from the two numerical values to their two true values costs at most 2τ . This proves the second inequality in (53). The first is the farthest-first separation argument of Lemma 7.3: if there is an unselected point, the selected points together with a farthest point are r_n -separated; transferring them to an arbitrary M -ball cover of the true set gives $r_n \leq 2e_n(M) + 2\tau$. If every point is selected, $r_n = 0$ and the inequality is immediate.

For an actual command, first compare the true current state with its true representative, at cost $L_n E_n$, and then replace the command by its grid code, at cost $G_n h$. Apply the just proved bound to the appended representative history. This gives (52). Compare a true query with its value at that representative, at cost $K_n E_n$, and then use the certified stored output, at cost τ . Iterating the recurrence finitely many times and using the first inequality in (53) proves the last assertion. At no step is an approximate vector passed to a transition map whose domain is the feasible state set. $\square$

The residual bound in Proposition 7.6 remains valid for an arbitrary table. It is a different assertion: its right side is measured from that table. The following comparison makes the logical distinction quantitative. The example and its role as a transition diagnostic were given in [26]; we include the argument because the distinction is part of the numerical theorem.

Proposition 8.3 (Residual validity and covering-order realization). *There is a fixed compact one-step Lipschitz system for which a sequence of M -label tables has exact outputs at all representatives and valid residual bounds, yet its worst-history squared prediction error divided by the squared optimal M -centre radius is $4M^2$. Thus these two properties, even together with knowledge of the optimal covering radius, do not imply the construction-to-covering conclusion. For the same system a table satisfying the nearest-centre construction has the covering-order bound of Theorem 8.2.*

Proof. Take $U = [1/4, 3/4]$, $s_0 = 1/2$, $T(s, u) = (s + u)/2$, and $Q(s) = s$. The reachable interval $[3/8, 5/8]$ has length $1/4$. Since M balls of radius e have total length at most $2Me$, its covering radius is at least $1/(8M)$; equally spaced midpoints attain that value. Select representatives by farthest-first on an endpoint-containing command grid whose first history gives $3/8$, and store their exact query values. Replace every transition target by this first index. The output for command $3/4$ is now $3/8$, instead of $5/8$. The maximum error is exactly $1/4$, so the asserted squared ratio is $(1/4)^2 / (1/(8M))^2 = 4M^2$. Its measured transition residual is also $1/4$, hence its residual bound is true. None of these computations changes the optimal radius of the experiment. For the correct table, (51) and Theorem 8.2 give error $O(e(M) + h)$; choose $h = O(e(M))$. For the dyadic grids with $8M$ segments used in the finite diagnostic, M a power of two, farthest-first gives maximum grid error at most $2e(M)$: after the endpoints, each new point bisects a longest interval between consecutive centres. Induction over the dyadic levels bounds its largest half-gap by $1/(4M)$; for $M = 1$ the endpoint error is $1/4$. $\square$

This proposition is not a limitation on finite approximation methods. It states exactly what a residual estimate does not assert about an arbitrary program. The additional input that identifies the sharp scale in this paper is the attained geometry of Theorem 5.3, not the error

recurrence itself. We next give a consequence of that geometry which also concerns imperfectly known experiments, rather than only a count of table entries.

8.2. Stability of attainable sets and their profiles. All state coordinates in this subsection are the formal raw coordinates; coincident labels are not removed. Two experiments have the same command and report alphabets and are compared on the same histories. Their moment tables contain every formal degree through N .

Lemma 8.4 (Paired-history stability). *Consider two positive monomial experiments with uniformly bounded cell coefficient norms and report factors bounded below by $\kappa > 0$. Suppose the maximum entrywise difference of their cell coefficients and formal moment tables is at most $\delta \leq 1$. There is a constant C_* , depending only on the fixed finite format and positivity bounds, such that, at every history of length $n < N$,*

$$(54) \quad \|s_n(w) - s'_n(w)\|_\infty \leq C_*\delta.$$

For a common probe menu with fixed formal coefficients the query probabilities satisfy the same assertion, with another fixed constant. In particular,

$$(55) \quad d_H(S_n, S'_n) \leq C_*\delta, \quad |e_n(M) - e'_n(M)| \leq C_*\delta \quad (M \geq 1).$$

No lower bound on a nonzero difference of exponent sums occurs here.

Proof. Let B bound the coefficient norm of a report factor and put $C_f = Jr$. For a fixed command, the difference of the two factor coefficient vectors has norm at most $C_f\delta$. Telescoping a product of n factors and then changing its moments bounds each integrated numerator and its evidence by

$$D_n\delta, \quad D_n = nC_f(B+1)^{n-1} + (B+1)^n.$$

The same calculation applies after multiplication by any formal raw monomial of the remaining degree. If A/Z and A'/Z' are the two raw moments, then $Z, Z' \geq \kappa^n$ and $0 \leq A'/Z' \leq 1$. Consequently

$$\left| \frac{A}{Z} - \frac{A'}{Z'} \right| = \frac{|A - A' + (A'/Z')(Z' - Z)|}{Z} \leq 2D_n\kappa^{-n}\delta.$$

Take the maximum over the finitely many stages. A fixed formal query is a bounded linear combination of raw moments, proving its assertion. These estimates compare two actual experiments; the formal moment entries need not satisfy the same coincidence identities in the two.

Every reachable point is obtained from some history. Use that identical history in the other experiment to obtain (54) in both directions. This proves the Hausdorff bound. An arbitrary M -centre cover of one set, with radius within ϵ of its infimum, covers the other with additional radius at most $C_*\delta$. Interchange the sets and let ϵ decrease to zero to finish. $\square$

Uniformity over priors must be specified, not inferred from full support alone. Fix a full-support Borel probability μ_0 on $[0, 1]$ and constants $0 < c_- \leq 1 \leq c_+ < \infty$. Write

$$\mathcal{D} = \{\mu \text{ a probability} : c_-\mu_0 \leq \mu \leq c_+\mu_0\}.$$

These are inequalities of measures. They do not require μ_0 , or any member of $\mathcal{D}$, to have a Lebesgue density.

Lemma 8.5 (Uniformity on a dominated prior family). *The constants in the intrinsic checkpoint and causal laws, and in Lemma 7.8, can be chosen uniformly for $(a, \mu) \in K \times \mathcal{D}$. The finite-horizon transition and finite-input estimates have the same uniformity.*

Proof. The family $\mathcal{D}$ is weakly compact. Indeed, probabilities on $[0, 1]$ are weakly compact, and each measure inequality is closed: pass to the limit against a nonnegative continuous function and then use regularity of finite Borel measures. Every member has full support. The ordinary and complete confluent moment entries used in Lemma 4.3 depend continuously on (a, μ) on this compact family. For a logarithmic entry, its positive exponent is bounded away from zero; hence $t^c(\log t)^j$, extended by zero at the origin, varies continuously in the uniform norm on the relevant compact parameter set. The zero-exponent constant block has multiplicity one.

There are finitely many formal orderings and confluent block types. For each, the mixed-moment pairing is strictly positive by the confluent positivity lemma for every $\mu \in \mathcal{D}$. Equivalently, in its determinant integration formula a product of k copies of μ dominates $c_-^k \mu_0^{\otimes k}$; the ordered separated intervals used to prove strict positivity therefore retain positive mass. Compactness and continuity give the uniform lower singular-value bounds in the normalized attainable flag argument. The derivatives used in its local inverse and volume minorization are uniformly bounded, since all factor coefficients are bounded and the evidence is at least κ^n . The acquisition-prefix probability has this same lower bound independently of the prior.

For the global cover, changing the moments only changes real coefficients in the bounded-format formula. Neither its format nor its attainable dimension changes. The bounding rectangle, evidence bound, and finite query maps are uniformly bounded. Thus both the local lower estimate and the global upper estimate used in Theorem 5.3 are uniform. The causal argument uses only these estimates and finitely many uniform transition bounds. Finally, the proof of Lemma 7.8 and the explicit moment error estimate use those same constants. This accounts for all prior-dependent inputs of the asserted estimates. $\square$

Corollary 8.6 (Stability of the collision profile). *Fix the detector and let $(a, \mu), (a', \mu') \in K \times \mathcal{D}$. If their formal moment tables differ by at most δ , then, uniformly for all $M \geq 1$,*

$$(56) \quad \Xi_N(M, a) \leq C(\Xi_N(M, a') + \delta^2), \quad \Xi_N(M, a') \leq C(\Xi_N(M, a) + \delta^2).$$

For a fixed prior, δ may be replaced by $C\|a - a'\|_\infty$. In particular, equality of additive sums may change without producing an instability above the scale of the moment error.

Proof. Lemmas 8.4 and 8.5 give $e(M) \leq e'(M) + C_*\delta$ and the reverse inequality, uniformly. Square these inequalities and use Lemma 7.8. For the calibration assertion, the constant formal label is independent of a . For every other label of degree at most N , its exponent is uniformly positive on K . The derivatives of $t^{\alpha \cdot a}$ with respect to positive-exponent coordinates are $\alpha_j t^{\alpha \cdot a} \log t$, uniformly bounded on $[0, 1]$. The line segment between two points of K retains a positive lower bound on the one-step exponents. Integrating the derivative along that segment proves the stated moment estimate for any probability prior. $\square$

8.3. One program from a common imperfect numerical name. Suppose a collection of compact systems shares its labelled history interface, its state-coordinate dimensions, and constants L_n, G_n, K_n . A *common numerical name of floor ρ* , $0 < \rho \leq 1$, supplies at scale $h = 2^{-b}$ one finite command grid and one numerical evaluation on its feasible histories. For every compatible system, the grid is an h -net and its state and query errors, including rounding, are at most

$$\tau_b = \max\{h, \rho\}.$$

The same data are used for all compatible systems. This quantifier is essential: a different exact prior hidden in a program is not a common numerical name. Write $e_{\mathcal{E}}(M)$ for the largest covering radius of a compatible system $\mathcal{E}$, and let $A = \max_n A_n$ be the history-net constant. Assume the state sets lie in fixed bounded cubes.

Theorem 8.7 (Stable common-advice realization). *From such a common name one can construct one table satisfying Definition 8.1, with at most M persistent labels at each stage, whose worst-history query error in every compatible system is at most*

$$(57) \quad C(e_{\mathcal{E}}(M) + \rho).$$

The construction does not select a compatible system. More precisely, put $c = A + 4$, and stop at the first dyadic scale for which

$$(58) \quad r_b \geq 4ch_b \quad \text{or} \quad h_b \leq \rho, \quad r_b = \max_{1 \leq n < N} r_{n,b}.$$

The stopping scale satisfies, simultaneously for every compatible system,

$$h_b \asymp \min\{1, e_{\mathcal{E}}(M) + \rho\}.$$

All constants depend only on the fixed horizon, bounds, and dimensions.

Proof. The second condition ensures finite termination. Until and including the first stopping scale, $\tau_b \leq 2h_b$: before crossing the floor it is h_b , and the first dyadic scale below ρ is greater than $\rho/2$ (with equality at the initial endpoint treated separately). For each compatible system, the finite-cover argument therefore gives

$$(59) \quad \max\{0, r_b/2 - 2h_b\} \leq e_{\mathcal{E}}(M) \leq r_b + (A + 4)h_b.$$

The numerical radius on both sides is common, although the true covering radius may depend on the system.

Suppose $b > 0$ is the first stopping index. At the previous scale both conditions failed. By (59) and $h_{b-1} = 2h_b$,

$$e_{\mathcal{E}}(M) < 10ch_b, \quad \rho < 2h_b.$$

Thus $h_b > (e_{\mathcal{E}}(M) + \rho)/(10c + 2)$. If the radius condition holds at b , use $r_b \leq 2e_{\mathcal{E}}(M) + 4h_b$ to obtain $h_b \leq e_{\mathcal{E}}(M)/(2c - 2)$. If it does not, the floor condition gives $h_b \leq \rho$. In both cases $h_b \leq e_{\mathcal{E}}(M) + \rho$. The initial stopping case follows from the fixed diameter bound and $\rho \leq 1$; if its radius condition holds, the same inequality bounds $e_{\mathcal{E}}$ below by a fixed positive number, and otherwise $\rho = 1$. This proves the asserted scale comparison.

The farthest-first selection and the independently checked nearest transitions use only the common numerical lists. In every compatible system Theorem 8.2 bounds the query error by $C(e_{\mathcal{E}} + h_b + \tau_b)$. At the stopping scale the last two terms are $O(e_{\mathcal{E}} + \rho)$, proving (57). Every state remains an index of an actual representative history for that system. Neither its true coordinates nor its model identity are consulted at runtime. $\square$

Corollary 8.8 (Finite moment uncertainty through collisions). *Fix a detector, K , N , and the prior envelope $\mathcal{D}$. Suppose a finite numerical moment table of degree at most N is within δ of every compatible $(a, \mu) \in K \times \mathcal{D}$. The cell coefficients may also be supplied to this numerical accuracy; the underlying detector is fixed. For all sufficiently small $\delta > 0$, one causal M -label filter constructed from these same data has maximum worst-history regret at most*

$$(60) \quad C(\Xi_N(M, a) + \delta^2)$$

in every compatible experiment. No collision decision or value of Ξ_N is supplied to the construction. Its selected scale is comparable to $\min\{1, \sqrt{\Xi_N(M, a)} + \delta\}$ for each such experiment.

Proof. Lemma 7.7 bounds the error of evaluating any feasible history from the supplied finite data by $C_0\delta$, with C_0 uniform by Lemma 8.5. Use $\rho = 2C_0\delta$, increasing C_0 if necessary to cover the fixed query coefficients. Exact rational evaluation of the supplied entries, followed by rounding at $b + 2$ binary places, has error at most $\rho/2 + h_b/8 \leq \max\{h_b, \rho\}$. Every compatible experiment therefore has the same valid numerical name. The rational command cube admits

the required grids. Apply Theorem 8.7, square its bound, and use Lemma 7.8, uniformly in the prior envelope. The physical probe map is the same fixed formal map for the fixed detector. The running program has the prescribed label budget throughout. The smallness restriction on δ merely ensures the printed positive-evidence and $\rho \leq 1$ bounds. $\square$

This is a robustness statement about one finite description, not an assumption that the approximate moment table is itself a positive moment sequence. It supplies a direct interpretation of the full collision law when even exact identities among supplied entries are unavailable. The additional δ^2 term is not only an artifact of propagating a norm bound.

Proposition 8.9 (Sharp advice uncertainty in a positive experiment). *There is a fixed positive detector with $r = 2$ and $N = 2$, and a dominated full-support prior family, for which the minimax regret of one filter required to serve all priors compatible with a common δ -accurate moment table has order*

$$M^{-2} + \delta^2,$$

for all $M \geq 1$ and all sufficiently small $\delta > 0$. The lower bound holds under the same independent command exploration and a fixed finite spanning probe menu. It applies to programs receiving that common advice; it is not a precision converse for programs given different exact priors.

Proof. Take $a_0 = 0, a_1 = 1$ and the three detector cells

$$k_0(t) = \frac{1}{4}, \quad k_1(t) = \frac{3}{8} + \frac{1}{8}t, \quad k_2(t) = \frac{3}{8} - \frac{1}{8}t.$$

They are positive, sum to one, and span $\text{span}\{1, t\}$. Let $\eta = 1/4$, and take as reference prior uniform measure on $[0, 1]$, with envelope constants $1/2$ and $3/2$. For $\epsilon = 6\delta \leq 1/2$ consider the two densities $1 \pm \epsilon(2t - 1)$. Their moments are

$$m_k^\pm = \frac{1}{k+1} \pm \epsilon \frac{k}{(k+1)(k+2)}.$$

Since $k/((k+1)(k+2)) \leq 1/6$ for every nonnegative integer k , the uniform-prior moments form one common δ -accurate numerical name, even if more moments than required are supplied.

Upon reporting accepted cell 0, its likelihood $g_0/4$ is constant in t , so the posterior remains the prior. The joint density of this report and the observed command is identical in the two experiments; its probability is at least $\eta/4$. The probe obtained with command $(1/2, 3/8, 5/8)$ is the event of rejection, whose function is $q(t) = 1/2 + t/32$. Its two posterior means differ by $\delta/16$. For any possibly randomized prediction Y using the common advice, the distribution of Y conditional on the accepted-cell-0 history is the same in the two experiments. If the two means are q_-, q_+ ,

$$\frac{\mathbb{E}(Y - q_-)^2 + \mathbb{E}(Y - q_+)^2}{2} = \mathbb{E}\left(Y - \frac{q_- + q_+}{2}\right)^2 + \frac{(q_+ - q_-)^2}{4} \geq \frac{\delta^2}{1024}.$$

Integrate over those identical command histories and multiply by the fixed positive probability of selecting this probe from the spanning menu. The larger of the two unconditional regrets is at least $c\delta^2$, independently of M .

For either fixed prior, the intrinsic checkpoint lower bound is $c'M^{-2}$, uniformly in this envelope. Indeed, $n = m = 1$ and the attainable dimension is one, so $\Xi_2(M, a) = M^{-2}$. A common-advice filter is an admissible M -label filter for that fixed experiment; supplying less information does not invalidate its memory lower bound. Combining the two lower bounds gives a constant times $M^{-2} + \delta^2$. Corollary 8.8 gives the matching upper bound simultaneously for all compatible priors in the envelope, proving the assertion. $\square$

8.4. Exact rational workspace.

Lemma 8.10 (Input accuracy and exact intermediate lengths). *For the finite moment evaluator at fixed horizon and fixed format, $b + O(1)$ -place dyadic inputs suffice for $O(2^{-b})$ evaluation error. If evaluated with exact rational operations, every intermediate numerator and denominator has $O(b + 1)$ bits, with a constant depending on the fixed arithmetic circuit. This statement does not give an additive $O(1)$ overhead in exact intermediate length. In fact, for $x_b = (2^b - 1)/2^b$, the exact fractional length of x_b^N is Nb .*

Proof. The input-accuracy assertion is the positive-evidence error bound of Lemma 7.7. For the exact length assertion, bound the lengths of the numerator and denominator of a reduced fraction by those of its unreduced representation. Multiplication or division adds the operand lengths, while addition forms two products and their sum, increasing the resulting length by at most one beyond those product bounds. The circuit for one fixed-degree moment quotient has a number of gates depending only on the fixed format. Induction over these gates gives $C(b + 1)$ bits for every intermediate. Its evidence is nonzero by the supplied certificate. Distances and comparisons of a fixed number of coordinates satisfy the same bound; selecting among more histories increases the number of operations and the index length, not the degree of a moment evaluation.

Finally, $2^b - 1$ is odd. The reduced denominator of x_b^N is thus exactly 2^{Nb} . This proves the stated exact length and explains why input/output accuracy and exact arithmetic storage cannot be identified. The rational-operation counts in the preceding sections exclude the cost of obtaining numerical advice; they are not bit-operation counts. $\square$

APPENDIX A. THE OBSERVATION ALGEBRA AND ASYMMETRIC PROFILES

For integral exponents define the standard graded algebra

$$\mathcal{R}_A = \mathbb{R}[zt^a : a \in A] \subset \mathbb{R}[z, t], \quad \deg(zt^a) = 1.$$

Its degree- m component has basis $\{z^m t^s : s \in mA\}$, so its Hilbert function is exactly $h_A(m)$. The general sumset–Hilbert-function connection is classical [2, 3]. Its role here is operational: $h_A(m) - 1$ is the number of independent future observable tests, and Theorem 3.3 proves when the attainable past can excite all of them. The algebra alone would not prove that transversality.

Proposition A.1 (Three-cell classification). *Let $A = \{0, a, b\}$ with integers $0 < a < b$, put $g = \gcd(a, b)$, $a' = a/g$ and $B = b/g$. Then*

$$(61) \quad h_A(m) = \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2}.$$

The homogeneous relation is generated by $x_1^B - x_0^{B-a'} x_2^{a'}$ under the map $(x_0, x_1, x_2) \mapsto (z, zt^a, zt^b)$. In particular,

$$d_A(n, m) = \min \left\{ 2n, \binom{m+2}{2} - \mathbf{1}_{\{m \geq B\}} \binom{m-B+2}{2} - 1 \right\}.$$

For nonnegative real exponents $0 < a < b$ with a/b irrational, $h_A(m) = \binom{m+2}{2}$ for every m .

Proof. A monomial $x_0^k x_1^i x_2^j$ of degree m maps to $z^{m+2ai+bj}$. Two exponent triples of the same degree have the same image exactly when their difference is an integer multiple of $(-(B - a'), B, -a')$: this follows by dividing $a(i - i') + b(j - j') = 0$ by g and using coprimality of a', B . After cancelling a common monomial, each corresponding binomial is a multiple of $x_1^B - x_0^{B-a'} x_2^{a'}$, by the difference-of-powers identity. The kernel is spanned by these binomials,

because monomials with distinct images are linearly independent and each fiber's coefficient sum must vanish. Thus this one relation generates the kernel.

The degree- m polynomials in three variables have dimension $\binom{m+2}{2}$. Multiplication by the displayed nonzero degree- B relation is injective in the polynomial ring, so the degree- m part of the ideal has dimension zero for $m < B$ and $\binom{m-B+2}{2}$ otherwise. Subtraction proves (61). If a/b is irrational, $a(i - i') + b(j - j') = 0$ forces both differences to vanish; all $\binom{m+2}{2}$ possible pairs are distinct. The operational formula now follows from Theorem 3.3. $\square$

Corollary A.2 (The sparse space that separates rank from future dimension). *For $A = \{0, 2, 5\}$, the first values of $h_A(m)$, starting at $m = 0$, are 1, 3, 6, 10, 15, 20, 25. At $n = 3, m = 2$ the exact information dimension is five, for every full-support prior. For a total budget of six the minimal causal dimensions, starting at $n = 0$, are*

$$0, 2, 4, 6, 5, 2, 0.$$

Thus the peak need not be the peak of a rank-based symmetric tent.

Proof. Apply (61) with $B = 5$ and then (1). In particular $2A = \{0, 2, 4, 5, 7, 10\}$, while the binomial witness at $n = 3$ has $B_{A,3} = \{0, 2, 5, 7, 10, 12, 15\}$. The mixed matrix has rank six and normalization leaves five. This proves the example uniformly over full-support priors without relying on a single large rational determinant. The uniform-prior three-cell example and its numerical witness first appeared in the source-pinned report [10]; the present corollary is an application of the general theorem, not a new attribution of that example. $\square$

There is also an exact finite computation for any integral A : start with $S_0 = \{0\}$, form $S_{m+1} = \{s + a : s \in S_m, a \in A\}$, and count distinct exponents. This handles all additive collisions, not just a nominal degree range. For real exponents the formula is equally exact as mathematics, but an effective implementation requires exact equality information about the supplied exponents. Neither fixed rank nor arbitrarily accurate floating-point exponents substitute for that information.

APPENDIX B. CONFLUENT OBSERVATION ALGEBRAS AND RESOLVED DIRECTIONS

The cardinality of a sumset is discontinuous when two of its elements coalesce. The corresponding experiment need not be discontinuous. We now identify the scales of the directions lost at such a collision. The parameter θ below indexes a *known calibration*; the unknown parameter of each experiment is still t .

B.1. Affine exponent families. Fix $0 = a_0 < a_1 < \dots < a_{r-1}$ and real numbers b_i , with $b_0 = 0$. On a sufficiently small compact interval $0 \leq \theta \leq \theta_0 \leq 1$ put

$$A_\theta = \{a_i + \theta b_i : 0 \leq i < r\}.$$

The order of these exponents is assumed unchanged. Take $I = [0, 1]$ and one fixed full-support prior μ . A fixed coefficient matrix specifies positive detector cells

$$(62) \quad k_j^\theta(t) = \sum_{i=0}^{r-1} c_{ji} t^{a_i + \theta b_i}, \quad \sum_j k_j^\theta = 1.$$

It has column rank r , and the cells have a common positive lower bound. The comparator cube is fixed. These hypotheses include any sufficiently small affine perturbation of a strictly positive calibration of full rank. No continuity of an unknown prior density is assumed.

For a fixed positive future length $m \geq 1$, let

$$\mathcal{B}_m = \left\{ \left(\sum_{\nu=1}^m a_{i_\nu}, \sum_{\nu=1}^m b_{i_\nu} \right) : 0 \leq i_\nu < r \right\}.$$

Repeated *pairs* are identified. Distinct pairs give distinct exponents $\lambda + \theta\gamma$ for $0 < \theta \leq \theta_0$; shrink θ_0 to exclude crossings if necessary. For each first coordinate λ , order its second coordinates as $\gamma_{\lambda,0} < \dots < \gamma_{\lambda,s_\lambda-1}$. Set $K_m = \sum_\lambda s_\lambda$. The cluster at $\lambda = 0$ consists only of $(0, 0)$.

Define the divided-difference functions

$$(63) \quad \psi_{\lambda,k}^\theta(t) = [\lambda + \theta\gamma_{\lambda,0}, \dots, \lambda + \theta\gamma_{\lambda,k}](x \mapsto t^x), \quad 0 \leq k < s_\lambda.$$

At $\theta = 0$ their continuous values are

$$(64) \quad \psi_{\lambda,k}^0(t) = t^\lambda (\log t)^k / k!.$$

For $\lambda > 0$ these functions have value zero at $t = 0$; the sole zero-exponent function is the constant one. The bracket in (63) is a divided difference in the exponent, not a new apparatus output. For a history h let $z_\theta(h)$ be its posterior expectations of these functions, omitting the constant coordinate.

List the orders k of the nonconstant functions in nondecreasing order:

$$0 = \nu_1 \leq \dots \leq \nu_d, \quad d = K_m - 1.$$

There are $|mA_0| - 1$ zero orders. The associated resolution function is

$$(65) \quad \Phi_\theta(M) = \max_{1 \leq \ell \leq d} \theta^{2(\nu_1 + \dots + \nu_\ell)/\ell} M^{-2/\ell}, \quad M \geq 1.$$

We set $0^0 = 1$ and $0^a = 0$ for $a > 0$ in this formula. In particular $\Phi_0(M) = M^{-2/(|mA_0|-1)}$.

Choose r attainable failure factors which form a basis in the formal monomials $t^{a_i + \theta b_i}$, with coefficients independent of θ . For example use $1/2$ and $1/2 + \delta t^{a_i + \theta b_i}$, $i > 0$, with δ sufficiently small. The fixed right inverse of the calibration matrix realizes these by fixed command labels. All ordered m -fold products form a common finite probe menu. Any other such fixed basis, and its full product menu, gives equivalent bounds below.

Theorem B.1 (Uniform resolved checkpoint law). *Suppose*

$$(66) \quad K_m - 1 \leq n(r - 1).$$

Supply independent uniform commands for the first n trials, then reveal an independent uniform query from the preceding product menu. Let $\bar{R}_{M;n,m}^\theta$ be the optimal unconditional squared-prediction regret of an arbitrary M -message checkpoint encoder. Let $R_{M;n,m}^{\theta,\max}$ be its minimax counterpart over admitted histories. There are constants $0 < c \leq C < \infty$, independent of M and θ , such that

$$(67) \quad c\Phi_\theta(M) \leq \bar{R}_{M;n,m}^\theta \leq R_{M;n,m}^{\theta,\max} \leq C\Phi_\theta(M), \quad 0 \leq \theta \leq \theta_0.$$

The encoders may be Borel and randomized as in Definition 2.4. At the common interior binomial command tuple the normalized map $h \mapsto z_\theta(h)$ is a submersion of rank d , including at $\theta = 0$. Its local inverse bounds and an unconditional minorization can be chosen uniformly in θ .

The upper bound in this theorem is a checkpoint bound. It does not assume that access to an exact prefix is free during repeated compression. Every streaming filter is a checkpoint encoder, so the lower bound applies to such filters as well. Section E proves a matching genuinely streaming bound across an entire singular family. Theorem D.1 below removes the full-future restriction by truncating the resolution flag, and Theorem D.2 proves the fixed-horizon causal

version for the affine class. When $m = 0$ there is no nonconstant future query and the prediction regret is zero; formula (65) is not used in that case.

B.2. Confluent positivity and the physical prediction metric. We give the details of the analytic and geometric steps in Theorem B.1.

The proof of Lemma 4.3 appears in the principal analytic inputs.

Lemma B.2 (Desingularization and observable scales). *The functions in (63), and the needed derivatives in θ , are uniformly bounded on $[0, 1] \times [0, \theta_0]$. The posterior vectors $z_\theta(h)$ therefore lie in a fixed box. For the common product probe menu let $p_\theta(h)$ be its probability vector. There is a fixed matrix G of full column rank such that*

$$(68) \quad p_\theta(h) - p_\theta(\tilde{h}) = G \operatorname{diag}(\theta^{\nu_i})(z_\theta(h) - z_\theta(\tilde{h})).$$

In particular the squared norm on the left is bounded above and below by positive constants times $\sum_i \theta^{2\nu_i} |z_i(h) - z_i(\tilde{h})|^2$, with constants independent of θ .

Proof. The divided difference of order k is the integral of $f^{(k)}(\sum_j w_j x_j)$ over the standard k -simplex, whose volume is $1/k!$. This identity follows by iterating the fundamental theorem of calculus, starting with $(f(y) - f(x))/(y - x) = \int_0^1 f'(x + s(y - x)) ds$. It also gives the repeated-node value. Here $f(x) = t^x$, so every integrand is $t^x (\log t)^k$. At a positive cluster the exponent is bounded below by some $a > 0$ uniformly on the fixed parameter interval. For every fixed j , $\sup_{0 < t \leq 1} t^a |\log t|^j < \infty$. The same domination applies after finitely many θ derivatives. There are no higher divided differences at zero. Posterior expectations of bounded functions are bounded independently of the history.

Newton's interpolation identity, proved by successive subtraction of the interpolating polynomials, is exact at each node:

$$(69) \quad t^{\lambda + \theta \gamma_{\lambda,j}} = \sum_{k=0}^j \theta^k \left(\prod_{v < k} (\gamma_{\lambda,j} - \gamma_{\lambda,v}) \right) \psi_{\lambda,k}^\theta(t).$$

The matrix multiplying $\operatorname{diag}(\theta^k)$ is fixed, triangular within each cluster, and has nonzero diagonal. The standard identities used here are classical divided-difference calculus [12]; the integral proof makes the uniformity at coalescence explicit.

The expansion matrix from the formal pair monomials in $\mathcal{B}_m$ to the probe products has full column rank. To see this, invert the fixed one-step basis change and multiply it m times. Every formal monomial product is a linear combination of the probe products. Identifying equal pairs is a surjective linear quotient, so the spanning statement survives that identification. Both this expansion matrix and the triangular matrix in (69) are independent of θ . Their product has full column rank. Subtracting two normalized posterior vectors removes its constant column and gives (68). The remaining columns are still independent. Their extreme singular values give the two constants. The equality extends to $\theta = 0$ by continuity; vanishing weights then remove precisely the unresolved jets. $\square$

Lemma B.3 (Uniform attainable patch with its evidence). *Under (66), there are $\rho, \beta > 0$, independent of θ , such that the subprobability distribution of z_θ on all-failure prefixes dominates β times uniform probability on a translate of $[-\rho, \rho]^d$.*

Proof. Use the binomial factors $(1 + c_i t^{D_\theta})/2$ of Lemma 3.2, with distinct small c_i and $D_\theta = a_{r-1} + \theta b_{r-1}$. Since the coefficient calibration is fixed, one common interior command tuple implements this family for every θ . Its unnormalized tangent has $n(r-1) + 1$ distinct monomials, also at zero. At positive θ , pairing with the K_m distinct future exponents has rank K_m by

Lemma 3.1. The divided-difference basis change is invertible there. At zero, Lemma 4.3 gives the same rank directly, without passing a rank assertion through a singular limit. The product itself lies in the tangent. The normalization map $v \mapsto v - pe_0v$ has kernel exactly $\text{span}\{p\}$ on this full image, as in (10). Thus the normalized jet map has rank d at every θ , including zero.

Here are uniform local details. Let q be the total number of command coordinates and g_* the common interior tuple. The matrices $J_\theta = D_g z_\theta(g_*)$ are continuous in θ and surjective. Their least row singular value has a positive minimum σ on the compact parameter interval. Choose an orthonormal basis E_θ of the row space and a complementary orthonormal basis F_θ of its kernel, and use coordinates $g = g_* + E_\theta u + F_\theta w$. The derivative at zero of

$$(u, w) \mapsto (z_\theta(g_* + E_\theta u + F_\theta w) - z_\theta(g_*), w)$$

is block diagonal with invertible upper block, least singular value at least σ , and lower block the identity. Its derivative norm and its second derivative have common finite bounds on a fixed neighborhood. This follows from Lemma B.2, finite product rules, and the prefix evidence lower bound κ^n .

Choose a ball inside the command cube on which the derivative differs from its central value by at most half its least singular value. This radius can be chosen uniformly. For a sufficiently small target ball, the map obtained by subtracting the nonlinear remainder and applying the inverse central derivative is a contraction of that command ball into itself. The contraction theorem gives a differentiable inverse. Hence a product of a d -cube and a $(q - d)$ -cube of common positive side length lies in its image, uniformly in θ . All inverse Jacobian determinants are bounded below by a positive constant: the forward derivative is uniformly bounded, so its determinant is uniformly bounded above. The orthogonal command-coordinate change has determinant of absolute value one. No continuity of the chosen orthonormal frames in θ is needed for these uniform bounds.

Uniform commands have density $(1 - 2\eta)^{-q}$. Their joint law with the all-failure word has density

$$(1 - 2\eta)^{-q} \mu \left(\prod_{i=1}^n F_{g_i} \right) \geq (1 - 2\eta)^{-q} \eta^n.$$

Change variables using the inverse just constructed and integrate over the complementary cube. This gives the asserted minorization, with $\beta > 0$ the product of this density bound, the inverse-Jacobian bound, and the two cube volumes. The convention for a zero-dimensional complement is volume one. This is a subprobability bound retaining the failure evidence, not conditioning away a rare prefix. $\square$

B.3. A scale-sensitive finite-state calculation.

Lemma B.4 (Rectangular quantization at every budget). *For $a_1 \geq \dots \geq a_d > 0$ set*

$$e_M(a) = \max_{1 \leq \ell \leq d} \left(\frac{a_1 \cdots a_\ell}{M} \right)^{1/\ell}.$$

A box with these side lengths has an at-most- M -cell rectangular partition of diameter at most $2\sqrt{d}e_M(a)$ in every cell. The optimal mean squared distortion of its uniform distribution using M arbitrary centers is at least $c_d e_M(a)^2$, with $c_d > 0$ depending only on d . Zero side lengths may be omitted.

Proof. Put $e = e_M(a)$ and take $N_i = \max\{1, \lfloor a_i/e \rfloor\}$ equal intervals in coordinate i . If ℓ coordinates satisfy $a_i \geq e$, then $\prod_i N_i \leq \prod_{i \leq \ell} a_i/e^\ell \leq M$. Every side length in a cell is at most $2e$: for $a_i/e \geq 1$ use $\lfloor a_i/e \rfloor \geq a_i/(2e)$, and otherwise use $a_i < e$. Choose an actual reachable

representative in each nonempty cell when covering a reachable subset of the box. The diameter bound remains valid.

For the lower bound project onto the first ℓ coordinates. The projected uniform density is $(a_1 \cdots a_\ell)^{-1}$ on its box. The union of M balls of radius r has probability at most $Mv_\ell r^\ell / (a_1 \cdots a_\ell)$, where v_ℓ is the volume of the unit ball. Taking $r = (a_1 \cdots a_\ell / (2Mv_\ell))^{1/\ell}$ shows that mean squared distance is at least $r^2/2$. Projection cannot increase distance. Apply this for every ℓ and take the smallest of the finitely many positive dimensional constants. Centers need not lie in the box. $\square$

Proof of Theorem B.1. Let $D_\theta = \text{diag}(\theta^{\nu_i})$; this temporary matrix notation is unrelated to the peak dimension $D_A(N)$. By Lemma B.2, $D_\theta z_\theta(h)$ lies in a translated box of side lengths at most $2B\theta^{\nu_i}$ for a fixed B . Lemma B.4 partitions this box into at most M cells with squared diameter at most $C\Phi_\theta(M)$. Choose a reachable representative in each nonempty cell. At the checkpoint, encode the cell of the exact history and decode its representative's probe probabilities. These are probabilities in $[0, 1]$. Equation (68) gives the uniform upper regret bound. Finite cells with deterministic boundary conventions give Borel encoders.

For the lower bound first average private decoder coins conditional on the message and query. Squared loss cannot increase, leaving at most M centers in probe space. Orthogonally project these centers onto the affine space $p_* + \text{range}(G)$; projection cannot increase distance to a true prediction. Invert G on its range. Its lower singular bound reduces the problem to distance from at most M arbitrary centers in the scaled z coordinates. The law there dominates β times the image of the cube from Lemma B.3; its side lengths are $2\rho\theta^{\nu_i}$. The lower half of Lemma B.4 gives $c\Phi_\theta(M)$ after division by the fixed number of queries. Encoder randomization cannot beat distance to the closest center at a fixed history. Fresh public randomness can be fixed and then averaged because it is independent of the entire prefix. At zero, omit zero-weight coordinates in this argument. The same constants, reduced if necessary over the finitely many possible dimensions, work throughout the compact parameter interval. This proves both criteria and the claimed submersion statement. $\square$

Thus the relevant finite-resolution data are the multiplicities of additive collisions and their confluent orders, not only the cardinality of the nearby sumset. The result is a uniform comparison up to constants. It neither asserts a new general quantization principle nor identifies a universal leading distortion constant.

APPENDIX C. FINITE-STATE FILTERING AFTER EVERY REPORT

Fix $N \geq 2$ and abbreviate $d_n = d_A(n, N - n)$ and $D_* = D_A(N) > 0$. The exact state from Theorem 3.4 will be denoted s_n . A clock n , calibration, prior moments and a precomputed finite program are read-only data. The charged resource is the persistent state between successive input symbols.

The resource model and loss are defined in Section 2.

Lemma C.1 (Uniform finite-horizon transition bounds). *On the compact exact state sets there are constants $L_n < \infty$ such that the update $T_n(s, g, x)$ satisfies*

$$\|T_n(s, g, x) - T_n(\tilde{s}, g, x)\| \leq L_n \|s - \tilde{s}\|$$

for every admitted g, x and $s, \tilde{s} \in \mathcal{S}_n$. Every fixed finite probe prediction vector is likewise Lipschitz in the state, with a finite constant independent of the history.

Proof. In factor coordinates an update before changeover appends the new factor $f/\mu f$. The old factor coordinates are unchanged. All unnormalized report probabilities lie in $[\kappa, 1]$; consequently

the normalized factors lie pointwise in $[\kappa, 1/\kappa]$. Their coordinates are bounded, since they are continuous images of a compact cube and finite alphabet. At changeover each needed moment is a ratio of polynomial expressions in these coordinates and the fixed prior moments, with denominator the integral of a product of normalized factors, bounded below by κ^{n+1} . The same lower bound holds on line segments between factor tuples, because each individual interpolated normalized factor is still at least κ . Finite differentiation and the mean value theorem give a uniform derivative bound there.

In moment coordinates the update is (11). Its denominator is at least κ at every state; on a line segment between two such moment vectors the corresponding posterior mixture has the same lower bound. Numerators, coefficients and moment coordinates range over compact sets. The quotient rule therefore gives a uniform bound. Probe predictions are linear in the relevant moment coordinates, or normalized polynomial expressions in the factor coordinates with the same positive denominator control. The argument gives their claimed Lipschitz constants. All constants use only finite algebra and the supplied moments; they may grow with N or deteriorate with κ . $\square$

Theorem C.2 (Finite-state streaming upper bound). *For every $M \geq 1$ there is a deterministic M -state filter, updated after every report, whose prediction regret at every checkpoint and every admitted command-report history is at most*

$$(70) \quad C_N M^{-2/D_*},$$

where $C_N < \infty$ depends on the fixed experiment, prior, probe lists and total horizon, but not on M or the input history. The representatives and read-only transition functions are supplied existentially from the exact model; this is not an effective synthesis theorem from finite-precision data.

Proof. Enclose $\mathcal{S}_n \subset \mathbb{R}^{d_n}$ in $[-B_n, B_n]^{d_n}$. For $d_n > 0$ partition each coordinate interval into $k_n = \lfloor M^{1/d_n} \rfloor$ intervals. In each nonempty cell choose one representative *in* $\mathcal{S}_n$. There are at most $k_n^{d_n} \leq M$ representatives, and assigning a state to its cell representative has error at most

$$\delta_n \leq 2B_n \sqrt{d_n}/k_n \leq 4B_n \sqrt{d_n} M^{-1/d_n} \leq CM^{-1/D_*}.$$

Here C is the maximum of the finite prefactors. Empty cells need no representative. For $d_n = 0$ use the unique state. Deterministic boundary and tie conventions give Borel quantizers Q_n .

The filter stores only the index of a representative $\hat{s}_n$. Its transition is exactly

$$(71) \quad \hat{s}_{n+1} = Q_{n+1}(T_n(\hat{s}_n, g_{n+1}, x_{n+1})).$$

Since each representative is reachable and every report has positive probability, its exact image lies in $\mathcal{S}_{n+1}$. Thus the quantizer is always applied inside its domain and no infeasible moment update is used. The finite representative choices and the functions in (71) are read-only program data. They may be obtained by selecting points in nonempty cells of the compact state sets; no efficient global synthesis or computability of unsupplied real prior moments is asserted.

Let $e_n = \|s_n - \hat{s}_n\|$. On the actual command-report stream, Lemma C.1 gives

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + CM^{-1/D_*}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C$. Induction proves $e_n \leq b_n M^{-1/D_*}$. This explicitly accumulates the errors from *all* previous lossy updates. At a query use the probe predictions of $\hat{s}_n$, which are actual probabilities. If the vector prediction map has Lipschitz constant K_n , (6) bounds regret by $(K_n^2/s_n^q)b_n^2 M^{-2/D_*}$. The maximum of these finitely many constants proves (70). No exact prefix was kept for the changeover: the factor-to-moment computation uses the representative factors already encoded by i_n . $\square$

Theorem C.3 (Matching exponent in one fixed streaming experiment). *Choose n_* attaining $D_* = d_A(n_*, N - n_*)$. For the first n_* trials supply independent uniform commands from $\mathcal{G}$, independent of the parameter, cartridges and filtering coins. The realized command is read once by the filter. At n_* reveal the independent probe query and execute its remaining trials. This is one fixed experiment, known before the filter is chosen. Its optimal unconditional regret over M -state filters satisfies*

$$(72) \quad 0 < c_N M^{-2/D_*} \leq \overline{\mathcal{R}}_{M,N}^{\text{stream}} \leq C_N M^{-2/D_*}.$$

The same power is sharp for the minimax streaming criterion taking the maximum conditional regret over checkpoints and input histories. Fresh private update and decoder randomization does not defeat the lower bound; a fresh public coding seed independent of the complete prefix may also be allowed. Thus for fixed N , the required persistent bits for this prediction task satisfy $B_(\varepsilon) = \frac{D_*}{2} \log_2(1/\varepsilon) + O_N(1)$.*

Proof. Write $n = n_*$, $d = D_*$, and $s = s_n^q$. Theorem 3.3 gives an interior command tuple g_0 where the probe map $p(g)$ on all-failure prefixes has rank d . Select d independent probe coordinates, and let E denote their coordinate projection, of operator norm one. Choose complementary command coordinates w so that $g \mapsto (Ep(g), w(g))$ has an invertible derivative at g_0 . Let Φ be its local inverse. There is a compact product $B_\rho^d \times W$ inside its domain, centered at the projected prediction point, such that Φ stays in the command interior and $|\det D\Phi| \geq j_0 > 0$. The complementary box W has positive volume, with unit-volume convention in zero dimensions.

If $k = Kn$ is the number of real command entries, their density is $(1 - 2\eta)^{-k}$. The joint density of commands and an all-failure prefix is this density times $Z(g) = \mu \prod_i F_{g_i} \geq \eta^n$. Change variables with Φ and integrate over W . The subprobability law of the projected predictions on this prefix dominates β times uniform probability on B_ρ^d , with

$$\beta = \frac{\eta^n j_0 |W| |B_\rho^d|}{(1 - 2\eta)^k} > 0.$$

In particular the lower bound retains the probability of the failure prefix. It does not assign an atom to an exact real command or assume a flat ball in the entire curved prediction image.

Every streaming filter is in particular an M -message encoder at this checkpoint. Given any message, average decoder coins conditional on each query; squared loss can only decrease. This gives at most M prediction vectors. Randomizing the encoder cannot beat the closest of these vectors at a fixed history. Projecting their centers by E cannot increase squared distance. For a uniform point X in a d -ball of radius ρ and any M centers, the union-of-balls estimate gives

$$\Pr\{\min_i \|X - z_i\| \leq r\} \leq \min\{1, M(r/\rho)^d\}.$$

Integrating its complementary tail yields

$$\mathbb{E} \min_i \|X - z_i\|^2 \geq \int_0^{\rho^2 M^{-2/d}} (1 - Mt^{d/2}/\rho^d) dt = \frac{d}{d+2} \rho^2 M^{-2/d}.$$

Multiply by β/s to obtain $c_N > 0$. Fresh public randomness can be fixed and then averaged because it is independent of the prefix. Private online coins may equivalently be drawn on an independent tape for this proof; fixing them makes a deterministic transducer and the same bound holds uniformly. That proof device supplies no tape to the implemented filter. The prescribed exploration is independent of that tape, a condition that is essential to this particular argument. An old command-generation seed is not fresh coding randomness and remains charged.

Theorem C.2 gives the upper bound in the same experiment. The minimax conditional criterion is at least its unconditional regret, while the universal upper filter works at all checkpoints. Finally put $M = 2^B$ and solve the two inequalities. The Euclidean covering mechanism is classical [4]; the identified sparse dimension and the actual repeated-update construction are the experiment-specific inputs. $\square$

Theorem C.4 (A separate finite-state control bound). *For the same finite detector and horizon, suppose stage rewards $r_n(g, x) \in [0, 1]$ are continuous in g for each report, and the objective is their expected sum. There is an M -state feedback controller whose loss relative to the full-history optimum is at most*

$$(73) \quad C'_N M^{-1/D_*}.$$

No matching lower power for arbitrary control tasks is asserted. An additional finite payoff automaton must be counted as task state, multiplying the number of states if it is included.

Proof. Exact dynamic programming on $\mathcal{S}_n$ is valid by Proposition 2.3. With terminal value zero, its recursion is

$$V_n(s) = \max_{g \in \mathcal{G}} \sum_{x \in \mathcal{X}} p_x(s, g) \{r_{n+1}(g, x) + V_{n+1}(T_n(s, g, x))\}.$$

Here p_x is a linear moment functional or a normalized factor expression. The positive denominator bounds, compact action set and continuous rewards make the displayed objective continuous, and the maximum is attained. By backward induction it is uniformly Lipschitz in s , since finite sums, products of bounded Lipschitz functions, and the updates in Lemma C.1 preserve that property. The maximum of functions with a common Lipschitz constant is again Lipschitz. Denote by H_n a uniform Lipschitz constant of the action objective, before its maximization.

At each representative choose an exact maximizing action and store that action in the finite output table. At the true state s_n , the loss from using the action optimal at $\hat{s}_n$ is at most $2H_n e_n$. Update the representative by (71). The true exact state is computed only for analysis along the controller's actual observations; the implemented controller still stores only its index. Bellman telescoping and conditional expectation give total loss at most $2 \sum_{n < N} H_n e_n \leq 2 \sum_{n < N} H_n b_n M^{-1/D_*}$. This proves (73). It uses no continuity of the optimizing action as a function of state, and it does not differentiate an optimizer. Its approximation principle is related to [8]; it is not claimed as a new general dynamic-programming principle. $\square$

C.1. Horizon and calibration remain part of the constants. If every factor in a probe menu satisfies $0 < F \leq v < 1$, then $H_{n,j} \leq v^{N-n}$ and every posterior prediction satisfies the same bound. The one-state decoder that always announces zero therefore has

$$(74) \quad \mathcal{R} \leq v^{2(N-n)}.$$

For the shared physical menu $F = c + \delta k_j$ one may take $v = c + \delta$. Thus a checkpoint implementation may choose, in advance, between the streaming filter and the one-state decoder and obtain

$$\mathcal{R} \leq \min\{C_N M^{-2/D_*}, (c + \delta)^{2(N-n)}\}.$$

At $c = 1/2$, $\delta = 1/8$, $N - n = 20$, the second bound is $(5/8)^{40} < 10^{-8}$. This does not contradict a fixed- N high-resolution exponent. Neither the new sparse rank theorem nor the streaming construction turns it into a horizon-uniform useful-memory law. We have not rescaled the loss or discarded rare histories to avoid this bound.

The upper constants involve state bounds, transition Lipschitz constants and repeated error propagation. The lower constants involve an attainable Jacobian, a chart radius, evidence and the prior pairing. They are not uniform over vanishing detector amplitudes, all full-support

priors or increasing horizons. With a finite fixed command alphabet and finite N there are finitely many histories and an exact finite code eventually suffices; the matching asymptotic lower bound above expressly uses the fixed continuous-command exploration protocol.

APPENDIX D. ATTAINABLE RESOLUTION FLAGS AND UNIFORM CAUSAL MEMORY

The full-future hypothesis in Section B makes every nonconstant future jet attainable. It is not needed for a resolution law, but it cannot simply be omitted from the formula there. When the past is shorter, the relevant list is an initial part of the future resolution flag. The lower bound requires transversality to this flag; the upper bound requires an estimate on the whole, generally curved, attainable image. We establish both and then prove their compatibility with every successive report update.

D.1. Statement of the attainable law. Retain the affine calibration and positivity assumptions of Section B, but do not impose (66). Fix $n, m \geq 1$. There are K_m formal sum pairs in $\mathcal{B}_m$; the zero pair is the constant, and the other $q_m = K_m - 1$ divided-difference coordinates have orders

$$\nu_{m,1} \leq \cdots \leq \nu_{m,q_m}.$$

Order equal orders by increasing limiting exponent, once and for all. Thus every initial part of the list includes, within each cluster, all jets preceding any jet that it contains. Set

$$(75) \quad p_{n,m} = \min\{n(r-1), q_m\}, \quad S_{m,\ell} = \sum_{i=1}^{\ell} \nu_{m,i}, \quad \Phi_{n,m}^{\text{att}}(M, \theta) = \max_{1 \leq \ell \leq p_{n,m}} \theta^{2S_{m,\ell}/\ell} M^{-2/\ell}.$$

Here $0^0 = 1$ and $0^s = 0$ for $s > 0$. If $n = 0$ or $m = 0$, define $\Phi_{n,m}^{\text{att}} = 0$ separately. No division by a zero dimension is then involved. Throughout, constants may depend on the fixed experiment, prior, horizon and calibration interval, but not on M or θ .

Theorem D.1 (Attainable confluent checkpoint law). *For every integer $M \geq 1$ and every calibration in the compact interval of Section B, the unconditional optimal M -message checkpoint regret under independent uniform commands and the minimax checkpoint regret both have order $\Phi_{n,m}^{\text{att}}(M, \theta)$, with positive uniform comparison constants. The upper code has at most M reachable representatives and works on every admitted history. The lower bound permits the independent encoder and decoder randomization specified in Definition 2.4.*

In particular, past limitation truncates the ordered ambient list at $p_{n,m}$; it does not introduce the remaining ambient orders into the rate. At zero the exponent is $2/\min\{n(r-1), |mA_0| - 1\}$, and at each fixed positive calibration it is $2/p_{n,m}$.

For the sequential assertion fix $N \geq 2$ and choose the compact calibration interval so that distinct formal pairs in $\mathcal{B}_m$, $1 \leq m < N$, do not meet at a positive calibration. Such an interval always exists after decreasing its positive endpoint: there are finitely many pairs and their exponent differences are affine. Put

$$(76) \quad \Omega_N(M, \theta) = \max_{1 \leq n < N} \Phi_{n, N-n}^{\text{att}}(M, \theta).$$

Use the same independent uniform command law at every acquisition time. For each selected checkpoint reveal its independent product query only after the index has been formed. As in Section C, only one checkpoint and its remaining query trials are executed in a run. Let $\mathfrak{R}_{M,N}^{\text{av}}$ be the infimum, over one streaming filter, of the maximum of its unconditional checkpoint regrets. Let $\mathfrak{R}_{M,N}^{\text{max}}$ be the corresponding infimum of the maximum conditional regret over checkpoints and histories.

Theorem D.2 (Uniform causal realization of the resolution flag). *In this fixed-horizon experiment there are $0 < c \leq C < \infty$ such that*

$$(77) \quad c\Omega_N(M, \theta) \leq \mathfrak{R}_{M,N}^{\text{av}}(\theta) \leq \mathfrak{R}_{M,N}^{\text{max}}(\theta) \leq C\Omega_N(M, \theta)$$

for every $M \geq 1$ and every admitted θ , including zero. For each known θ and each M , there exists a single deterministic M -state transducer attaining the upper bound after every report. Its codebooks and transitions may depend on θ and M ; the comparison constants do not. No joint-in-calibration measurable or computable selection of codebooks is asserted. It stores only a representative index; its transitions use physical moments, never division by a collision parameter. Read-only prior moments at the formal exponents in $\mathcal{B}_N$ suffice. The lower bound holds in the stated common exploration experiment, including for the independent randomized filters of Definition 2.4.

These are statements about finite persistent memory in the exact-real program model already defined, not about finite-precision synthesis. In particular the number of real coordinates used to describe a representative is not the persistent state count: the latter is the number of representative labels, at most M .

D.2. A global covering estimate. We first record the real-geometric input with its anisotropic consequence. A bounded semialgebraic format means a bound on the number of variables, polynomial relations and their degrees in a defining formula; arbitrary real coefficients are allowed. Quantified formulas of fixed format can also be used, since real quantifier elimination gives a bound depending only on this format. Neither a coefficient norm nor a lower bound on a nonzero coefficient is part of the definition.

The proof of Lemma 4.4 appears in the principal analytic inputs.

The bounded-format hypothesis is essential to a uniform statement of this kind. An unrestricted smooth image could develop arbitrarily many sheets. In the present experiment this hypothesis is a consequence of the finite product likelihood, not an extra regularity assumption on the prior or on an unknown attainable variety.

Lemma D.3 (The whole attainable image has bounded format). *Let $z_\theta(h) \in \mathbb{R}^{q_m}$ be the nonconstant divided-difference moment vector, in the ordered coordinates of (75), and put*

$$D_{m,\theta} = \text{diag}(\theta^{\nu_{m,i}}), \quad S_{n,m}^\theta = \{D_{m,\theta} z_\theta(h) : |h| = n\}.$$

The set $S_{n,m}^\theta$ is compact, has semialgebraic format bounded independently of θ , and has dimension at most $p_{n,m}$. It is contained in a rectangle with sides $[-B\theta^{\nu_{m,i}}, B\theta^{\nu_{m,i}}]$, with B uniform. At zero all positive-order coordinates are zero.

Proof. Fix the finite report word $x_1, \dots, x_n$ and write u for its command entries. Every report factor is affine in its current command. Thus the evidence $Z(u) = \mu \prod_{i=1}^n f_{x_i, g_i}^\theta$ and each numerator

$$N_j(u) = \mu \left(\psi_j^\theta \prod_{i=1}^n f_{x_i, g_i}^\theta \right)$$

are polynomials of degree at most n in u . Their coefficients are real prior integrals, well defined also at zero. The moment image is specified by the existence of u in its fixed cube with $Z(u)y_j = \theta^{\nu_{m,j}} N_j(u)$ for every j . Moreover $Z(u) \geq \kappa^n > 0$. This is a semialgebraic formula of fixed format, uniformly in its real coefficients. Finite union over the report words preserves a format bound, and continuity on compact command cubes proves compactness. There is no requirement that these real coefficients, as functions of θ , be semialgebraic.

For the dimension bound, divide each report factor by its prior integral. Its coefficient vector lies in the affine hyperplane $\mu f = 1$ of the r -dimensional one-step space. The constant monomial

allows its constant coefficient to be eliminated, leaving at most $r - 1$ coordinates. The product posterior and all displayed moments are rational functions of these $n(r - 1)$ coordinates. A rational map cannot increase semialgebraic dimension. Multiplication by $D_{m,\theta}$ cannot increase it either; the ambient dimension is q_m . Finally the uniform bounds for ψ_j^θ in Lemma B.2 give $|z_j| \leq B$ before any command integration: they are expectations under probability measures. This proves the rectangle assertion without an inverse power of the prefix evidence. $\square$

D.3. Transversality to every initial resolution flag.

Lemma D.4 (Attainment of the truncated flag). *At one common interior binomial command tuple, projection of $z_\theta(h)$ onto its first $p_{n,m}$ coordinates has full row rank for every calibration, including zero. Under independent uniform commands, the all-failure subprobability distribution of this projection dominates β times uniform probability on a translated cube of side 2ρ , where $\beta, \rho > 0$ are independent of the calibration.*

Proof. Choose distinct sufficiently small $c_1, \dots, c_n$ and realize the factors $(1 + c_i t^{D_\theta})/2$ by the fixed right inverse of the calibration matrix, as in Lemma 2.1. Their commands stay in a common interior neighborhood. Lemma 3.2 identifies their unnormalized tangent with a monomial space of dimension $n(r - 1) + 1$, for every calibration, including zero.

Adjoin the constant to the selected $p = p_{n,m}$ future tests. Within each limiting exponent cluster, the selected tests are exactly an initial block of its divided differences. At a positive calibration this block is an invertible linear transformation of the monomials at its first nodes. Their union consists of distinct monomials. The mixed pairing with the binomial tangent therefore has column rank $p + 1$, by Lemma 3.1. At zero these same blocks become complete initial logarithmic-jet blocks. Lemma 4.3 gives column rank $p + 1$ directly at zero. This is not an inference of rank at zero from rank on a punctured interval.

Let L be this pairing, let F be the product at the witness, and write $v = LF/\mu F$, with constant coordinate $e_0 v = 1$. The normalized differential has image

$$(\mu F)^{-1}\{LQ - v e_0 LQ : Q \text{ in the product tangent}\}.$$

Since L is onto $\mathbb{R}^{p+1}$, the bracket is precisely the p -dimensional hyperplane $e_0 = 0$. Deleting the constant proves the asserted row rank. The remaining, unselected future coordinates need not be independent and are not included in this submersion assertion.

For clarity we give the uniformity and probability argument. On the compact calibration interval, the least row singular value of the selected derivative has a positive minimum, by the rank just proved and continuity of all divided-difference derivatives. The second command derivatives have a common bound, since the factors stay positive and the test functions are uniformly bounded. Complete the row coordinates with orthonormal kernel coordinates. On a fixed small command ball the resulting square derivative differs from its value at the center by at most half its uniformly positive least singular value. The contraction proof of the inverse theorem gives a common product box in the selected output and complementary coordinates. Shrink it so its inverse stays in the command interior. Uniform forward derivative bounds give a positive lower bound for the absolute inverse Jacobian. These estimates use only orthogonality of the chosen frames; a continuous choice of frames is unnecessary.

If k is the number of command entries, their joint density is $(1 - 2\eta)^{-k}$. The density of commands together with the all-failure word is this density multiplied by its evidence, at least η^n . Change variables using the local inverse and integrate over the common complementary box. The resulting measure dominates a fixed positive multiple of Lebesgue measure on the selected p -cube. Absorb the cube volume into β . This proves the stated subprobability minorization; it does not condition away the probability of the failure word. A full-support

prior without a density is permitted throughout, since the continuous density used here is that of the commands. $\square$

Proof of Theorem D.1. The physical prediction identity of Lemma B.2 reads, after the fixed coordinate permutation,

$$(78) \quad p_\theta(h) - p_\theta(\tilde{h}) = G_m D_{m,\theta} (z_\theta(h) - z_\theta(\tilde{h})),$$

where G_m has full column rank and is independent of θ . Apply Lemmas 4.4 and D.3 to the complete attainable image. Since $0 \leq \theta \leq 1$, its ordered side lengths are $B\theta^{\nu_{m,i}}$. Equation (13) supplies at most M reachable representatives with squared covering radius at most $C\Phi_{n,m}^{\text{att}}$. The fixed norm of G_m and the finite query average convert this into the required uniform prediction upper bound. This is a cover of all report strata, not just the local patch used in the lower bound.

For that lower bound, first average decoder randomization for each message and query. The outputs form at most M centers. Orthogonally project them to the affine span $p_* + \text{im } G_m$ and apply the fixed left inverse of G_m . Distances can decrease only by a fixed factor. Project further onto the first $p_{n,m}$ coordinates. Lemma D.4 then gives a subprobability multiple of uniform measure on a translated rectangle of side lengths $2\rho\theta^{\nu_{m,i}}$, $i \leq p_{n,m}$. The lower half of Lemma B.4, projecting onto each first ℓ axes, gives $c\Phi_{n,m}^{\text{att}}$. Centers may be outside the image; randomized assignment cannot improve on the nearest center. Fixing and averaging an independent public coding seed gives the same bound. At zero, omit the zero-width axes. There are $\min\{n(r-1), |mA_0| - 1\}$ surviving axes, which also proves the stated limiting exponent. For positive calibration the last power $M^{-2/p_{n,m}}$ dominates as $M \rightarrow \infty$. The unconditional lower bound is also a minimax lower bound, completing the proof. $\square$

D.4. Physical moments and successive updates. For $b = (\lambda, \gamma) \in \mathcal{B}_m$, write

$$v_b(h) = \mu(P_h t^{\lambda+\theta\gamma}), \quad v_{(0,0)} = 1.$$

We keep the formal labels even if the functions coincide at zero. Let $v^{(m)}$ denote the nonconstant vector. Newton interpolation within each cluster, as in Lemma B.2, gives a fixed invertible matrix T_m with

$$(79) \quad v^{(m)} = T_m D_{m,\theta} z_\theta.$$

The constant cluster has no other member, so no deleted constant mixes with this equation. Both T_m and its inverse have fixed finite norms. At zero the equation still holds with the appropriate repeated raw moments. It is the analysis of this equation, not inversion of its vanishing diagonal, that identifies the metric scales.

Lemma D.5 (Uniform updates in the physical metric). *The map from reachable raw moments at remaining length m to those at length $m-1$, after a command and a report, is uniformly Lipschitz in the raw Euclidean metric, independently of calibration. Every reachable representative maps to a reachable next vector.*

Proof. Write $e_i = (a_i, b_i)$ and the current report factor as $f = \sum_i f_i(g, x) t^{a_i + \theta b_i}$. Its coefficients are uniformly bounded, and $f \geq \kappa > 0$. Addition of formal pairs gives $b + e_i \in \mathcal{B}_m$ whenever $b \in \mathcal{B}_{m-1}$. Consequently the exact update is

$$(80) \quad v'_b = \frac{\sum_i f_i(g, x) v_{b+e_i}}{\sum_i f_i(g, x) v_{e_i}}, \quad b \in \mathcal{B}_{m-1}.$$

Every denominator is the posterior expectation of f , hence at least κ . A line segment between reachable raw vectors represents a mixture of their posterior probability measures, so the same

bound holds along the segment. All numerators, coefficients and coordinates have uniform bounds. The quotient rule and the mean value theorem prove the Lipschitz assertion. No exponent separation occurs in a denominator. The formula is valid also when formal labels have equal values. Finally a history realizing the representative, followed by the specified input, realizes its updated vector; positivity admits every report. $\square$

Proof of Theorem D.2. For each $1 \leq n < N$, Lemmas 4.4 and D.3, followed by the fixed matrix T_{N-n} , give a raw-moment quantizer Q_n with at most M reachable representatives and radius

$$\delta_n \leq C_n \sqrt{\Phi_{n,N-n}^{\text{att}}(M, \theta)} \leq C_0 \sqrt{\Omega_N(M, \theta)}.$$

The maximum C_0 is over finitely many stages. At time zero the prior moment vector is one fixed state; at time N the remaining state is constant. Initialize at the prior vector and implement

$$\hat{v}_{n+1} = Q_{n+1}(U_n(\hat{v}_n, g_{n+1}, x_{n+1})),$$

where U_n is (80). The transition reads the old representative index and the current input only. Reachability makes its argument an element of the next quantizer's domain. The finite codebooks and these real-valued functions are read-only program data.

Let $e_n = \|v_n - \hat{v}_n\|$. With $e_0 = 0$, Lemma D.5 yields

$$e_{n+1} \leq L_n e_n + C_0 \sqrt{\Omega_N(M, \theta)}.$$

Set $b_0 = 0$ and $b_{n+1} = L_n b_n + C_0$. Induction gives $e_n \leq b_n \sqrt{\Omega_N(M, \theta)}$ after every update. Each query prediction is a linear form with fixed coefficients in the raw formal moments. Its Lipschitz constant is therefore uniform, and (6) proves the last inequality in (77). There is no exact-prefix recomputation: all intermediate vectors in this implementation are functions of a finite representative label and the current command-report pair.

For every streaming filter, its index at checkpoint n is an M -message encoder of the prefix. Theorem D.1 therefore bounds that checkpoint's unconditional regret below by $c_n \Phi_{n,N-n}^{\text{att}}$. The commands in all these assertions are prefixes of the same independent uniform acquisition rule. Taking the maximum over n , and then the infimum over filters, gives the first inequality with $c = \min_n c_n > 0$. Independent filtering coins can be fixed for this argument because they do not generate the acquisition commands; averaging restores the randomized statement. Finally a maximum conditional risk dominates its unconditional average, which proves the middle inequality. $\square$

Corollary D.6 (A uniform bit profile). *Let $B_N(\varepsilon, \theta)$ be the least number of persistent bits needed to make the minimax streaming regret at most ε . For $0 < \varepsilon < 1$, uniformly in the admitted calibration,*

$$(81) \quad B_N(\varepsilon, \theta) = \max \left\{ 0, \max_{\substack{1 \leq n < N \\ 1 \leq \ell \leq p_{n,N-n}}} \left(\frac{\ell}{2} \log_2 \frac{1}{\varepsilon} - S_{N-n,\ell} \log_2 \frac{1}{\theta} \right) \right\} + O(1).$$

At zero retain only terms with $S_{N-n,\ell} = 0$. The same formula holds for the maximum unconditional criterion.

Proof. For positive θ , the condition $\Omega_N(M, \theta) \leq \varepsilon$ is equivalent to all inequalities $M \geq \theta^{S_{N-n,\ell} \varepsilon^{-\ell/2}}$. Taking the largest right-hand side, adjoining $M \geq 1$, and then taking base-two logarithms gives the expression. The two fixed comparison constants in (77) change it by a bounded amount, since the possible ℓ are bounded by the fixed horizon. Rounding to a power of two changes it by at most one further bit. The zero-calibration statement follows by omitting vanishing terms. $\square$

D.5. A past-limited peak with two weak directions. The next consequence answers the concrete seven-trial boundary in the source-pinned v7 technical note [19]. It is a consequence of the attainable-image argument, rather than an extrapolation of the full-future formula.

Corollary D.7 (Seven-trial resolved memory). *For the cells (83), their fixed command cube and the product probes (84), take total horizon seven. For the two streaming criteria above, uniformly for $0 \leq \theta \leq 1/2$ and $M \geq 1$,*

$$(82) \quad \mathfrak{R}_{M,7}(\theta) \asymp \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The lower bound already holds at checkpoint four under the same uniform exploration. The crossover budget has order θ^{-6} and the crossover regret has order θ^2 .

Proof. Put $D = 2 + \theta$. For positive $\theta \leq 1/2$, the ten distinct three-fold exponents are

$$0, 1, 2, D, 3, D+1, D+2, 2D, 2D+1, 3D.$$

At zero their limiting clusters have multiplicities 1, 1, 2, 2, 2, 1, 1 at exponents 0, 1, 2, 3, 4, 5, 6. Thus the nonconstant order list has six zeros and three ones. At $n = 4, m = 3$, however, $p_{4,3} = 8$, and the selected list has six zeros and only two ones. Theorem D.1 gives

$$\Phi_{4,3}^{\text{att}} = \max\{M^{-1/3}, \theta^{2/7} M^{-2/7}, \theta^{1/2} M^{-1/4}\} = \max\{M^{-1/3}, \theta^{1/2} M^{-1/4}\}.$$

The middle term is the product of the first term to power 3/7 and the last to power 4/7, so cannot exceed their maximum. The minorized eight-dimensional flag uses, at zero, the tests

$$1, t, t^2, t^2 \log t, t^3, t^3 \log t, t^4, t^5, t^6$$

paired with the binomial tangent $\mathcal{P}_8$. Strict confluent positivity proves its rank for every full-support prior. The third ambient weak jet is not counted as an additional attainable dimension.

For completeness the extension of the upper bound to the whole stated calibration interval does not require excluding positive collisions in longer, unused sumsets. At stages $n = 1, 2, 3$, the raw reachable images have dimension at most $2n \leq 6$ and bounded format, and all raw coordinates lie in $[0, 1]$. Lemma 4.4 gives squared covering errors at most $CM^{-1/3}$ uniformly, whether or not some formal raw coordinates coincide. At stage four use the preceding anisotropic cover, valid throughout $[0, 1/2]$ because its ten exponents are distinct at every positive calibration in that interval. At stage five, $m = 2$ and the full-future cover has squared radius at most $C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\} \leq CM^{-1/3}$. Stage six has two coordinates and squared covering radius at most CM^{-1} . Formula (80) remains valid with all formal labels at every stage, even when values coincide. Its uniform Lipschitz estimate and the repeated-quantization argument prove the upper bound in (82) on the entire interval. The stage-four checkpoint lower bound proves the other direction.

Equating the two terms in (82) gives $M = \theta^{-6}$ and common order θ^2 . These are comparison orders, not a numerical integer threshold. At fixed positive calibration the rate is $M^{-1/4}$, consistently with eight attainable dimensions; the ambient nine-dimensional rate $M^{-2/9}$ does not occur. $\square$

APPENDIX E. A UNIFORM STREAMING LAW THROUGH AN ADDITIVE COLLISION

We apply the confluent law to a family in which exact dimension changes at a fixed horizon and at a fixed positivity margin. In this family all repeated quantization errors can be controlled in the physical prediction metric, uniformly at the collision.

E.1. One fixed five-trial protocol. Let $0 \leq \theta \leq 1/2$, $A_\theta = \{0, 1, 2 + \theta\}$, and let μ be any fixed full-support probability on $[0, 1]$. Define

$$(83) \quad \begin{aligned} k_1^\theta &= \frac{1}{3} + \frac{1}{12}t, & k_2^\theta &= \frac{1}{3} + \frac{1}{12}t^{2+\theta}, \\ k_0^\theta &= \frac{1}{3} - \frac{1}{12}(t + t^{2+\theta}). \end{aligned}$$

These cells are at least $1/6$, sum to one, and span W_{A_θ} . Use the fixed comparator cube $[1/4, 3/4]^3$; every report factor is at least $\kappa = 1/24$. The three fixed probe commands have failure factors

$$(84) \quad F_j^\theta = \frac{1}{2} + \frac{1}{8}k_j^\theta, \quad j = 0, 1, 2.$$

The command accepts cell j with probability $3/8$ and the other cells with probability $1/2$. Their sum is $13/8$, so these factors span W_{A_θ} . At a checkpoint n use the 3^{5-n} ordered product queries of length $5 - n$. Exactly one checkpoint and one query are used in a run; all five trials are counted.

For the unconditional experiment the first three commands are independent uniform vectors in the comparator cube. After their reports have been compressed, reveal the independent uniform two-step query. The controller knows θ and μ as read-only calibration, but has access to past commands and reports only through its persistent index. The input law and all query labels are independent of θ .

For a history write $M_s = \mu(P_h t^s)$ and use the five coordinates

$$(85) \quad w = (M_1, M_2, M_{3+\theta}, M_{4+2\theta}, M_{2+\theta} - M_2).$$

They determine every length-two future law. They are ordinary posterior expectations of physical tests and their differences. For analysis put

$$(86) \quad z = (w_1, w_2, w_3, w_4, w_5/\theta) \quad (\theta > 0), \quad z_5|_{\theta=0} = \mu(P_h t^2 \log t).$$

This fifth coordinate at zero is an analytic continuation, not an extra observable supplied to the filter. The zero-calibration future law uses only the first four coordinates.

Theorem E.1 (Sharp two-parameter streaming order). *Set*

$$(87) \quad \Psi_\theta(M) = \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}, \quad M \geq 1,$$

where the second term is zero at $\theta = 0$. There are constants $0 < c \leq C < \infty$, depending on the fixed prior and protocol but not on M or $\theta \in [0, 1/2]$, with the following properties.

There is a deterministic M -state filter, updated after every report, whose conditional prediction regret at every checkpoint and every admitted history is at most $C\Psi_\theta(M)$. In the prescribed three-trial exploration experiment, every M -state filter has unconditional regret at least $c\Psi_\theta(M)$. The same lower bound therefore holds for the minimax criterion over histories and checkpoints. The lower assertion permits the independent randomization of Definition 2.4.

In particular the two regimes are

$$(88) \quad \overline{\mathcal{R}}_{M,5}^{\theta,\text{stream}} \asymp \begin{cases} M^{-1/2}, & 1 \leq M \leq \theta^{-4}, \\ \theta^{2/5} M^{-2/5}, & M \geq \theta^{-4}, \end{cases}$$

with uniform comparison constants. This is an order-sharp crossover, not a claim of an exact transition budget or a leading distortion constant. The family and its base-model perturbation estimate were introduced in the source-pinned technical note [14]. The conclusion here is the matching, uniform two-parameter law and its online implementation.

E.2. Attainment of the fifth desingularized direction.

Lemma E.2 (A nondegenerate limiting patch). *At a common interior three-command tuple the map to z in (86) has rank five for every $\theta \in [0, 1/2]$. Its all-failure subprobability law dominates a fixed positive multiple of uniform probability on a translated five-dimensional cube of fixed side length, uniformly in θ .*

Proof. The formal two-fold sum pairs give exponents

$$0, 1, 2, 2 + \theta, 3 + \theta, 4 + 2\theta.$$

There is just one double cluster, at exponent two. Its divided difference is $(t^{2+\theta} - t^2)/\theta$. The nonconstant orders are $(0, 0, 0, 0, 1)$. The condition in (66) is $5 \leq 3(3 - 1) = 6$.

For an explicit common attainable tuple take $c_1 = 1/64$, $c_2 = 2/64$, $c_3 = 3/64$ and failure factors $(1 + c_i t^{2+\theta})/2$. They are realized by rejection probabilities

$$(1 - g_0, 1 - g_1, 1 - g_2) = (1/2 - 2c_i, 1/2 - 2c_i, 1/2 + 4c_i),$$

strictly inside $[1/4, 3/4]^3$. At zero their product tangent is the entire polynomial space of degree at most six. The future confluent space has basis

$$1, \quad t, \quad t^2, \quad t^2 \log t, \quad t^3, \quad t^4.$$

Lemma 4.3 makes its mixed pairing with that tangent surjective. Normalization removes one dimension, not two. For example, for the uniform prior, the minor with rows t^i , $0 \leq i \leq 5$, is exactly

$$(89) \quad \det \left[\frac{1}{i+1}, \frac{1}{i+2}, \frac{1}{i+3}, -\frac{1}{(i+3)^2}, \frac{1}{i+4}, \frac{1}{i+5} \right]_{i=0}^5 = \frac{1}{338751673344000000} > 0.$$

This rational identity is only a diagnostic of the preceding all-prior argument.

For every positive $\theta \leq 1/2$ the six displayed exponents are distinct and the binomial tangent has seven distinct exponents. Strict mixed-moment positivity gives surjectivity throughout this interval. All derivatives in the desingularized coordinates extend continuously to zero, as proved in Lemma B.2. Compactness and the uniform inverse construction in Lemma B.3 therefore apply on the *whole* interval $[0, 1/2]$, not only on an unspecified punctured neighborhood. The exact change of order in (86) is a fixed permutation of the general jet coordinates and changes no rank or minorization conclusion. $\square$

E.3. Uniform causal quantization.

Proof of Theorem E.1. At times one and two retain, exactly for the moment, the chronological normalized factors $f_i/\mu f_i$, with two coefficient coordinates per factor. At time three change to w in (85). At time four retain $(M_1, M_{2+\theta})$, and at time five use the unique terminal state. These states are sufficient for the remaining observable future law. All their reachable sets are compact.

The first two factor sets lie in fixed boxes of dimensions two and four, uniformly in θ . Indeed report coefficients are uniformly bounded, $\mu f \geq \kappa$, and the coefficient hyperplane $\mu f = 1$ can be coordinatized by the two nonconstant coefficients, its constant coefficient then being $1 - \sum_{a>0} f_a \mu(t^a)$. The required prior moments are fixed read-only data for each known calibration. Uniform boundedness of these affine formulas on $[0, 1/2]$ is immediate.

At time three the first four coordinates of w belong to $[0, 1]$. For $0 < t \leq 1$, the fundamental theorem of calculus in the exponent gives

$$|t^{2+\theta} - t^2| \leq \theta t^2 |\log t| \leq \theta.$$

This extends to zero. Consequently $-\theta \leq w_5 \leq 0$ for every posterior, with no evidence denominator in this bound. At time four both retained moments lie in $[0, 1]$. Lemma B.4, with reachable representatives in nonempty cells, now gives at most M representatives at each time and squared covering errors

$$(90) \quad \delta_1^2 \leq C_1 M^{-1}, \quad \delta_2^2 \leq C_2 M^{-1/2}, \quad \delta_3^2 \leq C_3 \Psi_\theta(M), \quad \delta_4^2 \leq C_4 M^{-1}.$$

The C_i are independent of θ . At zero simply omit the last, constant coordinate of the third state. Every bound in (90) is at most $C_0 \Psi_\theta(M)$.

It remains to verify that repeated updates do not divide by θ . The factor-to-factor update appends the current normalized factor. The factor-to- w update uses the product of the two retained factors and the new report factor. It is a ratio of polynomial coefficient expressions and bounded prior integrals. Along line segments between factor tuples every interpolated normalized factor is at least κ , so the evidence denominators and derivatives have positive uniform bounds. Each physical test in w is uniformly bounded; differentiating in factor coordinates therefore gives a uniform Lipschitz constant for this changeover.

For the only nontrivial shrinking update, write the new report factor as $f = a + bt + ct^{2+\theta}$. In terms of w set

$$Z_f = a + bw_1 + c(w_2 + w_5).$$

The next two moments are exactly

$$(91) \quad M'_1 = \frac{aw_1 + bw_2 + cw_3}{Z_f}, \quad M'_{2+\theta} = \frac{a(w_2 + w_5) + bw_3 + cw_4}{Z_f}.$$

Coefficients are uniformly bounded and $Z_f \geq \kappa$. A line segment between reachable w vectors represents a posterior mixture, so the same lower bound holds along that segment. The quotient rule proves a uniform Lipschitz estimate in the *physical* w coordinates. There is no divided difference and no factor $1/\theta$ in (91). The next, terminal update is constant.

At each time the filter stores only a representative index and applies the exact transition to that representative and the current input, then requantizes. A reachable representative followed by any admitted report has a reachable image, so every quantizer is used inside its domain. If e_n is the state error in the appropriate Euclidean physical coordinates, the preceding bounds give

$$e_0 = 0, \quad e_{n+1} \leq L_n e_n + C_0^{1/2} \Psi_\theta(M)^{1/2},$$

with L_n independent of θ . Iterate this recurrence through the five times. At times one and two each query prediction is a normalized polynomial expression in retained factors; at times three and four it is a fixed linear expression in the retained physical moments. These prediction maps have uniform Lipschitz constants, again by the positive evidence bounds. Squaring the resulting prediction error proves the conditional upper bound at every checkpoint. The representatives and transition functions are supplied existentially from the exact model, not synthesized from finite-precision exponent data. Only their index is persistent mutable memory.

For the lower bound at time three apply Lemma E.2 and the physical-metric argument in the proof of Theorem B.1, now to the nine common queries in (84). Their coefficient change from any other fixed spanning basis is invertible before taking products, so its full product expansion has constant full column rank. Four physical directions have scale one and the last has scale θ . Lemma B.4 gives the maximum of the four-dimensional lower bound $cM^{-1/2}$ and the five-dimensional lower bound $c\theta^{2/5}M^{-2/5}$. The constants are uniform on $[0, 1/2]$. Every streaming filter is an M -message encoder at this time, and the retained all-failure evidence is at least $(1/4)^3$. The randomization argument of Theorem B.1 applies unchanged. Finally,

minimax regret is at least the unconditional regret in this one prescribed experiment. This completes the proof. $\square$

Corollary E.3 (Uniform bit law and the lower-constant scale). *For sufficiently small $\varepsilon > 0$, the least persistent bits for unconditional regret at most ε in the five-trial protocol, and likewise for its checkpoint-uniform minimax criterion, satisfy*

$$(92) \quad B_*(\varepsilon, \theta) = \log_2 \max\{\varepsilon^{-2}, \theta\varepsilon^{-5/2}\} + O(1),$$

where the $O(1)$ is uniform in $\theta \in [0, 1/2]$. If

$$c_{\text{opt}}(\theta) = \inf_{M \in \mathbb{N}, M \geq 1} M^{2/5} \overline{\mathcal{R}}_{M,5}^{\theta, \text{stream}},$$

then $c_{\text{opt}}(\theta) \asymp \theta^{2/5}$ for $0 < \theta \leq 1/2$, with constants independent of θ .

Proof. Solve the two necessary and sufficient inequalities from Theorem E.1 for M , obtaining, up to constant factors, $M \geq \varepsilon^{-2}$ and $M \geq \theta\varepsilon^{-5/2}$. Rounding to $M = 2^B$ changes B by a bounded amount. The same bounds apply to the minimax criterion. The lower bound for c_{opt} follows from the five-dimensional term. For its upper bound let M tend to infinity in the matching upper estimate, so $M^{-1/10}$ disappears. No limiting leading quantization constant is needed. $\square$

The exact profiles remain $(0, 2, 4, 4, 2, 0)$ at zero and $(0, 2, 4, 5, 2, 0)$ at positive θ . The new assertion is that the extra direction is resolved at distortion scale θ^2 and state-budget scale θ^{-4} , with matching estimates on both sides. It sharpens the base-model bound $C(M^{-1/2} + \theta^2)$ without contradicting it.

APPENDIX F. THE VALUE OF A RESOLVED DIRECTION IN ONE SEQUENTIAL DECISION

We next use the same exploration histories, the same two-step query menu and the same eventual Bernoulli event as in Section E. Only the terminal decision is changed. This gives a common-payoff comparison between keeping and erasing the newly resolved direction, rather than comparing two separately normalized prediction tasks.

F.1. An independently priced observable ticket. After the time-three state has been formed, reveal the query J and a price U , independent and uniform on $[0, 1]$. The controller chooses a binary action $A \in \{0, 1\}$. Execute the two query trials and let Y be the same all-failure indicator as before. The payoff is

$$(93) \quad A(Y - U).$$

The price and the query are new inputs, not historical storage. The payoff belongs to $[-1, 1]$ and depends only on observable apparatus outcomes and the chosen action. Accepting the ticket changes the payoff, not the detector law or the five-trial budget. The exploration command program is prescribed; it is not optimized by the controller.

Let $p_j(h) = \mathbb{P}(Y = 1 \mid h, J = j)$. Write V_*^θ for the full-history optimal expected payoff and V_M^θ for the supremum over all M -state streaming policies, including randomized transitions and terminal actions. At the terminal decision the policy reads only its index, the new query and the new price.

Proposition F.1 (Exact decision–prediction identity). *For every fixed streaming encoder with index I , its optimal terminal rule accepts exactly when $U \leq \mathbb{E}[p_J(h) \mid I, J]$. Its value gap is*

$$(94) \quad V_*^\theta - V^\theta(I) = \frac{1}{2s} \sum_{j=1}^s \mathbb{E} \left[(p_j(h) - \mathbb{E}[p_j(h) \mid I])^2 \right], \quad s = 9.$$

Consequently

$$(95) \quad c\Psi_\theta(M) \leq V_*^\theta - V_M^\theta \leq C\Psi_\theta(M)$$

with positive constants independent of M and $\theta \in [0, 1/2]$.

Proof. Conditional on the full history and $J = j, U = u$, action one has expected payoff $p_j - u$ and action zero has payoff zero. The optimum is the threshold rule and its average over price is $\int_0^{p_j} (p_j - u) du = p_j^2/2$. Given only an index, replace p_j by $\bar{p}_{I,j} = \mathbb{E}[p_j | I]$, since price and query are independent of the prefix and encoder. A randomized action is a convex combination of the two payoffs and cannot improve the threshold optimum. Thus

$$V_*^\theta = \frac{1}{2s} \sum_j \mathbb{E} p_j^2, \quad V^\theta(I) = \frac{1}{2s} \sum_j \mathbb{E} \bar{p}_{I,j}^2.$$

Conditional-variance decomposition gives (94). Conditional expectation is also the optimal squared-loss decoder for the same encoder. Taking the infimum over the same streaming encoders therefore makes the optimal value gap exactly one half the optimal prediction regret. Theorem E.1 proves (95). The upper implementation may also threshold the representative's prediction directly: conditional on a history, integrating the mistaken decisions over the interval between that prediction and p_j gives half their squared difference. $\square$

This proposition is a decision identity, not a new general comparison principle. Its use here is to retain the entire resolved streaming law in a bounded, binary-action task and then compare a specific erased observation in that *same* task.

F.2. Erasing the weak coordinate. Let

$$(96) \quad T_\theta(h) = (M_1, M_2, M_{3+\theta}, M_{4+2\theta})$$

be the first four coordinates of (85). This is a deterministic, calibration-dependent postprocessing of the history. It is not a coarsening of a raw one-step detector symbol, and it is not claimed to be the best possible four-dimensional encoding. Let V_T^θ be the optimal payoff of a terminal policy given the *entire exact* statistic T_θ , with no memory restriction, and then the same J, U . This comparison gives the erased controller more computational and storage power than any M -message compression of T_θ .

Theorem F.2 (Uniform value of the weak direction). *There are constants $c_e, C_e, K > 0$, depending on the fixed prior and protocol but independent of $0 < \theta \leq 1/2$, such that*

$$(97) \quad c_e \theta^2 \leq V_*^\theta - V_T^\theta \leq C_e \theta^2.$$

Moreover, whenever $M \geq K\theta^{-4}$,

$$(98) \quad V_M^\theta - V_T^\theta \geq \frac{1}{2} c_e \theta^2 > 0.$$

In particular an M -state streaming policy beats every policy whose only historical input is an M -message compression of T_θ , even though V_T^θ allows that comparator uncompressed access to the four coordinates. Both values concern the same actual calibration, exploration law, query, price, target, payoff and five-trial budget. At $\theta = 0$ the statistic T_0 is sufficient and $V_^0 = V_T^0$.*

Proof. Put $Z_\theta = w_5/\theta$ for positive θ , with its continuous value at zero as in (86). The common product query expansion has the form

$$(99) \quad p_j(h) = a_j + \ell_j(T_\theta(h)) + \theta b_j Z_\theta(h),$$

where a_j, b_j and the linear forms ℓ_j are independent of θ . They are obtained by expanding the two factors in (84) into the six formal monomials and replacing $M_{2+\theta}$ by $M_2 + \theta Z_\theta$. The full

coefficient matrix has rank six, as proved in Lemma B.2. Hence $\sum_j b_j^2 > 0$; otherwise its weak column would be zero.

Apply the conditional-variance identity from Proposition F.1 to the sigma-algebra generated by T_θ . With $b_* = (2s)^{-1} \sum_j b_j^2 > 0$ one obtains the exact formula

$$(100) \quad V_*^\theta - V_T^\theta = b_* \theta^2 \mathbb{E} \text{Var}(Z_\theta \mid T_\theta).$$

The random variable Z_θ is bounded uniformly, since $|(t^{2+\theta} - t^2)/\theta| \leq t^2 |\log t| \leq 1$ before taking any posterior expectation. Thus the conditional variance is uniformly bounded above.

For the lower bound, Lemma E.2 gives a fixed positive subprobability multiple β of uniform measure on a translated cube of side 2ρ in the coordinates (T_θ, Z_θ) . For every Borel function v of the first four coordinates,

$$\mathbb{E}|Z_\theta - v(T_\theta)|^2 \geq \beta \int |z - v(t)|^2 d \text{Unif}_{\text{cube}}(t, z) \geq \beta \rho^2 / 3.$$

The last inequality integrates the variance of the independent uniform last coordinate; it is unaffected by the translation. Taking the infimum over v proves $\mathbb{E} \text{Var}(Z_\theta \mid T_\theta) \geq \beta \rho^2 / 3$. This argument uses domination of an actual distribution, not an assumed uniform posterior on a formal tangent space. Equation (100) proves (97) with uniform positive constants.

By (95),

$$V_M^\theta - V_T^\theta \geq c_e \theta^2 - C \max\{M^{-1/2}, \theta^{2/5} M^{-2/5}\}.$$

For $M \geq K\theta^{-4}$ the maximum on the right is at most $\theta^2 \max\{K^{-1/2}, K^{-2/5}\}$. Choose $K \geq 1$ large enough that $C \max\{K^{-1/2}, K^{-2/5}\} \leq c_e/2$. This proves (98). A policy observing only a compression of T_θ cannot outperform the one observing all of T_θ : conditional on that statistic its randomization is independent of the latent history, and averaging produces an admissible randomized T_θ -policy. Finally at zero the weak column in (99) vanishes, proving sufficiency and equality of the full and erased values. $\square$

The absolute advantage vanishes quadratically as the exponents collide. What remains nondegenerate is the advantage after normalization by the actual weak-direction scale θ^2 , once the state budget resolves that scale. The theorem is not a lower bound for arbitrary exploration objectives. It is an optimal-policy comparison in one prescribed sequential acquisition experiment, with no change of Bayes baseline and no substitution of the separate mechanical sign experiment.

Corollary F.3 (Necessary and sufficient budget order). *Define*

$$M_{\text{beat}}(\theta) = \min\{M \in \mathbb{N} : V_M^\theta > V_T^\theta\}.$$

For $0 < \theta \leq 1/2$ this minimum exists and $M_{\text{beat}}(\theta) \asymp \theta^{-4}$, with uniform constants. More explicitly, for some $0 < k < K < \infty$,

$$\begin{aligned} 1 \leq M \leq k\theta^{-4} &\implies V_M^\theta - V_T^\theta \leq -C_e \theta^2 < 0, \\ M \geq K\theta^{-4} &\implies V_M^\theta - V_T^\theta \geq \tfrac{1}{2} c_e \theta^2 > 0. \end{aligned}$$

The first assertion is vacuous if its integer range is empty.

Proof. Write $E_\theta = V_*^\theta - V_T^\theta$ and $L_M^\theta = V_*^\theta - V_M^\theta$. Theorems already proved give $E_\theta \leq C_e \theta^2$ and $L_M^\theta \geq c_s M^{-1/2}$. Their exact difference uses the same experiment and baseline, hence

$$V_M^\theta - V_T^\theta = E_\theta - L_M^\theta \leq C_e \theta^2 - c_s M^{-1/2}.$$

Choose $k = (c_s/(2C_e))^2$, decreasing it if necessary so that $k < K$. The first inequality follows. The second is Theorem F.2, with K increased if needed. The admissible filter classes are nested

as M increases because extra states can be unused. Thus the first-crossing budget exists; the two bounds and integer rounding give its stated order. $\square$

This consequence was extracted in the source-pinned v7 referee technical note [19]. It concerns beating the exact, uncompressed T_θ comparator. It does not locate an exact integer transition, and it does not assert equality of the full and erased experiments at smaller *equal compressed* budgets. At zero T_0 is sufficient, so no finite-state policy has value exceeding $V_T^0 = V_*^0$.

APPENDIX G. A COLLISION BIT AT A COMMON FINITE-MEMORY BUDGET

This section compares total risks for one fixed target, not regrets against two different Bayes forecasters. It concerns an uncensored raw detector, so that erasure is literal postprocessing. It does not identify this operation with erasure *before* a commanded comparator.

G.1. The counted mechanical realization. Let the triangular lattice have fundamental area $\mathfrak{a} = \sqrt{3}/2$, let $I_R = [9/20, 47/100]$, and put a reflecting disk of radius R at each lattice point. In each cartridge prepare a uniform position in the ambient torus, a uniform velocity direction, and independent uniform apparatus coordinates. A guarded insertion fails when the position is inside a disk. Otherwise a unit-speed particle is released. Every attempted insertion consumes one cartridge.

Fix $0 < T < 3/50$ and $\epsilon > 0$ with $\epsilon(47/100)^2 < 1$. Distinct lifted disks have separation at least $3/50$; an outgoing reflected ray cannot reenter the same convex disk before another collision. Thus each released particle has at most one collision by time T . At a collision let z be its impact coordinate relative to the disk center. The fixed detector reports $+$ when its prepared uniform coordinate is at most $1/2 + \epsilon|z|^2/\pi$, and otherwise reports $-$. The controller never receives z or the uniform seed. This is an internally exact-coordinate apparatus, not a lower bound on the complexity of physical measurement.

For completeness, incoming one-collision states are parameterized by

$$q = Rn_\alpha - sv^-, \quad v^- = -\cos\phi n_\alpha + \sin\phi t_\alpha, \quad 0 < s < T, \quad |\phi| < \pi/2.$$

The backward segment remains outside its own disk by convexity and cannot reach a distinct disk within T by separation. The forward reflected segment has the same exclusion. Every regular colliding initial state has its unique impact triple. At fixed incoming laboratory angle the position Jacobian is $R \cos\phi$, and the angular coordinate change has determinant of absolute value one. Integration against ambient preparation $dq d\vartheta/(2\pi\mathfrak{a})$ gives collision mass

$$\frac{R}{2\pi\mathfrak{a}} \int_0^T \int_0^{2\pi} \int_{-\pi/2}^{\pi/2} \cos\phi d\phi d\alpha ds = \frac{2TR}{\mathfrak{a}}.$$

Tangencies are null, not an excluded positive-mass neighborhood. Since $|z| = R$ at impact, the four uncensored raw probabilities are

$$(101) \quad \begin{aligned} k_\partial &= \pi R^2/\mathfrak{a}, & k_0 &= 1 - \pi R^2/\mathfrak{a} - 2TR/\mathfrak{a}, \\ k_\pm &= TR/\mathfrak{a} \pm 2T\epsilon R^3/(\pi\mathfrak{a}). \end{aligned}$$

They are positive on I_R and sum to one. For k_0 , positivity follows already at $R = 47/100$, $T = 3/50$, using $\mathfrak{a} > 0.86$ and $\pi < 22/7$; the other inequalities follow from the stated bounds. Their coefficient matrix in degrees zero through three is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & -2T/\mathfrak{a} & T/\mathfrak{a} & T/\mathfrak{a} \\ \pi/\mathfrak{a} & -\pi/\mathfrak{a} & 0 & 0 \\ 0 & 0 & 2T\epsilon/(\pi\mathfrak{a}) & -2T\epsilon/(\pi\mathfrak{a}) \end{pmatrix},$$

with determinant of absolute value $4T^2\epsilon/\alpha^3$. A bounded lookup comparator therefore realizes $A = \{0, 1, 2, 3\}$. Merging the signs gives the three probabilities $k_\partial, k_0, k_+ + k_-$, which span $A = \{0, 1, 2\}$. This supplies the dense special cases of Theorem 1.3 from one counted apparatus. The stronger complete-record response, positive filter, boundary-control and surrogate derivations are preserved in the source companion [9]; the collision calculation above uses their whole-preparation normalization rather than conditioning away insertion failures.

G.2. One target and all state budgets. Consume two cartridges without censoring. Let $X \in \{\partial, 0, +, -\}$ be the first raw report, and let Y be the placement-failure indicator on the second cartridge. After the first report the controller stores one of M states, then announces its forecast $a \in [0, 1]$ and incurs $(a - Y)^2$. The erased experiment observes $e(X) \in \{\partial, 0, H\}$ with $e(+)=e(-)=H$. Both experiments use the same prior, both ambient attempts, and the same second-cartridge variable Y . There is no acceptance charge or saved cartridge. This is a separate uncensored experiment; it is not subject to the lower acceptance margin of the streaming theorem.

Put $h(R) = \pi R^2/\alpha$ and define

$$p_x = \mu k_x > 0, \quad z_x = \mu(k_x h), \quad b_x = z_x/p_x, \quad p_B = \sum_{x \in B} p_x, \quad z_B = \sum_{x \in B} z_x.$$

For a nonempty subset B set

$$(102) \quad \delta(B) = \sum_{x \in B} \frac{z_x^2}{p_x} - \frac{z_B^2}{p_B} = \frac{1}{p_B} \sum_{\{x,y\} \subset B} p_x p_y (b_x - b_y)^2.$$

All these quantities are finite combinations of prior moments through order five and the exact physical constants.

Theorem G.1 (Complete common-risk finite-state comparison). *For either alphabet $\mathcal{X}$ (four raw symbols or three erased tags), its optimal M -state risk, allowing independent randomization, is*

$$(103) \quad R_M(\mathcal{X}) = \mu h - \max_{\substack{\mathcal{Q} \text{ partition of } \mathcal{X} \\ |\mathcal{Q}| \leq M}} \sum_{B \in \mathcal{Q}} \frac{z_B^2}{p_B}.$$

A maximizing partition, its block index and block forecasts z_B/p_B give an attained finite-state implementation.

Let R_M^+ and R_M^0 denote the raw and erased risks and $\Gamma_M = R_M^0 - R_M^+$. Then $\Gamma_M \geq 0$ for every M , $\Gamma_1 = 0$, and, with $\Delta = \delta(\{+, -\})$,

$$(104) \quad \Gamma_3 = \Delta - \min_{x \neq y} \delta(\{x, y\}), \quad \Gamma_M = \Delta \quad (M \geq 4).$$

The case $M = 2$ is exactly (103): it compares the seven two-block raw partitions and the three two-block erased partitions (one-block partitions are dominated). Equivalently, if D_M^+ and D_M^0 are the minimum sums of merge costs in the two alphabets, then

$$(105) \quad \Gamma_M = \Delta + D_M^0 - D_M^+.$$

These are common total Bayes risks, not two separately normalized memory regrets.

Proof. Conditional on a block message, the optimal forecast is z_B/p_B . Expanding squared loss gives block contribution $z_B - z_B^2/p_B$; summing proves the risk of every deterministic partition. For arbitrary random encoders and decoders, first replace a randomized forecast at each message by its conditional mean. Holding these at most M forecasts fixed, each raw symbol can be assigned deterministically to a forecast minimizing its conditional squared loss. This cannot

increase risk. Reoptimizing each resulting block forecast also cannot increase risk. Thus a deterministic partition is optimal, and the finite maximum is attained. A fresh public seed independent of X, Y merely averages such risks and does not improve the optimum.

The identity (102) is obtained by expanding the weighted variance. The lossless raw risk is $\mu h - \sum_x z_x^2/p_x$; the lossless erased risk exceeds it by Δ . Adding the optimal merging distortion in each alphabet gives (105). Any erased code is a raw code after the deterministic map e , proving $\Gamma_M \geq 0$ directly. With one state both forecasts equal μh . With at least four states both alphabets are losslessly encoded. With three states the erased alphabet is lossless, and an optimal raw partition can be refined to exactly three blocks without increasing risk. Such a partition merges exactly one pair. This proves (104). The partition counts for $M = 2$ are the numbers of nontrivial subsets modulo complementation, namely $(2^4 - 2)/2 = 7$ and $(2^3 - 2)/2 = 3$. $\square$

Write $\mu_j = \mu(R^j)$, let $\mu_H(dR) = R\mu(dR)/\mu_1$, and put

$$v = \text{Var}_{\mu_H}(R^2), \quad \bar{c} = 1/2 + (\epsilon/\pi)\mu_H(R^2).$$

The exact lossless gain retained from [9] is

$$(106) \quad \Delta = \frac{2T\mu_1}{\mathbf{a}} \frac{(\epsilon v/\mathbf{a})^2}{\bar{c}(1-\bar{c})} > 0.$$

Indeed conditional on a first hit the sign probability is $c(R) = 1/2 + \epsilon R^2/\pi$, and $\text{Cov}_{\mu_H}(h, c) = \epsilon v/\mathbf{a}$. The two conditional-mean shifts are respectively this covariance divided by $\bar{c}$ and its negative divided by $1 - \bar{c}$. Their weighted squared sum is the covariance squared over $\bar{c}(1 - \bar{c})$. Multiplication by the hit probability proves (106), which is also precisely the pair cost $\delta(\{+, -\})$. Full support makes $v > 0$. This is an explicit conditional-variance calculation, not a new general comparison-of-experiments principle [1].

Corollary G.2 (A genuine finite-resolution threshold). *Fix $T > 0$ and a full-support prior on I_R . There exists $\epsilon_* > 0$ such that, for $0 < \epsilon < \epsilon_*$,*

$$(107) \quad \Gamma_M = 0 \quad (M = 1, 2, 3), \quad \Gamma_M = \Delta > 0 \quad (M \geq 4).$$

Thus a four-state, two-bit controller realizes the strict gain, whereas all smaller state budgets have exactly zero gain in this amplitude range. This does not identify exact algebraic dimension with useful accuracy.

Proof. At $\epsilon = 0$, $b_+ = b_- = b_H$. The three distinct tag means satisfy

$$b_0 < \mu h < b_H < b_\partial.$$

The first strict inequality follows because k_0 is strictly decreasing and h strictly increasing, so $\text{Cov}_\mu(k_0, h) < 0$. For example this sign follows by integrating $(k_0(R) - k_0(R'))(h(R) - h(R'))/2$ against two independent prior draws. The second inequality uses the positive covariance of R and h . The last follows in the hit-tilted prior from the same positive covariance of R and h , since the placement posterior is its additional R -tilt. Full support makes all these covariances strict.

For three states the unique zero-distortion merging pattern at amplitude zero is to keep the three distinct tag means separate and merge the two signs. Every other pair has positive merge cost. For two states, no optimal partition at amplitude zero splits the equal-mean signs between its two blocks. To prove this, use the optimal block means as centers. Equal-mean points assigned to different centers must be equidistant from them; otherwise reassignment to the closer center strictly improves risk. If the centers are distinct, b_H is their midpoint. Each block must then contain a different extreme tag to place its mean strictly on the corresponding

side of b_H . Move one sign into the other block, holding centers fixed; the cost is unchanged. The vacated-sign block still contains its extreme tag, so reoptimizing its center strictly reduces the cost, a contradiction. If the two centers agree, the achieved risk is the one-state risk, strictly worse than separating one extreme tag. Thus an optimal two-state partition never splits the signs.

There are finitely many partitions. At amplitude zero every partition that splits the signs and uses at most two states has a strictly positive risk gap to the optimum, as does every non-sign-merging three-state partition. All their risks depend continuously on ϵ because the cell masses stay positive. Therefore, for a sufficiently small common positive ϵ_* , none of these partitions is optimal for $M = 2, 3$. Every remaining partition keeps the signs together and has exactly the risk of the corresponding erased partition, because both $p_+ + p_-$ and $z_+ + z_-$ are independent of ϵ . This proves equality of risks for $M = 2, 3$; $M = 1$ and $M \geq 4$ follow from Theorem G.1 and (106). $\square$

The threshold is checkable from finitely many moment expressions: enumerate the partitions in (103), find their positive gaps at zero, and bound their continuous rational derivatives while cell masses stay bounded away from zero. No prior-uniform or horizon-uniform threshold is claimed. In particular (106) can be extremely small for a concentrated prior. The threshold theorem gives an exact resource comparison, not a promise of a practically large gain.

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